



GE HealthCare

SIGNA™ Prime MAX / SIGNA™ Prime Elite

Magnet Handling Manual

5951109-1EN
Revision 3

General Service Documentation

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Contents

- Chapter 1 Getting started 5**
 - 1.1 Overview5
 - 1.2 Examining packages for damage in transportation6
 - 1.3 Truck loading specifications7
 - 1.4 Predelivery instructions10
- Chapter 2 Unloading and moving the magnet 13**
 - 2.1 Shipping and crate configurations13
 - 2.2 Equipment requirements15
 - 2.3 Removing subsystem crates18
 - 2.4 Handling the crated magnet in transit with a crane18
 - 2.5 Handling the crated magnet in transit with a forklift18
 - 2.6 Removing the shipping cage20
 - 2.7 Removing the shipping crate26
 - 2.8 Unloading and moving the uncrated magnet with a forklift32
 - 2.9 Unloading the uncrated magnet with a crane to ground35
 - 2.10 Lifting the magnet with a crane through the opening in the exterior wall38
- Chapter 3 Magnet storage conditions, pending ramp 41**
 - 3.1 Converting an air magnet shipment to warehouse configuration41
 - 3.2 Connecting the magnet to the compressor48
 - 3.3 Connecting the magnet to the SVU54
 - 3.4 Monitoring the magnet57
- Chapter 4 Doing in-transit service and magnet system checks 59**
 - 4.1 Doing a magnet system check59
- Chapter 5 Preparing the scan room 61**
 - 5.1 Preparing the scan room for leveling magnet61
- Chapter 6 Installing a nonseismic vibroacoustic damping mat and leveling the magnet 69**
 - 6.1 Equipment overview69
 - 6.2 Positioning a nonseismic mounted vibroacoustic damping mats70
 - 6.3 Preparing to move the magnet71
 - 6.4 Moving the magnet to the MR suite75
 - 6.5 Preparations for centering the magnet75
 - 6.6 Centering the magnet in the MR suite84
 - 6.7 Lowering the magnet into position85

6.8 Leveling the magnet.....	87
Chapter 7 Installing a seismic vibroacoustic damping mat and leveling the magnet	91
7.1 Equipment overview	91
7.2 Positioning a seismic mounted vibroacoustic damping mat.....	92
7.3 Preparing to move the magnet.....	95
7.4 Moving the magnet to the MR suite.....	98
7.5 Lowering the magnet into position	98
7.6 Adding leveling shims	99
7.7 Leveling the magnet.....	100
7.8 Seismic mounting of a vibroacoustic damping mat to the floor.....	104
Chapter 8 Restoring the front and rear endbell	109
Revision history	121

Chapter 1 Getting started

1.1 Overview

Safety
Before working in any GE Healthcare MR suite or performing any GE Healthcare service procedure, you must: • Have read and understood all hazard conditions and safety requirements in the latest revision of the GE Healthcare MR Service Safety Manual (5452735). • Have successfully completed all relevant GE Healthcare Environmental Health and Safety (EHS) courses (or for non-GE employees, equivalent workplace training courses). • Comply with all site-specific training and workplace safety requirements. If you have any safety concerns at any time, do not begin work or immediately stop work and move to a safe location. Immediately contact your supervisor or site safety officer for instructions on how to proceed.

This Magnet Handling Manual addresses moving, storage, delivery, and installation of a JXM series (as defined by the three characters of the magnet serial number) zero boil-off magnet. The sequence of events involved in magnet delivery and installation, along with functions, responsibilities, and associated documentation, are shown in the table below.

Table 1-1 Magnet handling functions and responsibilities

	Function	Responsibility	Reference document
1	MR Magnet Safety Document Review and Compliance	GE Project Manager of Installation (PMI), GE Field Engineer and Rigger	The MR Service Safety Manual (5452735).
2	Site Delivery and Review: • Access and Route • Clearances	GE Project Manager of Installation (PMI), GE Field Engineer and Rigger	This Magnet Handling Manual and the appropriate Preinstallation Manual (5941479-8EN).
3	Magnet Transportation	Transportation Team, Rigger	Section 1.3 Truck loading specifications on page 7 in this manual.
4	Magnet Delivery	Rigger	Unloading and moving the magnet in this manual.
5	Prepare the Scan Room: • Clear the Room • Put the Positioning Template in Position • Set Up the Lasers • Mark the Room for Reference	GE Field Engineer	Chapter 5 Preparing the scan room on page 61 in this manual.
6	Put Vibroacoustic Damping Mats in Position	Rigger	Chapter 6 Installing a nonseismic vibroacoustic damping mat and leveling the magnet on page 69 or Chapter 7 Installing a seismic vibroacoustic damping mat and leveling the magnet on page 91 in this manual.

Table 1-1 Magnet handling functions and responsibilities (Table continued)

	Function	Responsibility	Reference document
7	Moving Magnet to MR Suite	Rigger	Chapter 6 Installing a nonseismic vibroacoustic damping mat and leveling the magnet on page 69 or Chapter 7 Installing a seismic vibroacoustic damping mat and leveling the magnet on page 91 in this manual.
8	Magnet Leveling and Bolt Down	Rigger and GE Field Engineer	Chapter 6 Installing a nonseismic vibroacoustic damping mat and leveling the magnet on page 69 or Chapter 7 Installing a seismic vibroacoustic damping mat and leveling the magnet on page 91 in this manual.
9	Magnet Cryocooler Connections pending room installation as soon as possible but no later than 6 hours	GE Field Engineer	Magnet storage conditions, pending ramp in this manual.
10	Magnet Conversion to Operating Configuration	GE Field Engineer	Installation/Conversion sections (for the defined magnet style) in the Magnet and Cryogen Manual (5951110-8EN).
11	Exhaust Gas Vent Connection as soon as possible - must be done prior to ramping magnet	GE Field Engineer	Exhaust Gas Vent Connection section (for the defined magnet style) in the Magnet and Cryogen Manual (5951110-8EN).
12	Continue with Magnet Commissioning using the Setup and Calibration chapter (for the defined magnet style) in the Magnet and Cryogen Manual (5951110-8EN).		

Photos and images contained within this manual are representative of the system(s) and configuration(s) shipped. The system received may vary slightly.

The latest release of this manual can be obtained through the SIMS Content Viewer or through your GE Healthcare Field Service Representative. Before using this document, make sure you are using the most current released version of this document.

1.2 Examining packages for damage in transportation

About this task

Examine all packages closely at delivery. If damage is apparent, do the following:

Procedure

1. Make sure the notation *damage in shipment* is written on all copies of the freight or express bill before delivery is accepted or signed for by a GE HealthCare representative or a hospital receiving agent.
2. Whether noted or concealed, you must report damage to the carrier immediately upon discovery, or in any event, within 14 days after receipt, and hold the contents and containers for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this 14-day period.

- 3. File a report of the damage. Contact your local service coordinator for more information on this process.

1.3 Truck loading specifications

Prerequisites



NOTICE

EQUIPMENT DAMAGE RISK

Improper transportation can result in damage to the magnet.

Review guidelines with the carrier prior to transporting the magnet. Any other shipping configuration must be demonstrated through testing and be approved by GE Healthcare.

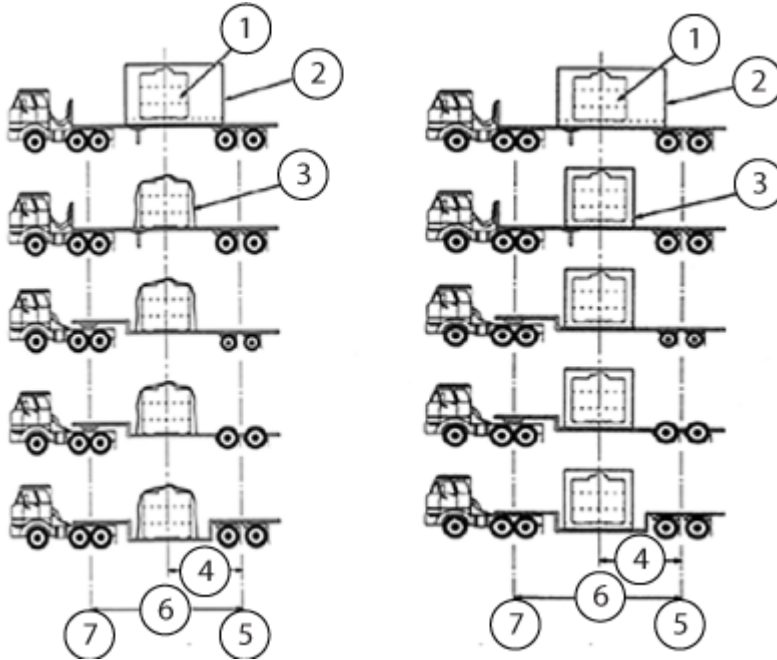
Required conditions
Air ride trailers must be used.
The magnet must be centered on the trailer with the magnet bore <u>perpendicular</u> to the truck.
The magnet must not be positioned over the trailer axles.
Extreme care must be used during forklift use. The magnet crate must be picked up from the sides only. The forks must be put directly under the four magnet feet. The magnet can be identified by the steel plates attached to the pallet. See Unloading and moving the magnet for more detailed handling requirements pertaining to forklift and/or crane moves.
Crate and/or frame top and sides must not be used to secure the magnet to the trailer. Magnets should be secured to the trailer using crate/magnet base.

Required conditions

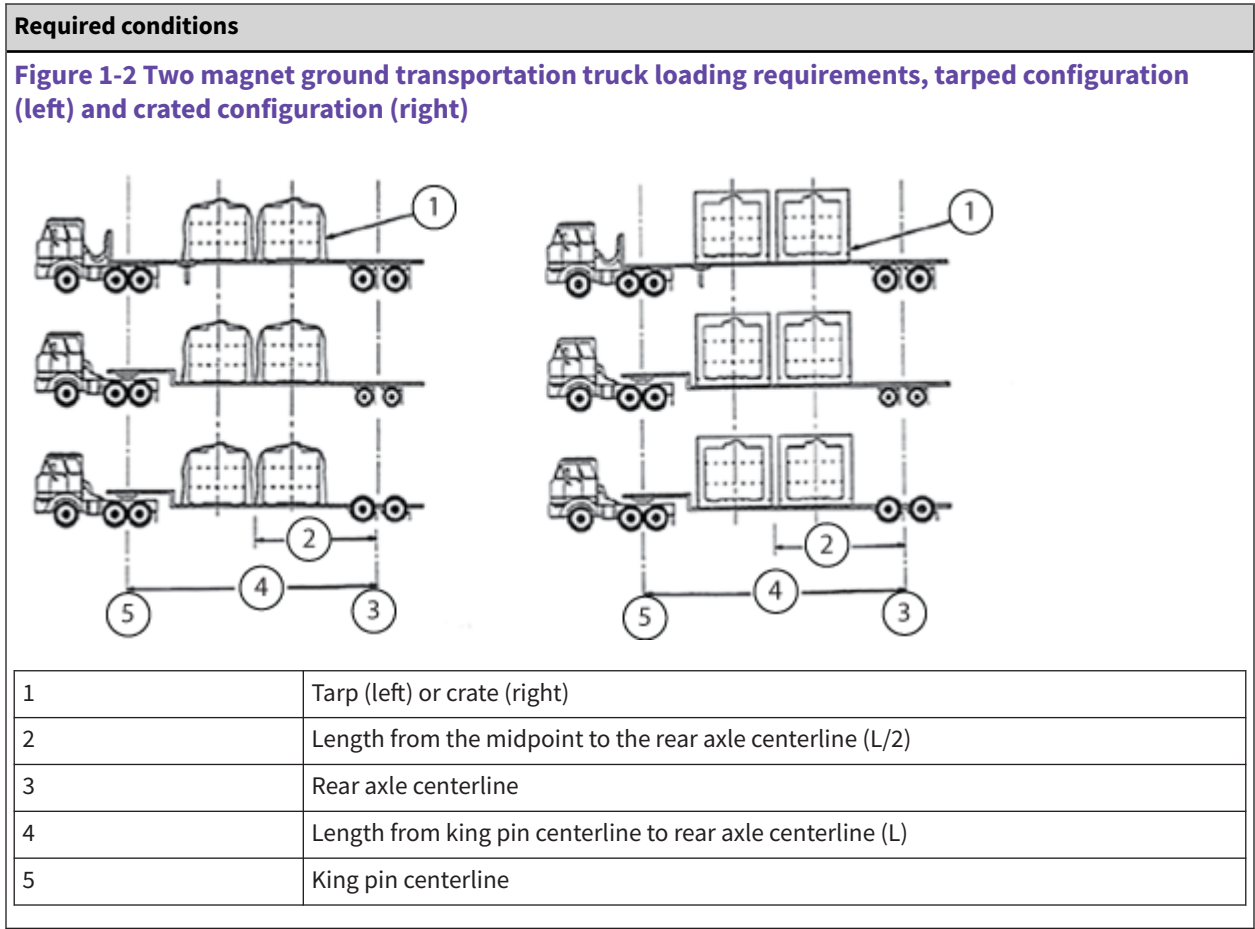
A maximum of two magnets per trailer is allowed. Acceptable dual load configurations are as follows:

- Two 1.5T magnets
- One 1.5T magnet and one 3.0T magnet

Figure 1-1 Single magnet ground transportation truck loading requirements, tarped configuration (left) and crated configuration (right)



1	Magnet
2	ISO box container
3	Tarp (left) or crate (right)
4	Length from the midpoint to the rear axle centerline ($L/2$)
5	Rear axle centerline
6	Length from king pin centerline to rear axle centerline (L)
7	King pin centerline



About this task

Table 1-2 Magnet loading specifications

Max tilt when suspended	30°
Allowable shipping modes	- Airplane (any plane that has openings large enough to accept a magnet) - Air ride trailer (see Figure 1-1 Single magnet ground transportation truck loading requirements, tarped configuration (left) and crated configuration (right) on page 8 and Figure 1-2 Two magnet ground transportation truck loading requirements, tarped configuration (left) and crated configuration (right) on page 9) - Boat or ocean-going ship
Forklift capability	Yes
Shipping temperature	-30 to 50°C (-22 to 122°F)
Maximum shock load	1.5g

1.4 Predelivery instructions

Prerequisites

NOTICE

EQUIPMENT DAMAGE RISK

Impacts/jolts/drops to the magnet while lifting/moving/lowering the magnet can cause expensive internal magnet damage.

The rigger is responsible for actual equipment/procedures used to lift and move a magnet into the customer facility, including through a raised opening in an exterior wall. The following EXAMPLE procedure only outlines the concept of one method.

Lift/move/lower the magnet smoothly. Do not let the magnet bump or hit anything forcefully. Avoid tilting the magnet more than the maximum tilt (30° from horizontal level). Do not apply any force to the magnet enclosures. Protect all customer surfaces during any and all move operations.

NOTICE

IMAGE QUALITY RISK

Improperly located anchors can cause image quality issues.

Make sure that all equipment anchors are located in conformance with the site's architectural drawings and are installed and tested per the *Anchor Hardware Requirements for MR Equipment inside RF Shield Room* section of the appropriate *Preinstallation Manual* (5941479-8EN).

Procedure

1. Before magnet delivery, the GE PMI and rigger must do the following:
 - 1.1. Visit the magnet site with the rigging foreman before magnet delivery to plan the move. GE PMI must hand-deliver a copy of this document to the rigging foreman.
 - 1.2. Review the guidelines for shipping/handling and for equipment/tools with the rigging foreman per Unloading and moving the magnet in this manual.
 - 1.3. Caution the rigger that the magnet is extremely fragile. Sudden jolts can damage the magnet. (Make riggers aware of the cost of a magnet replacement. This usually promotes more care while handling the magnet.)
 - 1.4. Make sure all routes and paths leading to the magnet room are level and free from obstacles and holes. (The rigger will be required to construct platforms where needed.)
 - 1.5. Make sure that clearance dimensions along the path to the magnet room meet the requirements stated in the *Preinstallation Manual* (5941479-8EN). In the case of a magnet being lifted by a crane through a raised opening, make sure the opening is of sufficient size to fit the magnet and all rigging equipment without interference.
 - 1.6. If roller dollies are to be used, have the rigger bring steel plates to put along the delivery route.

- 1.7. The rigger must take actions necessary to ensure that walls, floors, and so on along the transportation route/path are be protected from potential damage.
2. Before magnet delivery, the GE PMI and Shield Room Vendor must do the following:
 - 2.1. Make sure that floor levelness specifications stated in the *Preinstallation Manual* (5941479-8EN) is met after the finished flooring is installed.
 - 2.2. Make sure that the vent is located according to the specifications stated in the *Preinstallation Manual* (5941479-8EN).
 - 2.3. Make sure that markings are present on the magnet room floor in accordance to the specifications stated in the *Preinstallation Manual* (5941479-8EN).

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Chapter 2 Unloading and moving the magnet

2.1 Shipping and crate configurations

Safety	
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	 DANGER POTENTIAL ASPHYXIATION HAZARD Loss of magnet vacuum will result in the rapid expulsion of helium gas, which can cause asphyxiation in enclosed areas. Use extreme caution and do not contact or damage the vacuum vessel during magnet transit or siting.
	 WARNING POTENTIAL INJURY Magnet is an unbalanced load. Tilting can result in a hazardous condition that can cause magnet damage or serious personal injury. Do the following to avoid tilting: • Make sure that the lifting apparatus (crane, spreader beam, and so on) meets the specifications stated in this manual. • Adjust the lifting cables/slides and spreader beam (if used) lift point to level the magnet before fully lifting the magnet off the surface. • Make sure that the entire area where lifting will occur is free of obstructions and unauthorized personnel. • Make sure the surface where the magnet will be put after lifting is flat. • Do not crane lift a magnet during dangerous weather conditions.

Safety**WARNING**

RISK OF DEATH, PERSONAL INJURY, OR EQUIPMENT DAMAGE

The use of used and/or damaged bolts, lock washers, flat washers, and nuts could result in equipment and/or component damage, death, or serious physical injury!

Use new, undamaged mounting hardware.

**NOTICE**

COMPONENT DAMAGE RISK

Improper transportation can result in damage to the magnet.

Tasks in this section are to be done by riggers, not by GE HealthCare Personnel.

Weights referenced in section [2.2 Equipment requirements on page 15](#) are rigging weights (rounded up from actual weights) and are not actual shipping weights. Refer to magnet shipping documents for actual weights of the magnet and crate.

Do not use a crane to lift a magnet that is on a pallet or inside a crate. Crane lifting can only be done using the magnet lifting holes, which are not accessible while the magnet is inside the crate.

Domestic Shipping Crate (5151624)

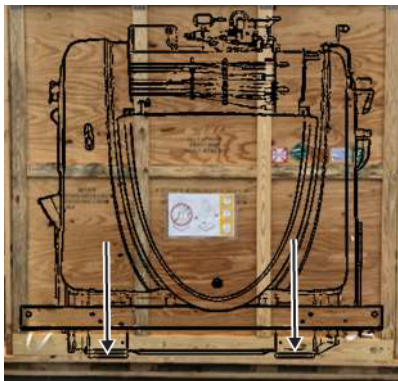
Figure 2-1 Shipping crate



- Approved for use on truck transportation only. (Not approved for flight or ocean usage.)
- Cage/frame positioned over/around magnet after magnet is loaded onto truck.
- Cage must be removed prior to magnet unload from truck (making lifting holes and lifting angle brackets accessible).

Platform International Shipping Crate (5806856-3)

Figure 2-2 Fork position



-  **NOTE**
- Magnet image superimposed on crates is for reference only. Magnet image shown is representative and may differ from the configuration shipped.
- Approved for use on truck, ocean, or air transportation.
 - Forklift approved moves only. (Crane movement of magnet in crate is not permitted.)
 - Crate is designed to put forklift forks directly beneath magnet feet - lifting from the side of the magnet, perpendicular to the magnet bore.
 - Minimum distance between forks = 1016 mm (40 inches).

2.2 Equipment requirements

Table 2-1 Forklift requirements*

Item and quantity	Equipment specification/rating		Furnished by	Function
Forklift quantity = 1	Magnet weight only (as shipped configuration with gradient, does not include crate)	5012 kg (11,050 lbs)	Rigger	Unloading or moving magnet
	Magnet only - minimum distance between forks using lifting angle bracket	1635 mm (64 inches)		
	Minimum fork length	2010 mm (79 inches)		
*Forklift and forklift equipment must be rated for the referenced magnet loads.				

Table 2-2 Crane requirements*

Item and quantity	Equipment specification/rating		Furnished by	Function
Crane quantity = 1	Magnet weight only (as shipped configuration with gradient, does not include crate)	5012 kg (11,050 lbs)	Rigger	Unloading or moving magnet
Lifting ring quantity = 4	The minimum force	4.75T		
Slings, hoists, bridles, shackles	Requirement to be determined by rigger.  NOTE Refer to magnet weights noted above.			

*Crane and crane equipment must be rated for the referenced magnet loads.

Table 2-3 Miscellaneous equipment and tools*

Item	Equipment/tool required	Responsible	Function
Magnet mechanical inter-face drawing (bare magnet)  NOTE Interface drawings may not contain shipping configurations and are for reference only	See clearance dimensions in 6.3 Preparing to move the magnet on page 71 .	Project manager of installation (PMI)	Identifying magnet dimension and features
Magnet weight only (as shipped configuration with gradient, does not include crate)	5012 kg (11,050 lbs)	Reference	Moving the magnet
Hydraulic or toe jack	Must support one end of magnet on two jacks or both ends of magnet on four jacks.  NOTE Refer to magnet weights noted above.	Rigger	Raising the magnet for roller dollies or leveling plates
Roller dollies	Must support magnet on at least three dollies. (Four dollies must be installed; however, the dollies must be able to support full weight on three dollies in case one foot is not touching.)  NOTE Refer to magnet weights noted above.	Rigger	Moving the magnet
Magnet aluminum shim pads, 3mm	5926243-301-39	GE Field Engineer	Leveling the magnet
Magnet aluminum shim pads, 1mm	5926243-301-40		
PH Alignment and Magnet Leveling Kit	5897979	GE Field Engineer	Leveling the magnet
*Miscellaneous equipment and tools must be rated for the referenced magnet loads.			

2.3 Removing subsystem crates

Procedure

1. Carefully inspect all packaging for damage that may have occurred during shipping.
2. Remove all subsystem crates, except the magnet crate, from the trailer/transport using a crane or forklift. To handle the magnet crate in transit from the truck or plane, see [2.5 Handling the crated magnet in transit with a forklift on page 18](#).
3. Inspect all crates for visible damage. Report any damage you find in conformance with the [1.2 Examining packages for damage in transportation on page 6](#) procedure.
4. Move subsystem crates to a receiving location protected from the weather, preferably close to and at the same level as the MR suite/magnet room.

2.4 Handling the crated magnet in transit with a crane

About this task

Do not use a crane to move the magnet if it is still in the crate. Using a crane requires clear access to top of the magnet, which are not accessible while the magnet is inside the crate.

2.5 Handling the crated magnet in transit with a forklift

Prerequisites

Safety
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Safety



NOTICE

EQUIPMENT DAMAGE RISK

Improper transportation during forklift operations can result in magnet damage.

Forklift must meet the minimum capacity and dimension requirements stated in section [2.2 Equipment requirements on page 15](#).

The magnet must be picked up from the magnet side orientation only with the forks inserted into the designated slots on the crate.

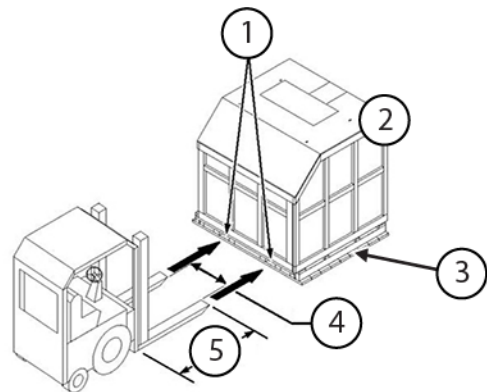
Avoid sudden jolts. Do not allow the crate/pallet to bump anything forcefully.

Avoid tilting the magnet/crate/pallet package more than the maximum tilt specified (30° from the horizontal level).

Procedure

- Put a forklift in position beside the magnet crate/pallet meeting the requirements stated in section [2.2 Equipment requirements on page 15](#).
- Carefully insert the forklift forks completely into the holes in the long side of the shipping pallet.

Figure 2-3 Forklift lifting points of magnet/crate/pallet package



1	Insert forks here only
2	Crate
3	Pallet
4	Crated magnet - minimum distance between forks: Platform International Shipping Crate (5806856-3): 1016 mm (40 inches)
5	Minimum fork length, 2363 mm (93 inches)

- Smoothly lift the crate/pallet with the forklift, and move the crate/pallet to the desired location.



NOTE

Magnet lift must be perpendicular to magnet bore. Crate slots for forklift are designed to ensure proper lift orientation.

- Lower the entire package to rest on a flat, smooth surface. Do not rest the magnet/crate/pallet on any surface that is not flat or strong enough to support the magnet/crate/pallet package.

- 5. Avoid tilting the magnet; every effort should be made to minimize tilt during transport or movement. Tilt must not exceed 30° from the horizontal level.

2.6 Removing the shipping cage

Prerequisites

Safety	
Before working in any GE HealthCare MR suite or doing any GE HealthCare service procedure, you must: • Have read and understood all hazard conditions and safety requirements in the latest revision of the GE HealthCare <i>MR Service Safety Manual</i> (5452735). • Have successfully completed all relevant GE HealthCare Environmental Health and Safety (EHS) courses (or for non-GE HealthCare employees, equivalent workplace training courses). • Comply with all site-specific training and workplace safety requirements. If you have any safety concerns at any time, do not begin work or immediately stop work and move to a safe location. Immediately contact your supervisor or site safety officer for instructions on how to proceed.	
	NOTICE EQUIPMENT DAMAGE RISK Improper removal of the shipping cage can result in damage to the magnet. Do not use a crane to move the magnet if it is still in the cage. Remove the magnet's shipping cage while the full magnet package is on the trailer/transport.

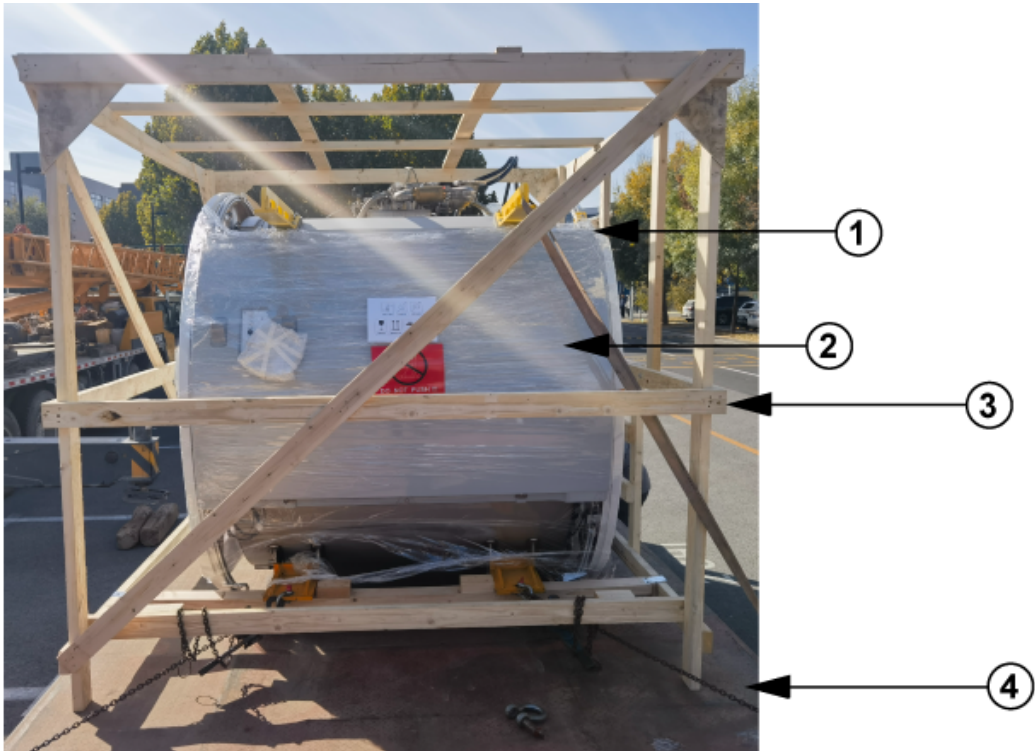
Safety

NOTICE



EQUIPMENT DAMAGE RISK

Care must be taken not to scrape or hit the sides of the magnet. The magnet is shipped inside plastic bubble wrap. The bubble wrap should be left intact until the magnet is set down inside the magnet room. Take care not to nick or cut any electrical cables during removal of the plastic bubble wrap. The sample shipping cage configuration (domestic shipping option) below shows intact bubble wrap.



1	Plastic bubble wrap
2	Magnet
3	Cage
4	Pallet

Procedure

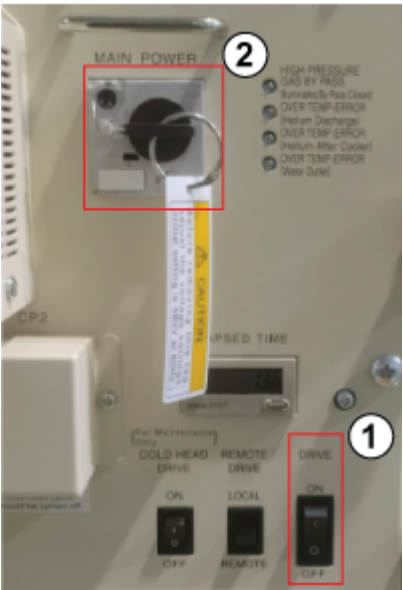
- 1. Power off the SVU by pressing the button.

Figure 2-4 SVU switch



- 2. Turn off the compressor:
 - 2.1. Turn off the compressor drive switch.
 - 2.2. Turn off the compressor main power switch.

Figure 2-5 Turn off the compressor

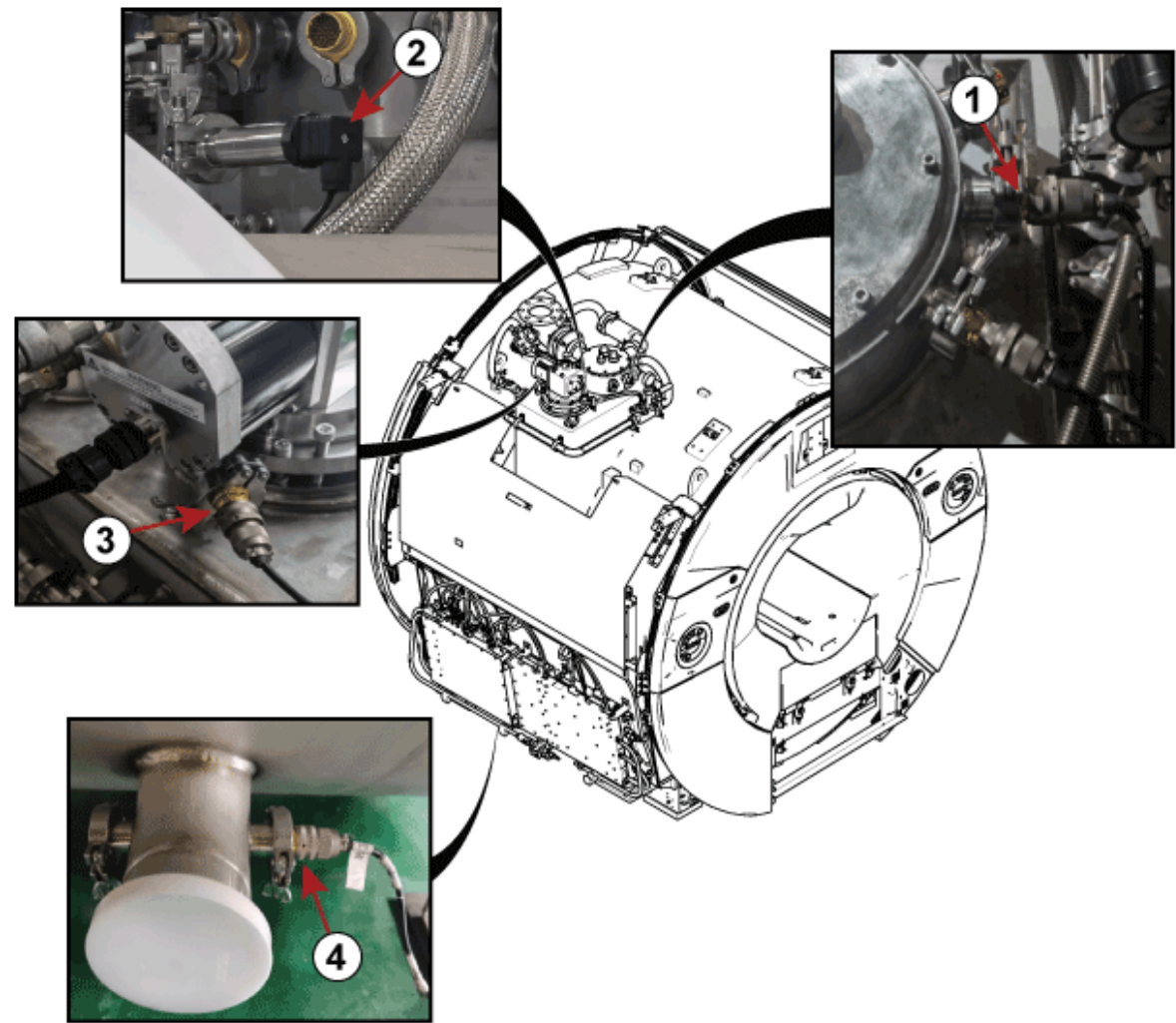


1	Compressor drive switch at the OFF position
2	Compressor main power switch at the OFF position

- 3. Disconnect the cables from the magnet.

! ATTENTION
A Phillips screw needs to be removed when disconnecting the Magnet P. Store the screw properly and DO NOT lose it.

Figure 2-6 Magnet B, D, E, and P cables



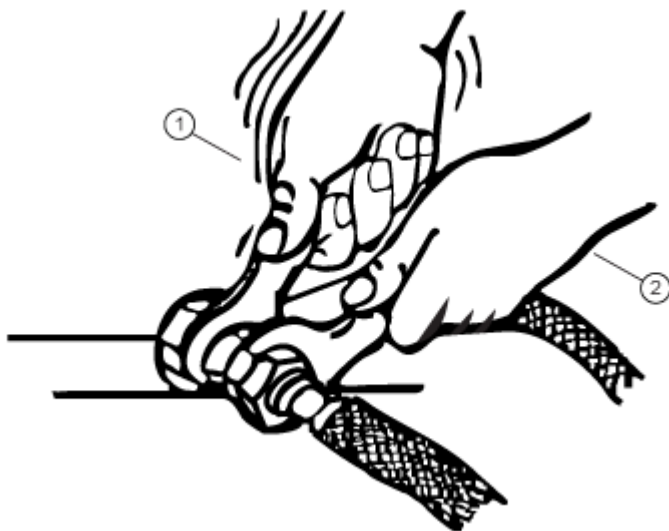
Item	Description
1	Magnet B
2	Magnet P
3	Magnet D
4	Magnet E

4. Using two wrenches (one to hold the backside adapter connection, and one to loosen), disconnect the coldhead compressor lines in the following order:

**NOTE**

The hold wrench maintains the backside adapter connection, making gas leaks in the system less likely.

Figure 2-7 Using two wrenches to loosen a connection

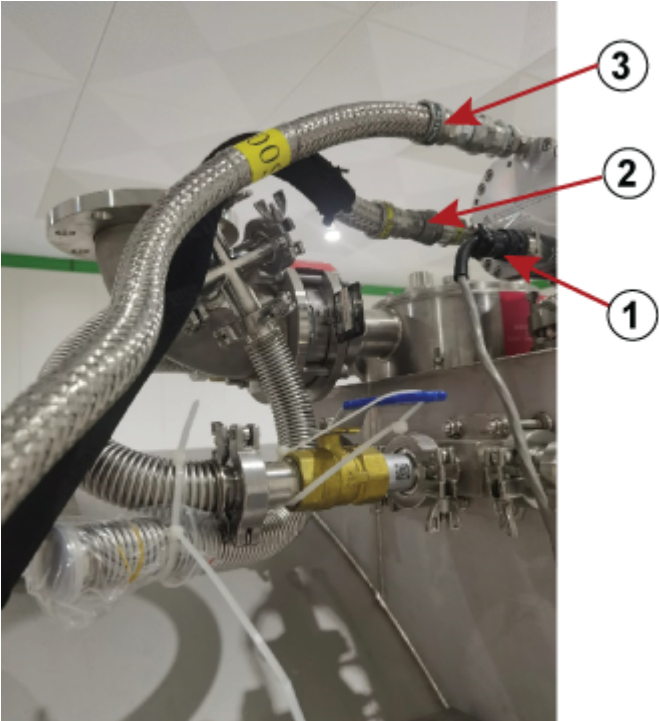


1	Loosen
2	Hold fast

**NOTE**

Do not put any bending force on Aeroquip fittings while connecting/disconnecting helium flexlines. A bending force will create difficulty in the rapid engagement/disengagement required to prevent helium loss and system contamination. Support the gas lines when connecting/disconnecting them to the compressor.

Figure 2-8 Coldhead compressor connections



1	First disconnection - coldhead power cable
2	Second disconnection - coldhead helium supply line
3	Third disconnection - coldhead helium return line

- 5. Remove the driver-supplied tarp from the shipping cage.
- 6. Unstrap the cage if any straps were applied. Then unchain the magnet from the flatbed.
- 7. To remove the shipping cage from the magnet, follow the steps below for one of the options:

Using a crane	1. Strap the frame in four locations. 2. Lift the cage straight up and move it away from the magnet.
Not using a crane	1. Unbolt the boards on one end of the cage. 2. Move the cage away from the magnet in the other direction.

2.7 Removing the shipping crate

Prerequisites

Safety

Before working in any GE HealthCare MR suite or doing any GE HealthCare service procedure, you must:

- Have read and understood all hazard conditions and safety requirements in the latest revision of the GE HealthCare *MR Service Safety Manual* (5452735).
- Have successfully completed all relevant GE HealthCare Environmental Health and Safety (EHS) courses (or for non-GE HealthCare employees, equivalent workplace training courses).
- Comply with all site-specific training and workplace safety requirements.

If you have any safety concerns at any time, do not begin work or immediately stop work and move to a safe location. Immediately contact your supervisor or site safety officer for instructions on how to proceed.



WARNING

EQUIPMENT DAMAGE RISK

Improper removal of the shipping crate can result in damage to the magnet.

Do not use a crane to move the magnet if it is still in the crate.

Remove the magnet's shipping crate while the magnet/crate/pallet package is on the trailer/transport.

Damage to the magnet and/or magnet enclosures may result if the crate is removed while the crate's side panels are in their closed (shipping) position, or if the crate is disassembled while the magnet is inside the crate.

Do not remove any lag screws on the crate or pallet until the crane is in position, and the slings/cable bridles are attached to the lifting rings on the crate.

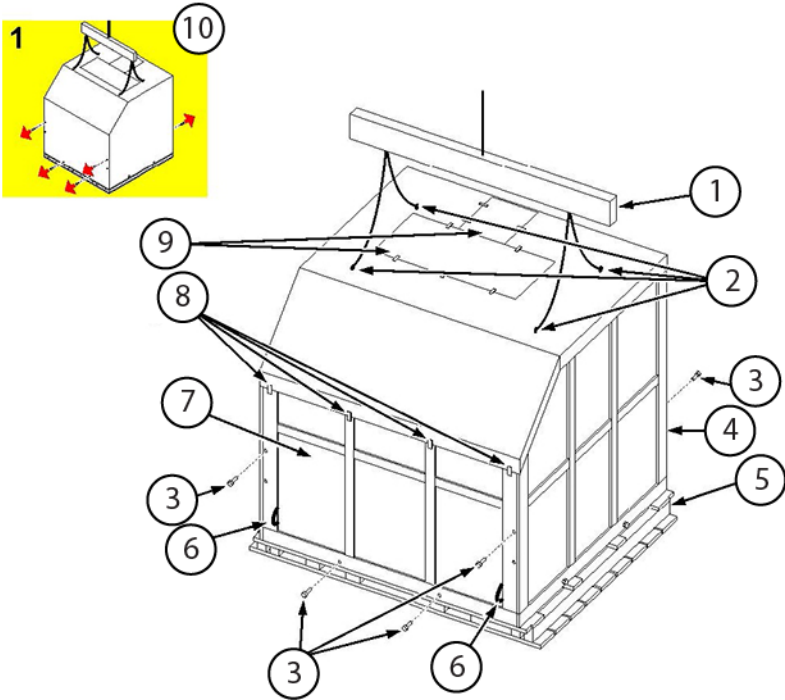
Procedure

1. Remove the driver-supplied tarp from the shipping crate.
2. Unchain the crate from the flatbed.
3. Put a crane meeting the specifications in Table 2 above the center of the crate.

- 4. Attach the slings/cable bridles of a spreader beam to the lifting rings on top of the crate.

NOTE  Either a hook/shackle and slings or a spreader beam and sling/cable bridles may be used during crate removal.

Figure 2-9 Crane positioning, lag screw removal to open crate side panels

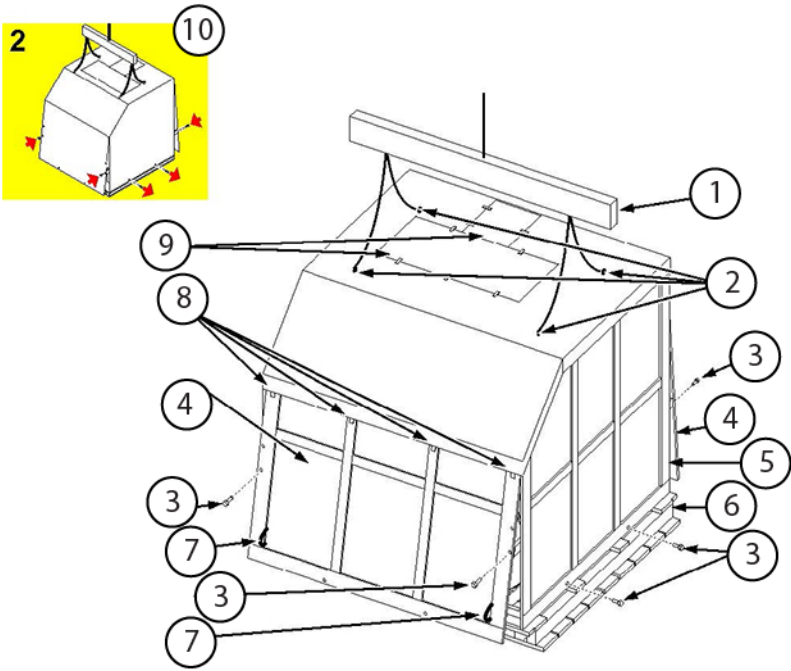


Item	Description	Item	Description
1	Spreader beam with slack lifting straps/ cables	6	Rope handle
2	Four lifting rings for lifting crate	7	Crate side panel
3	Lag screw	8	Side panel hinges
4	Front of crate	9	Access panels
5	Front of pallet	10	Graphic on crate

- 5. Remove the lag screws along the lower edges of the crate's left and right side panels that secure those panels to the pallet. Do not remove the lag screws securing the crate's front and back end panels to the pallet until after the crate's side panels are secured in their open position.
- 6. Remove the lag screws securing the crate side panels to the crate end panels. Do not remove the hinges along each crate side panel's upper edge.

- 7. Pull open the crate's left and right side panels using the rope handle on each side panel. The crate's left and right side panels are hinged along their upper edges.

Figure 2-10 Securing crate side panels in open position

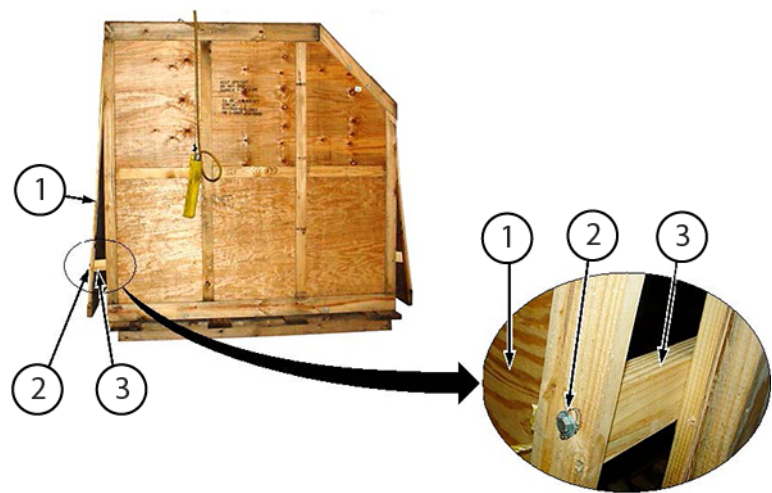


Item	Description	Item	Description
1	Spreader beam with slack lifting straps/ cables	6	Front of pallet
2	Four lifting rings for lifting crate	7	Rope handle
3	Lag screw	8	Side panel hinges
4	Crate side panel	9	Access panels
5	Front of crate	10	Graphic on crate

NOTE This figure shows the use of a spreader beam and cable bridles/slings. You can also use a hook or shackle and slings. All equipment must meet or exceed the size and loading specifications stated in this manual.

- 8. Swing the wooden block found inside each vertical corner of the crate outward.

Figure 2-11 Rear view of shipping crate with side panels raised

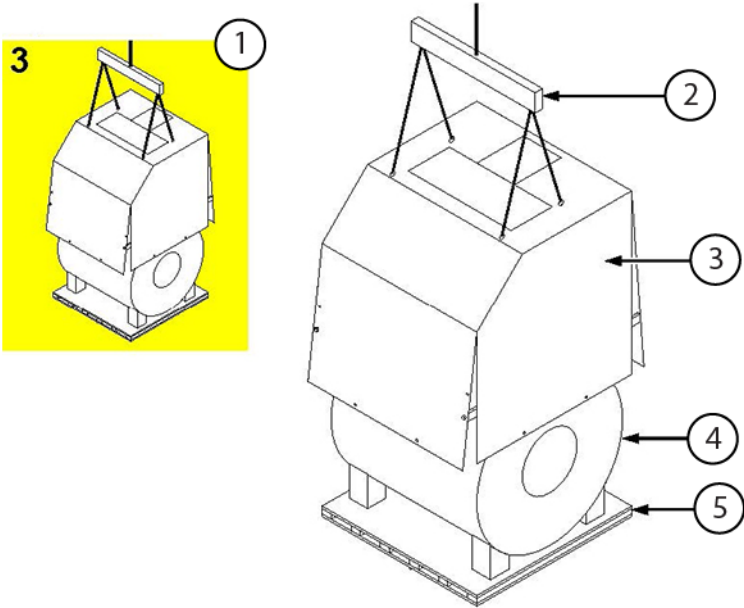


1	Side panel
2	Lag screw
3	Wooden block

- 9. Attach the loose end of each block to the adjoining side panel using one of the lag screws removed previously.
- 10. Remove the lag screws along the lower edges of the crate’s front and back end panels that secure those panels to the pallet. Refer to [Figure 2-10 Securing crate side panels in open position on page 28](#).

- 11. Tighten the slings/cable bridles and carefully begin lifting the crate. (Refer to graphic 3 on the shipping crate.) If the crate does not remain approximately level, carefully lower the crate and adjust the lifting configuration, then lift the crate again.

Figure 2-12 Lifting the crate using a crane



Item	Description	Item	Description
1	Graphic on crate	4	Magnet
2	Spreader beam with slack lifting straps/cables	5	Pallet
3	Crate	-	-

 **NOTE** This figure shows the use of a spreader beam and cable bridles/slings. You can also use a hook or shackle and slings. All equipment must meet or exceed the size and loading specifications stated in this manual.

- 12. Using the crane, lift the crate top off the magnet's shipping pallet and fully above and clearing the magnet.
- 13. Put the crate in a clear area.

14. Unbolt and remove the four bolts (one per magnet foot) securing the magnet feet to the pallet.

Figure 2-13 Unbolting the magnet from a shipping pallet



15. Visually examine the witness marks on each bolt and lifting bracket at each of the four magnet feet. Look for witness mark misalignment between the nut/bolt head and the lifting bracket.
 - If witness marks are aligned correctly, continue with [Step 16](#).
 - If witness marks do not align as shown above, complete these steps:
 - 15.2.1. Make sure the brackets are installed against the magnet feet at all four locations.
 - 15.2.2. Look inside of the magnet foot. If a through bolt is used (new process), make sure the lock washer is compressed under the bolt head. If it is not, tighten the bolt until the lock washer is compressed.
-  **NOTE** If the nut turns (new process only) but does not tighten, then the bolt hole on the magnet foot was drilled out and the head of the bolt (which is inside the magnet foot) will need to be held by a second wrench while torquing the nut.
- 15.2.3. Use a permanent marker to add a new witness mark only on the bolt or nut that was just torqued.
 - 15.2.4. Continue with [Step 16](#).
16. Rig and lift the magnet off the pallet in conformance with section [2.8 Unloading and moving the uncrated magnet with a forklift on page 32](#) or section [2.9 Unloading the uncrated magnet with a crane to ground on page 35](#) of this manual.

2.8 Unloading and moving the uncrated magnet with a forklift

Prerequisites

Safety

Before working in any GE HealthCare MR suite or doing any GE HealthCare service procedure, you must:

- Have read and understood all hazard conditions and safety requirements in the latest revision of the GE HealthCare *MR Service Safety Manual* (5452735).
- Have successfully completed all relevant GE HealthCare Environmental Health and Safety (EHS) courses (or for non-GE HealthCare employees, equivalent workplace training courses).
- Comply with all site-specific training and workplace safety requirements.

If you have any safety concerns at any time, do not begin work or immediately stop work and move to a safe location. Immediately contact your supervisor or site safety officer for instructions on how to proceed.

CAUTION



POTENTIAL INJURY HAZARD

Pushing magnet enclosures may result in bodily injury to personnel.

Do not push magnet enclosures. Follow the directions on the six "Do Not Push" signs (one on each side, and two on each end).



Safety**NOTICE****EQUIPMENT DAMAGE RISK**

Improper transportation during forklift operations can result in magnet damage.

Forklift must meet the minimum capacity and dimension requirements stated in section [2.2 Equipment requirements on page 15](#).

The magnet must be picked up from the magnet front orientation only with the forks inserted into the designated slots on the crate.

Forklift forks can damage the magnet enclosure or components. Use protective padding around the forks.

Avoid sudden jolts. Do not let the crate/pallet bump or hit anything forcefully.

Avoid tilting the magnet/crate/pallet package more than the maximum tilt specified (30° from the horizontal level).

**NOTICE****EQUIPMENT DAMAGE RISK**

Improper transportation can result in magnet damage.

Do the following to prevent magnet damage:

- Do not apply any force to the magnet enclosures.
- Only use equipment/tools that meet the specifications stated in section [2.2 Equipment requirements on page 15](#).
- Do not move the magnet to the MR Suite while the magnet is on its shipping pallet. Before moving the magnet to the MR Suite, refer to Unloading and moving the magnet.

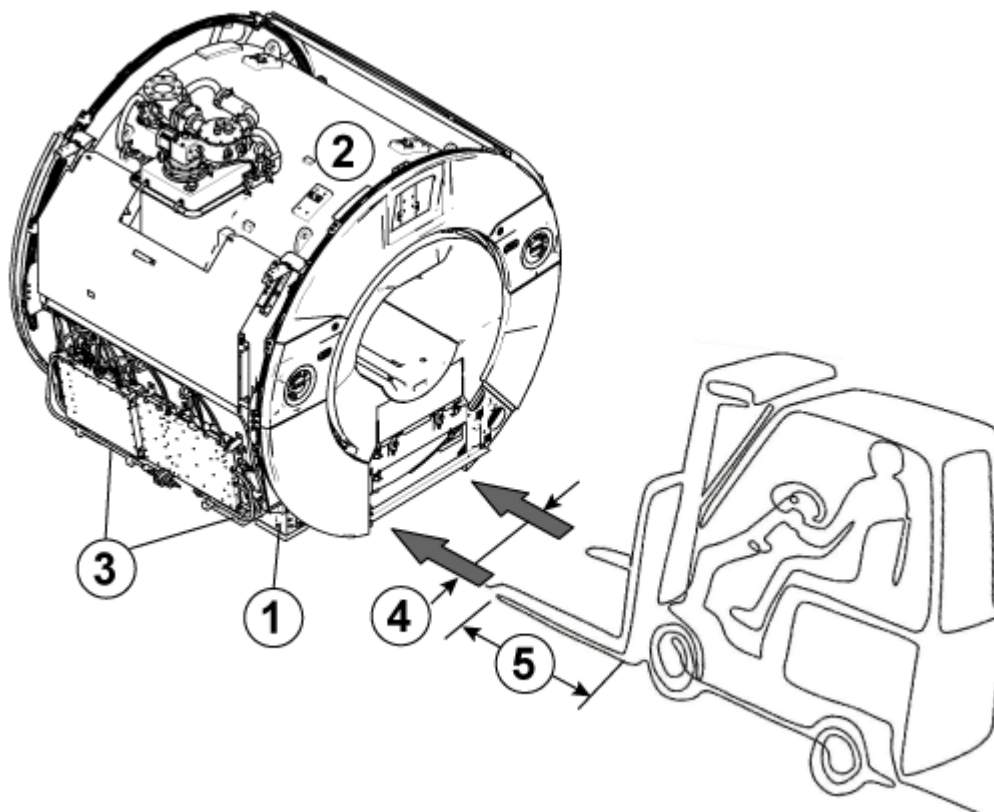
About this task**NOTE**

The rigger is responsible for actual equipment/procedures used to lift and move a magnet into the customer facility, including through a raised opening in an exterior wall. The following procedures are guidelines only. It is the responsibility of the rigger to safely move the magnet into the facility.

Procedure

1. Position a forklift that meets the specifications in [section 2.2 Equipment requirements on page 15](#) at the front of the magnet facing the magnet. Put the forks under the Lifting angle bracket to the outside of the magnet feet.

Figure 2-14 Forklift under Lifting angle bracket



Item	Description	Item	Description
1	Magnet feet	4	1635mm (64 inches) inside forks, minimum
2	Magnet (protective wrap not shown)	5	2010mm (79 inches) minimum
3	Lifting angle brackets	-	-



NOTE

Figure is not to scale. Put the forks under the Lifting angle bracket, outside of the magnet feet.

2. Carefully drive the forklift until the forks are completely under both Lifting angle bracket in the areas shown in [Figure 2-14 Forklift under Lifting angle bracket on page 34](#).



NOTE

The forks must be positioned under and fully cover all four lifting angle brackets to avoid tilting or falling.

3. Lift the forks to front below the Lifting angle bracket, adjust the distance between forks so that padded forks lightly touch the enclosure, and finish raising the forks to the Lifting angle bracket.

4. Lift the magnet with the forklift.
5. Smoothly move the magnet to the desired location, and carefully lower to rest on a flat surface.

2.9 Unloading the uncrated magnet with a crane to ground

Prerequisites

Safety	
	 WARNING POTENTIAL SERIOUS INJURIES Moving a magnet through a raised opening in an exterior wall can result in serious injuries and magnet/equipment damage. Do the following before moving the magnet: • Make sure a complete walk-through of the crane lift process was done prior to the actual event to make sure all process details are covered. • Make sure all necessary equipment is on-site and inspected for safety and load ratings. • Make sure all necessary personnel are trained and ready. • Make sure that the entire area where lifting will occur is free of obstructions and unauthorized personnel. • Make sure the surface where the magnet will be put after lifting is flat. • Do not crane lift a magnet during dangerous weather conditions.
	 WARNING POTENTIAL INJURY HAZARDS Improper lifting of the magnet may be hazardous to personnel and can result in damage to the magnet and equipment. Make sure that the entire area where lifting will occur is free of obstructions and unauthorized personnel. Make sure that the surface where the magnet will be put after lifting is flat.

Safety**NOTICE****EQUIPMENT DAMAGE RISK**

Improper transportation can result in expensive internal damage to the magnet.

Avoid sudden jolts. Lift/move/lower the magnet smoothly. Do not allow the magnet to bump or hit anything forcefully.

Avoid tilting the magnet more than the maximum tilt (30° from the horizontal level).

Do not apply any force to the magnet enclosures.

Do not crane lift a magnet that is on a pallet or inside a cage or crate.

Crane lifting can only be done using the magnet lifting holes, which are not accessible while the magnet is inside a cage or crate.

**NOTICE****EQUIPMENT DAMAGE RISK**

Improper crane transportation can result in damage to the magnet.

Do not crane lift crated/palleted magnets. Crated/palleted magnets must be handled in conformance with the other sections in Unloading and moving the magnet of this manual.

Only a rigger should unload and move the magnet.

About this task**NOTE**

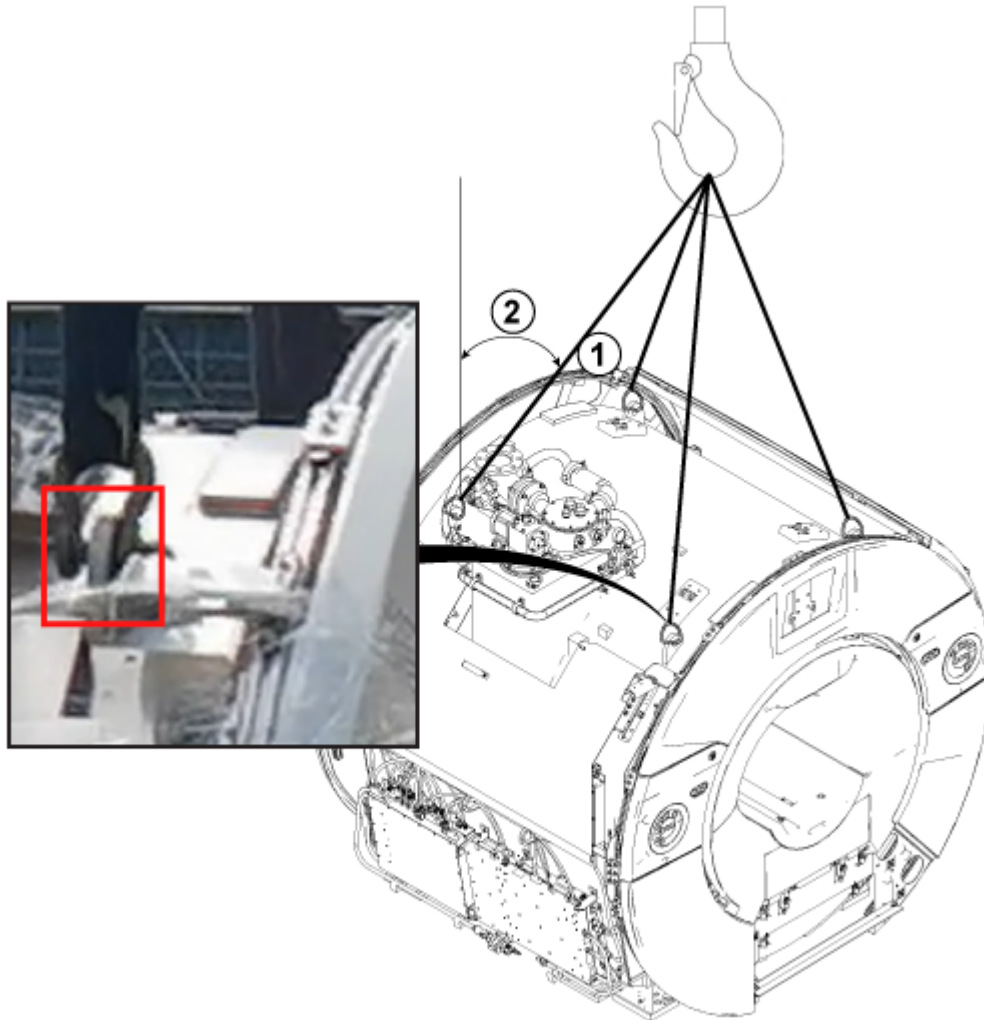
The rigger is responsible for actual equipment/procedures used to lift and move a magnet into the customer facility, including through a raised opening in an exterior wall. The following procedures are guidelines only. It is the responsibility of the rigger to safely move the magnet into the facility.

Procedure

1. Make sure that there are no obstructions in the area where lifting will occur and that a flat surface is available to position the magnet after lifting.
2. Attach 4 lifting rings to the 4 lifting holes.

3. Position the hook of a crane, which meet the specifications stated in section [2.2 Equipment requirements on page 15](#), centrally over the magnet to ensure a vertical lifting force on the lifting rings. Make sure that the strap length and crane lifting height are long enough so that the magnet can reach the desired height.

Figure 2-15 Crane lift configuration



Item	Description
1	The strap length > 1m
2	The strap tilt angle < 60°

4. Attach the rigging to the lifting rings at top of the magnet.



NOTE

Wire rope is recommended for better length match and stretch control.

5. When beginning to lift the magnet, if it does not remain close to level, carefully lower the magnet and adjust the lifting configuration, and then lift the magnet again.

6. Smoothly move the magnet to the desired location, and carefully lower so it rests on a flat surface.



NOTE

Do not load on the ground without any shims. The front panel may collapse.

2.10 Lifting the magnet with a crane through the opening in the exterior wall

Prerequisites

Safety	
	WARNING POTENTIAL SERIOUS INJURIES Moving a magnet through a raised opening in an exterior wall can result in serious injuries and magnet/equipment damage. Do the following before moving the magnet: • Make sure a complete walk-through of the crane lift process was done prior to the actual event to make sure all process details are covered. • Make sure all necessary equipment is on-site and inspected for safety and load ratings. • Make sure all necessary personnel are trained and ready. • Make sure that the entire area where lifting will occur is free of obstructions and unauthorized personnel. • Make sure the surface where the magnet will be put after lifting is flat. • Do not crane lift a magnet during dangerous weather conditions.
	WARNING POTENTIAL INJURY HAZARDS Improper lifting of the magnet may be hazardous to personnel and can result in damage to the magnet and equipment. Make sure that the entire area where lifting will occur is free of obstructions and unauthorized personnel. Make sure that the surface where the magnet will be put after lifting is flat.

Safety**NOTICE****EQUIPMENT DAMAGE RISK**

Improper transportation can result in expensive internal damage to the magnet.

Avoid sudden jolts. Lift/move/lower the magnet smoothly. Do not allow the magnet to bump or hit anything forcefully.

Avoid tilting the magnet more than the maximum tilt (30° from the horizontal level).

Do not apply any force to the magnet enclosures.

Do not crane lift a magnet that is on a pallet or inside a cage or crate.

Crane lifting can only be done using the magnet lifting rails, which are not accessible while the magnet is inside a cage or crate.

NOTICE**EQUIPMENT DAMAGE RISK**

Improper crane transportation can result in damage to the magnet.

Do not crane lift crated/palleted magnets. Crated/palleted magnets must be handled in conformance with the other sections in Unloading and moving the magnet of this manual.

Only a rigger should unload and move the magnet.

About this task**NOTE**

The rigger is responsible for actual equipment/procedures used to lift and move a magnet into the customer facility, including through a raised opening in an exterior wall. The following procedures are guidelines only. It is the responsibility of the rigger to safely move the magnet into the facility.

Procedure

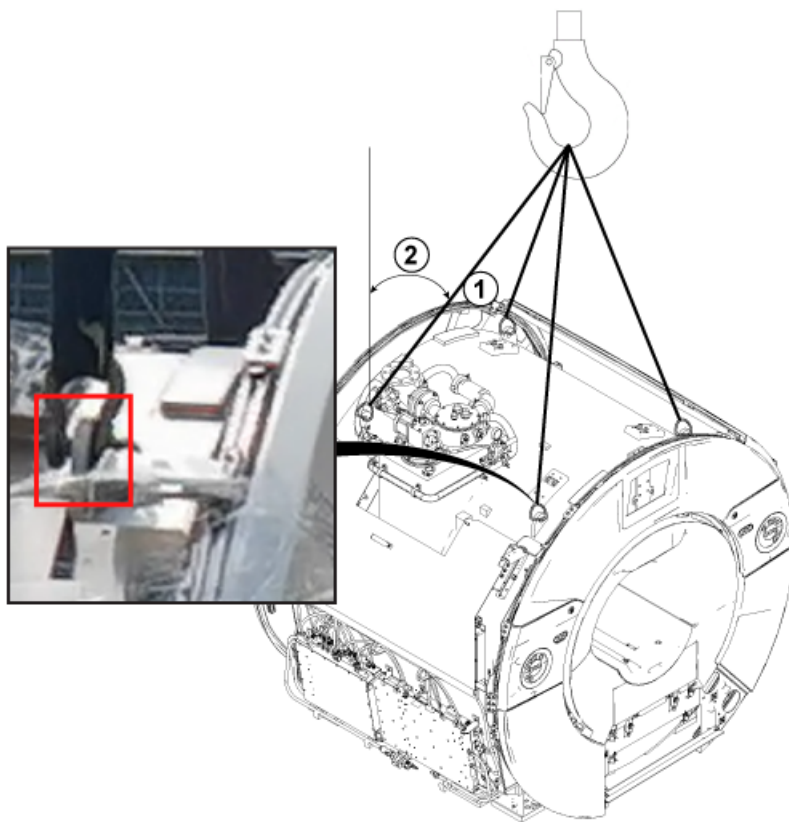
1. Make sure that the opening is at least 2439 mm (96 inches) wide and 2591 mm (102 inches) tall, minimum. The magnet should pass through the opening side first without hitting the opening.

**NOTE**

A larger opening will make both the operation easier and accidental magnet damage less likely.

2. Position the hook of a crane, which meet the specifications stated in section [2.2 Equipment requirements on page 15](#), centrally over the magnet to ensure a vertical lifting force on the lifting rings. Make sure that the strap length and crane lifting height are long enough so that the magnet can reach the desired height.

Figure 2-16 Crane lift configuration



Item	Description
1	The strap length > 1m
2	The strap tilt angle < 60°

3. Attach the lifting straps to the top of magnet to face toward the building.



NOTE

Wire rope is recommended for better length match and stretch control.

4. When beginning to lift the magnet, if it does not remain close to level, carefully lower the magnet and adjust the lifting configuration and then lift the magnet again.
5. Smoothly move the magnet through the raised opening in the exterior wall, and carefully lower so it rests on a flat surface.

Chapter 3 Magnet storage conditions, pending ramp

3.1 Converting an air magnet shipment to warehouse configuration

Prerequisites

Personnel requirements			
Required persons	Preliminary requirements	Procedure	Finalization
2	15 minutes	30 minutes	-

Tools and test equipment			
Item	Quantity	Part number	Manufacturer
Cryogen Safety Wear Kit	1 per person	46-271137G1	-
Heat Gun	1	-	-
Helium Regulator and Hose Kit	1	46-306734G1	-
Oxygen Sensor, O2 Low Alarm 19.5% Volume, High Alarm 23% Volume	1 per person	5393536	-
Pair: Nonferrous Safety Shoes	1 per person	-	-
Pair: Nonferrous Safety Shoes	1 per person	-	-
Set: Nonabsorbent Protective Clothing (long sleeve shirt and pants)	1 per person	-	-
Aluminum Platform Ladder, 1,206.5 mm (47.5 inches)	1	5155291	-

Replacement parts			
Item	Quantity	Part number	Manufacturer
Pressure transducer, FRU, for JXM magnet	1	5926243-302-2	-

Safety

Before working in any GE HealthCare MR suite or doing any GE HealthCare service procedure, you must:

- Have read and understood all hazard conditions and safety requirements in the latest revision of the GE HealthCare *MR Service Safety Manual* (5452735).
- Have successfully completed all relevant GE HealthCare Environmental Health and Safety (EHS) courses (or for non-GE HealthCare employees, equivalent workplace training courses).
- Comply with all site-specific training and workplace safety requirements.

If you have any safety concerns at any time, do not begin work or immediately stop work and move to a safe location. Immediately contact your supervisor or site safety officer for instructions on how to proceed.

 DANGER
**POTENTIAL ASPHYXIATION**

This procedure releases large quantities of odorless, colorless helium gas, which will displace oxygen in the air and potentially cause asphyxiation.

Make sure sufficient ventilation exists in the magnet room to dispel large quantities of helium gas. Make sure the magnet room vent exhaust fan is on and the magnet room door (or the hatch, if you are in a mobile van) is propped open.

 CAUTION
**POTENTIAL COLD BURN**

Exposure to cryogenics and contact with cold objects is possible during this procedure.

Wear protective clothing, nonabsorbent gloves, and goggles while doing this procedure.

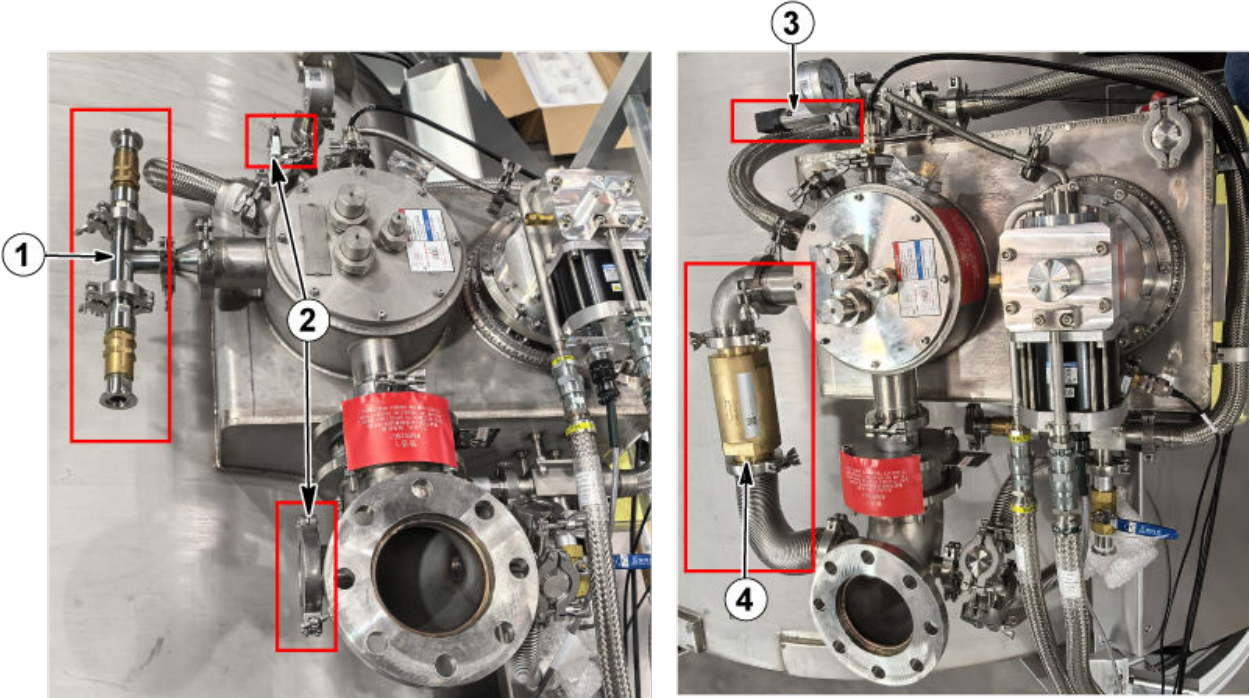
About this task

A magnet may arrive at the installation site or warehouse in its air-shipping configuration. For air shipment, the check valve and pressure transducer are removed, and the system is installed with a 7psi air-shipping kit. Once in the warehouse, before connecting the magnet to the compressor, the pressure transducer must be installed.

**NOTE**

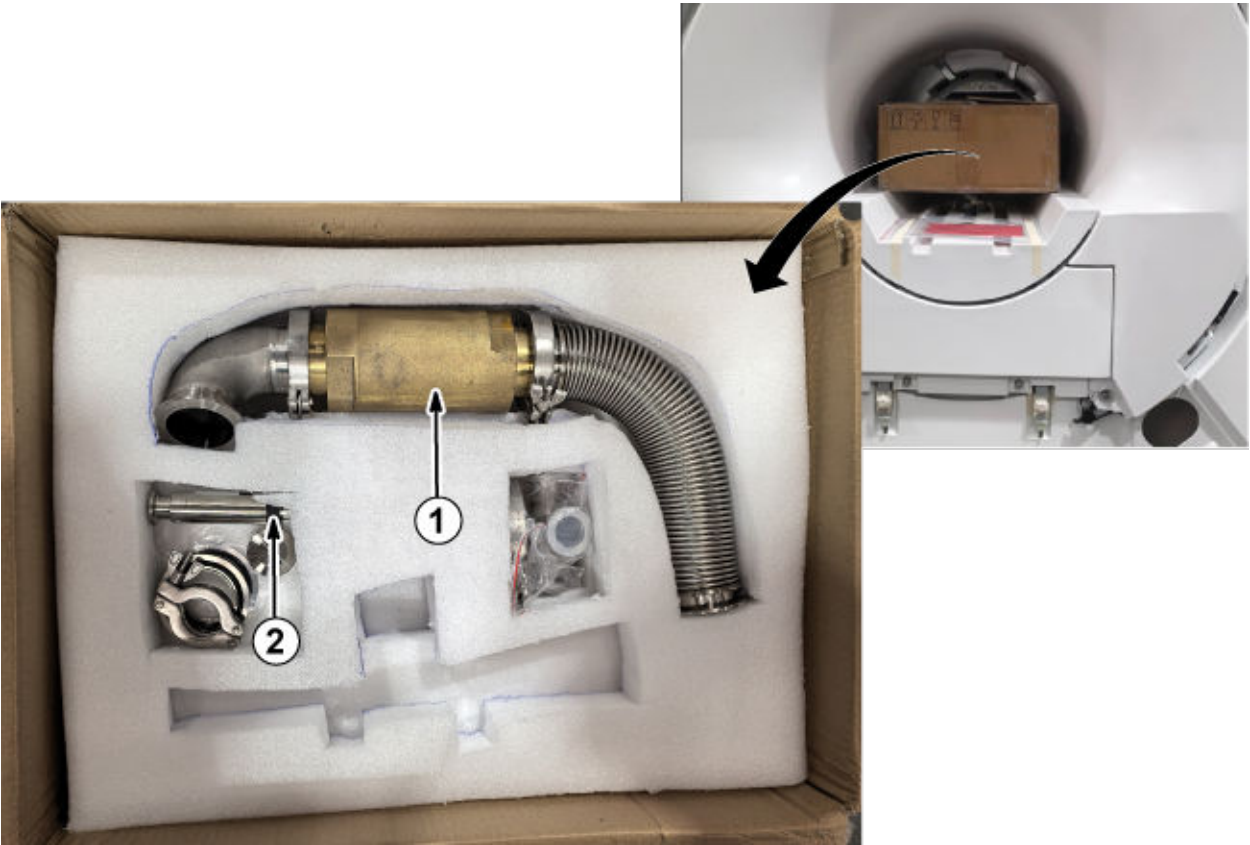
Full site configuration should be completed at the customer site, while storage warehouses require only the installation of the pressure transducer.

Figure 3-1 Air-shipping configuration (left) vs. Site configuration (right)



Item	Description
1	7psi air-shipping kit
2	Ports covered with blind plates
3	Pressure transducer
4	3psi check valve

Figure 3-2 Items shipped in the magnet bore



Item	Description
1	3psi check valve  IMPORTANT SHOULD NOT be installed in warehouses.
2	Pressure transducer, FRU, for JXM magnet

Procedure

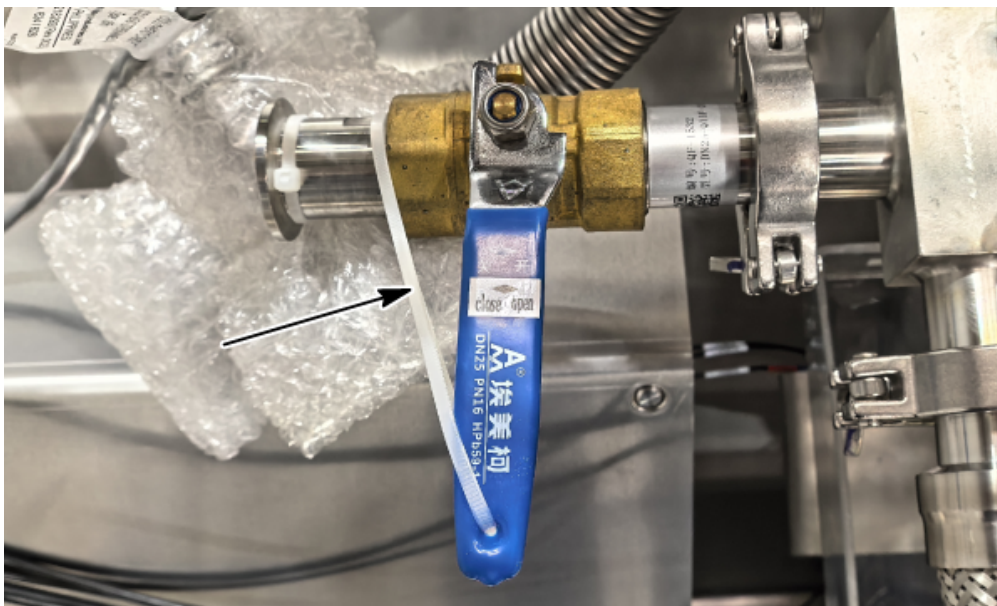
1. Confirm the magnet pressure gauge shows positive pressure. Resolve any negative readings before moving on.

Figure 3-3 Pressure gauge on magnet



2. Cut off the zip-tie securing the relief valve.

Figure 3-4 Relief valve zip-tie



3.

⚠ DANGER**POTENTIAL ASPHYXIATION**

This procedure releases large quantities of odorless, colorless helium gas, which will displace oxygen in the air and potentially cause asphyxiation.

Make sure sufficient ventilation exists in the magnet room to dispel large quantities of helium gas. Make sure the magnet room vent exhaust fan is on and the magnet room door (or the hatch, if you are in a mobile van) is propped open.

⚠ CAUTION**POTENTIAL COLD BURN**

Exposure to cryogenics and contact with cold objects is possible during this procedure.

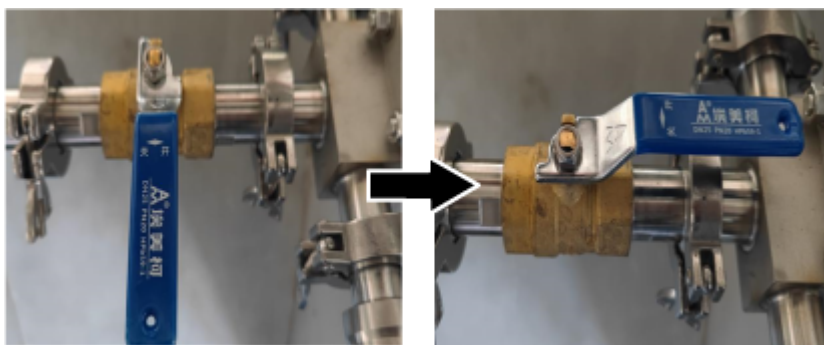
Wear protective clothing, nonabsorbent gloves, and goggles while doing this procedure.

Open the relief valve and vent the magnet pressure until there is no visible helium gas coming out of the port. Then, shut the valve.

**NOTE**

Make sure the magnet pressure remains positive for the entire time.

Figure 3-5 Open the relief valve



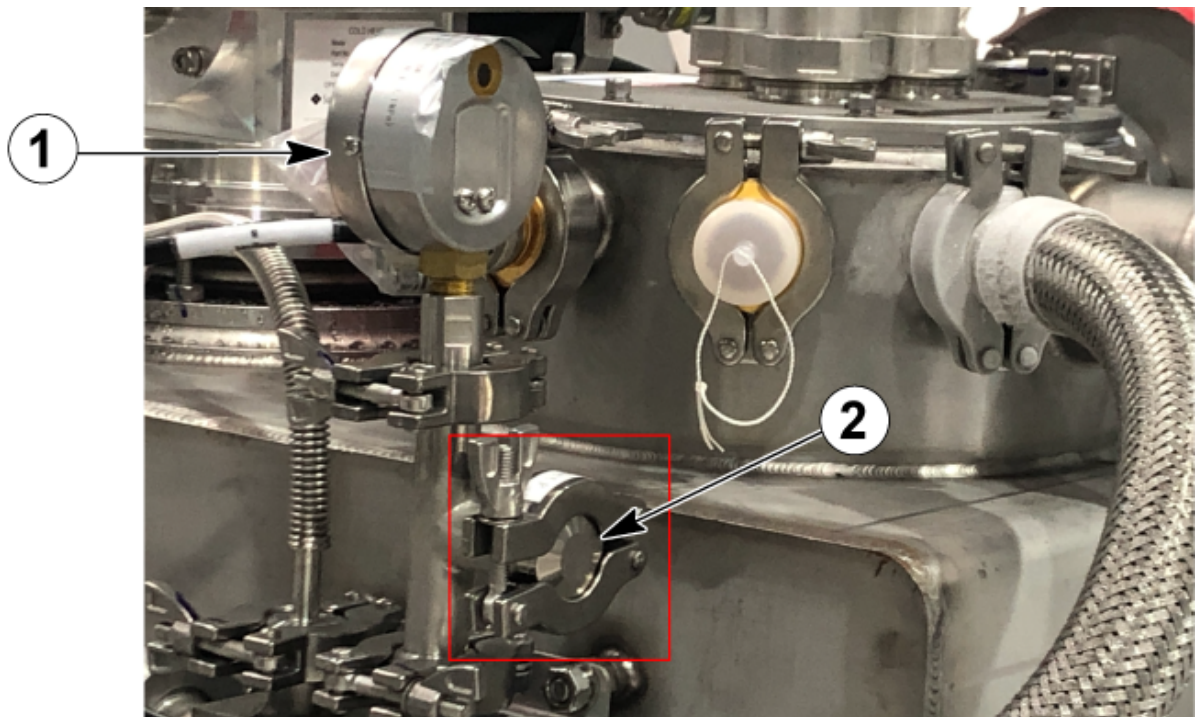
4.

**ATTENTION**

Be careful to not drop the O-ring positioning ring and the O-ring!

Locate and remove the blind plate under the pressure gauge.

Figure 3-6 Removing the blind plate



Item	Description
1	Pressure gauge
2	Blind plate for the pressure transducer port

5. Install the pressure transducer and tighten the clamp.

Figure 3-7 Installing the pressure transducer



6. Close the relief valve and secure the handle with a zip-tie.

Finalization

Do a leak test. Refer to the corresponding section of *SIGNA™ Prime Elite & SIGNA™ Prime MAX Magnet Cryogen Manual* (5951110-8EN), available through the online document library.



ATTENTION

Restore the system to its **air-shipping configuration** before the magnet leaves the warehouse for delivery to the customer by air.

3.2 Connecting the magnet to the compressor

Prerequisites

Safety	
	WARNING POTENTIAL COLD BURN OR ASPHYXIATION HAZARD Gaseous helium (odorless, colorless gas) is discharged from the magnet venting activities, and can cause cold burns or asphyxiation. Wear protective clothing, nonabsorbent gloves, and goggles when venting the magnet. Do not position yourself on or near the venting region.
	WARNING ELECTRIC SHOCK Contact with connectors leading to an energized compressor can cause electrical shock. Disconnect input power to the compressor and follow LOTO procedures to make sure power is not supplied to the compressor.
	NOTICE EQUIPMENT DAMAGE RISK Improper operation/maintenance of equipment can result in equipment damage. For detailed information about the operation, function, and maintenance of the cold-head and compressor, refer to the appropriate cryocooler and/or compressor vendor technical operating manual (Vendor Manuals CD32ZZ-226), available through the support documentation library at gehealthcare.com or your local GE Healthcare Service Representative. For detailed information about the operation, function, and maintenance of the Magnet Supervision Unit (SVU), please refer to the Magnet and Cryogen Manual (5951110-8EN), available through the support documentation library at gehealthcare.com or your local GE Healthcare Service Representative.

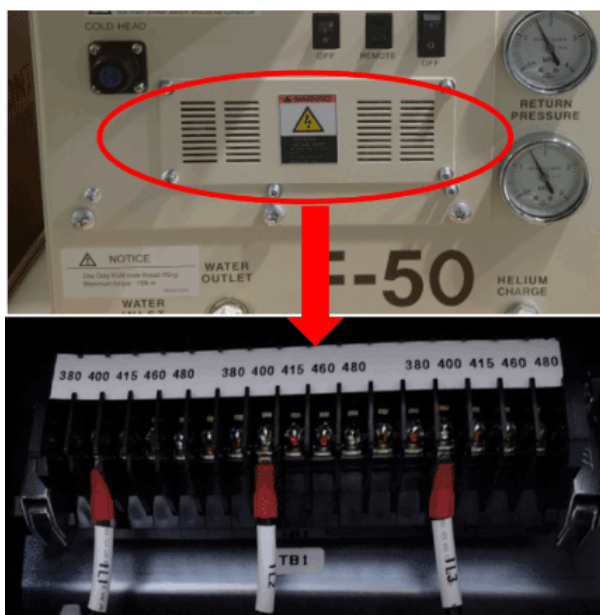
About this task

This magnet has superconducting coils immersed in a liquid helium vessel that is surrounded by an insulating cryostat. Due to site readiness delay, sometimes it is necessary to store the magnet in its shipping configuration at a warehouse. During this storage period, the magnet cryocooler must be connected to the compressor and the SVU system to maintain proper temperature and pressure, and minimize helium loss as well as risk for internal icing issues.

Procedure

1. Notify an electrician or other individual who has earned the electrician certificate to measure the site voltage, and the default running voltage of the compressor is 400V. If any changes are needed, remove the cover with a screwdriver, unscrew the nuts on three running voltage config, replace them to corresponding pressure-wires (with similar voltage) and tighten the screws, as shown in the figure below.

Figure 3-8 Compressor cover removed



2. Make sure that the compressor supply static pressure is within the range for the site's compressor:

Table 3-1 Compressor information

1.60MPa to 1.65MPa

Figure 3-9 F-50 helium compressor unit, supply pressure



3. Make sure that the O-rings are in the coldhead supply and coldhead return line connectors.
4. Connect the cooling water for the compressor (or do a check of the connection, if already connected).
5. Make sure that the compressor is on "REMOTE" mode. FE or warehouse staff must not change this.

Figure 3-10 Compressor on "REMOTE"

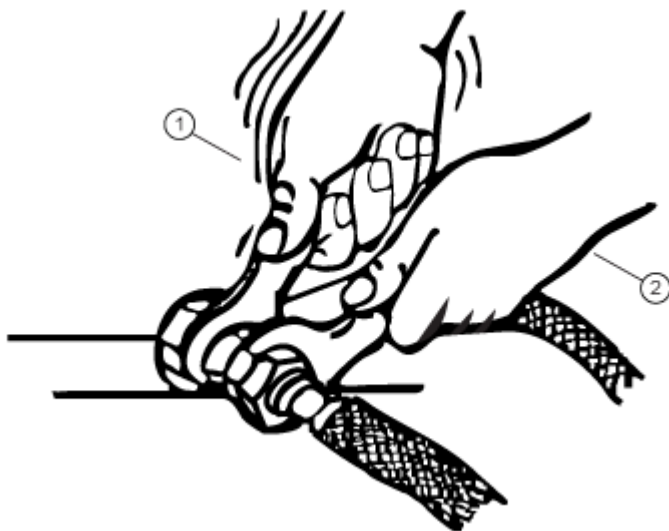


6. Using two wrenches (one to hold the backside adapter connection, and one to tighten), connect the coldhead compressor lines in the following order:

**NOTE**

The hold wrench maintains the backside adapter connection, making gas leaks in the system less likely. Do not over-tighten.

Figure 3-11 Using two wrenches to tighten a connection



1	Tighten
2	Hold fast

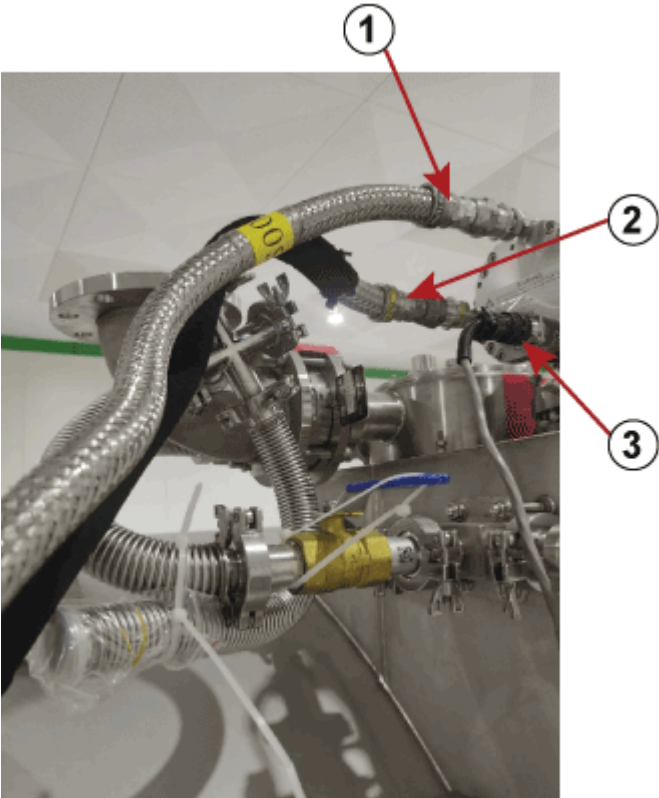
**NOTE**

Do not put any bending force on Aeroquip fittings while connecting/disconnecting helium flexlines. A bending force will create difficulty in the rapid engagement/disengagement required to prevent helium loss and system contamination. Support the gas lines when connecting/disconnecting them to the compressor.

- 6.1. Attach the coldhead return line.
- 6.2. Attach the coldhead supply line.

6.3. Attach the coldhead power cable.

Figure 3-12 Coldhead compressor connections



1	First attachment - coldhead helium return line
2	Second attachment - coldhead helium supply line
3	Third attachment - coldhead power cable

7. Complete the following steps for the compressor:

Option	Description
To turn on the compressor	1. For new compressor installations, make sure the input power voltage is connected to the compressor. If the input power voltage is 380V at 50Hz, no change on the voltage tap setting is required. Otherwise, set up the terminal wiring, located at the voltage tap area, per the input power voltage. Refer to vendor manuals for this procedure. 2. Turn the compressor main power switch to the ON position.

Option	Description
	Figure 3-13 Compressor main power switch 3. Press the compressor drive switch to the ON position, and wait 1 to 3 second for the compressor to start. Figure 3-14 Compressor drive switch 4. Do a check for warnings or alarms. ! IMPORTANT Whenever the compressor shows an alarm or warning, investigate the cause and solve the problem before resetting the alarm and running the compressor.
To turn off the compressor	1. When removing a unit from the compressor, do the above process in reverse order. 2. Disconnect and lockout/tagout (LOTO) input power to the compressor. Use a digital voltmeter (DVM) or equivalent measuring device to make sure that no voltage is present. 3. To prevent contamination of gas lines and the compressor, make sure the gas lines are removed immediately after power is disconnected. NOTE Do not put any bending force on Aeroquip fittings while connecting/disconnecting helium flexlines. A bending force will create difficulty in the rapid engagement/disengagement required to prevent helium loss and system contamination. Support the gas lines when connecting/disconnecting them to the compressor.

3.3 Connecting the magnet to the SVU

About this task

When connecting the cables, do not route the cables in a way that causes unnecessary strain or bending on the cables. Hand-tighten the screws on each connector; loose connections can cause reading errors.

Procedure

1. Examine the cables used to monitor the magnet's pressure and helium level before connecting them to the magnet. Make sure that all pins are straight and almost level with the connector casing (for example, not pushed back). The cables should also not have any broken insulation. If you see any damage, get a new cable and replace the old one as soon as possible.



NOTE

Each SVU should have a dedicated cable set, which is firmly attached to the unit. Swapping cables between different SVUs can cause early failure of the connectors.

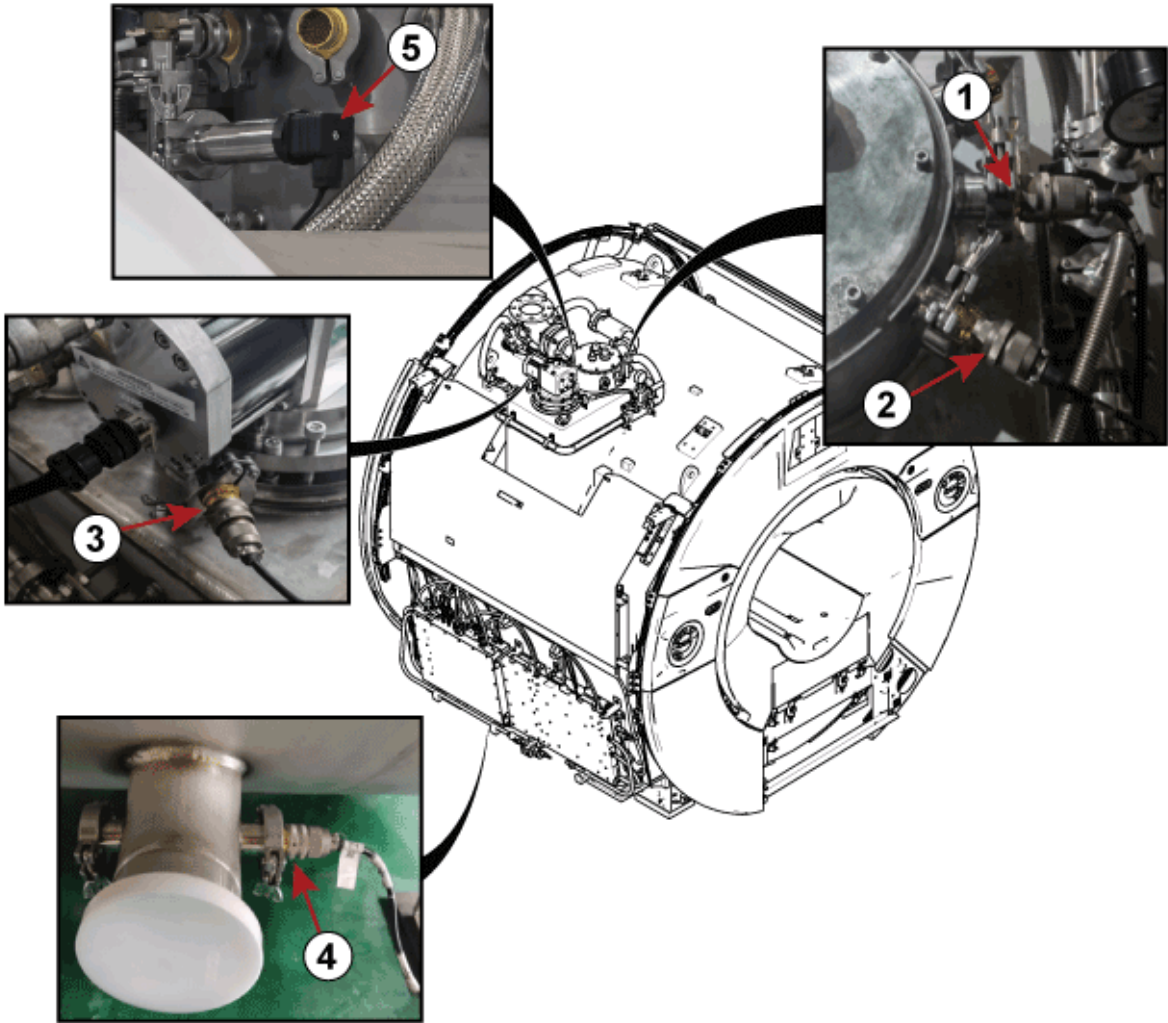
2. Connect the remote control cable(E3010/E3509) from compressor to SVU. Only after the remote control cable is connected and the SVU is powered on, the compressor will start running.
3. Connect cables(M3051/M3052) to the magnet.



ATTENTION

A Phillips screw needs to be tightened when connecting the Magnet P cable. Handle the screw properly and DO NOT drop it.

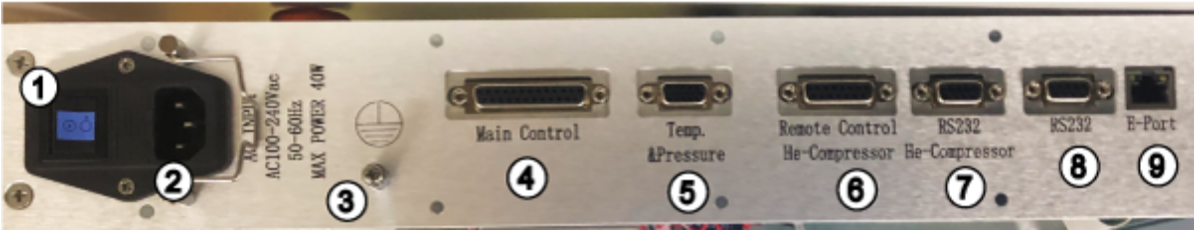
Figure 3-15 Connecting the cables



Item	From	To
1	Magnet B	Penetration Panel J48
2	Magnet C	Penetration Panel J48
3	Magnet D	Penetration Panel J66
4	Magnet E	Penetration Panel J66
5	Magnet P	Penetration Panel J66

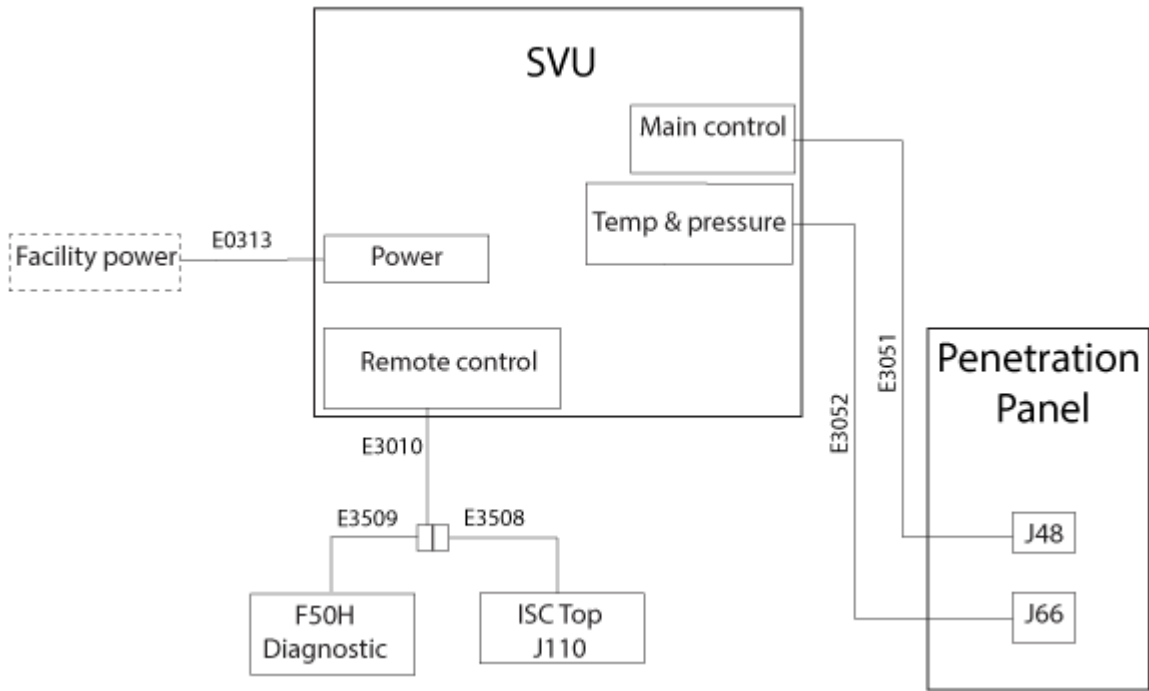
4. Connect the cables to SVU side panel.

Figure 3-16 SVU connect panel



Item	Description	Connections
1	SVU switch	/
2	Power outlet	E0313
3	Ground protection	/
4	Main control port	E3051
5	Temperature & Pressure port	E3052
6	Compressor remote control port	E3010
7	Not-used	/
8	Not-used	/
9	Ethernet Port	Ethernet cable for magnet monitoring data back-office connection

Figure 3-17 SVU connections



3.4 Monitoring the magnet

About this task

Using the SVU front panel user interface, monitor the magnet pressure and helium level at least once per day to minimize loss of liquid helium and minimize internal icing situations.

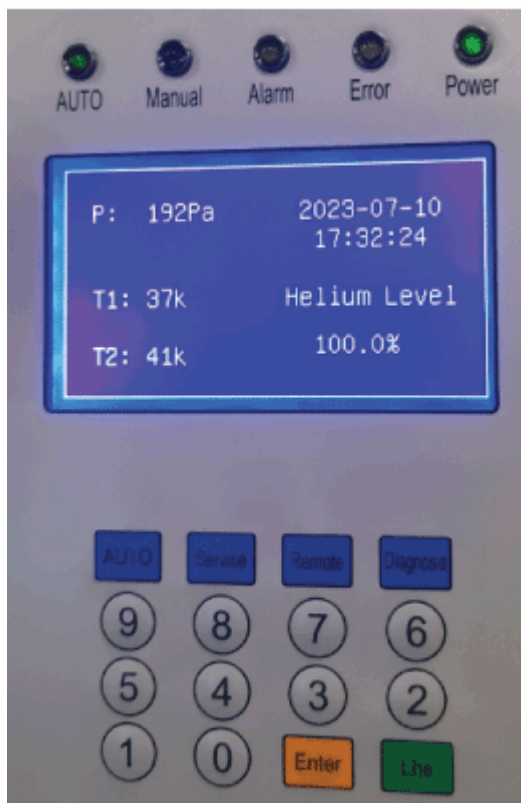
The front panel of the SVU is made up of the following elements:

- LEDs showing the system status.
- LCD display for user interaction
- 16-button soft keypad

Procedure

1. Press the **Auto** button to change the display to usual operating mode. This mode shows the following information:

Screen	Display	
1	Helium Pressure xxxx Pa	
2	Date	Time
3	Helium Level xx.xx% or xx L	
4	Shield Temperature (T1) Bore Tube Temperature (T2)	

Figure 3-18 SVU front panel user interface

2. At the time of helium level monitoring, press the **Lhe** button to update reading a helium level.

**NOTE**

It can take up to 60 seconds for the display to update and show the new level(s). Helium level requirements can vary according to purchase agreements. Work with your Logistics Team and/or Field Service Contact to determine helium level requirements.

3. Check the SVU helium pressure readings from the Home screen. The system should maintains on helium pressure 2000 ± 300 Pa, T1 on 30-55K, and T2 on 35-60K in normal working state.

**NOTE**

The upper limit of pressure displayed on the SVU is 10000Pa, and the lower limit is 0Pa. It is necessary to verify the magnetic pressure by checking the mechanical pressure gauge once the magnet connected. Check the data displayed on the SVU after pressure drops.

Chapter 4 Doing in-transit service and magnet system checks

4.1 Doing a magnet system check

Prerequisites

NOTICE

EQUIPMENT DAMAGE RISK

Improper transportation can result in damage to the magnet.

In-transit service and magnet electrical checks must be done by qualified personnel only, after the magnet is sited and in strict conformance with the *Doing a liquid helium fill* procedure stated in the appropriate manual:

- Magnet and Cryogen Manual (5951110-8EN).
- *MR Service Safety Manual* (5452735)

These documents are available through the support documentation library at gehealthcare.com or your local GE Healthcare Service Representative.

About this task



NOTE

Magnet system checks are a physical inspection done by GE Service.

Procedure

1. Locate the Pre-Delivery Information Package shipped with the magnet. It contains the Bill of Material for the magnet system delivered.
2. Make sure that all boxes indicated on the Bill of Material are included.
3. Make sure that the contents of each box matches its packing list when the boxes are brought into the MR site.
4. Inspect the magnet for physical damage and icing/condensation on the body, and do one of the following:
 - If no problems are found, unload the magnet.
 - If there is damage, report the damage in conformance with the procedure [1.2 Examining packages for damage in transportation on page 6](#). Report all problems found to the regional Magnet & Cryogenics (MAC) Team Leader.

5. If there is icing or condensation on the exterior or the bore of the magnet, check the liquid helium level before unloading. Refer to the Section 11.3 "Installation" of the SVU in the Magnet and Cryogen Manual (5951110-8EN).

**NOTE**

If the magnet has been sitting for a period of time with the coldhead inoperative, the magnet may be depleted of cryogens. Contact the logistics or field service team for further directions.

Chapter 5 Preparing the scan room

Procedure

1. Remove any debris from the magnet room floor where the mats will be positioned.
2. Make sure that there are no RF seams under the vibroacoustic damping mats.
3. Move the vibroacoustic damping mats to the MR suite.



NOTE

The vibroacoustic damping mats may arrive in a cold state preventing compression. Put the vibroacoustic damping mats in the magnet room prior to moving the magnet to the MR suite to let their temperature stabilize.

5.1 Preparing the scan room for leveling magnet

Prerequisites

Tools and test equipment			
Item	Quantity	Part number	Manufacturer
PH Alignment and Magnet Leveling Kit	1	5897979	-

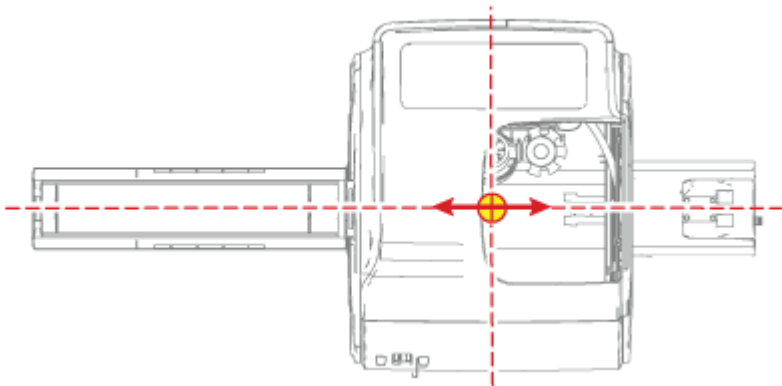
PH Alignment and Magnet Leveling Kit (5897979)			
Item	Quantity	Part number	Manufacturer
Laser Level PLS 180R Z	2	5831466	-
Tripod, Vanguard Alta Pro 263AB 100	2	5829559	-
Folding Ruler, 2 meters, FM-DELA.401.00	1	5897974	-

Accessories of JXM magnet (5926243-301)			
Item	Quantity	Part number	Manufacturer
JXM Series Magnet Positioning Template	1	5926243-301-31	-

Procedure

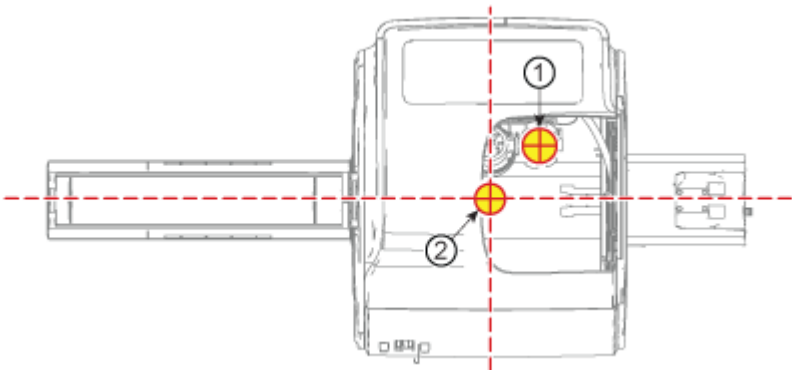
1. Make sure that markings are present on the magnet room floor in either of the following locations.
 - Magnet geometric isocenter location and z-axis vector (longitudinally through the bore).

Figure 5-1 Isocenter location and z-axis vector



- Magnet geometric isocenter location and center of vent pipe.

Figure 5-2 Isocenter location and vent pipe location

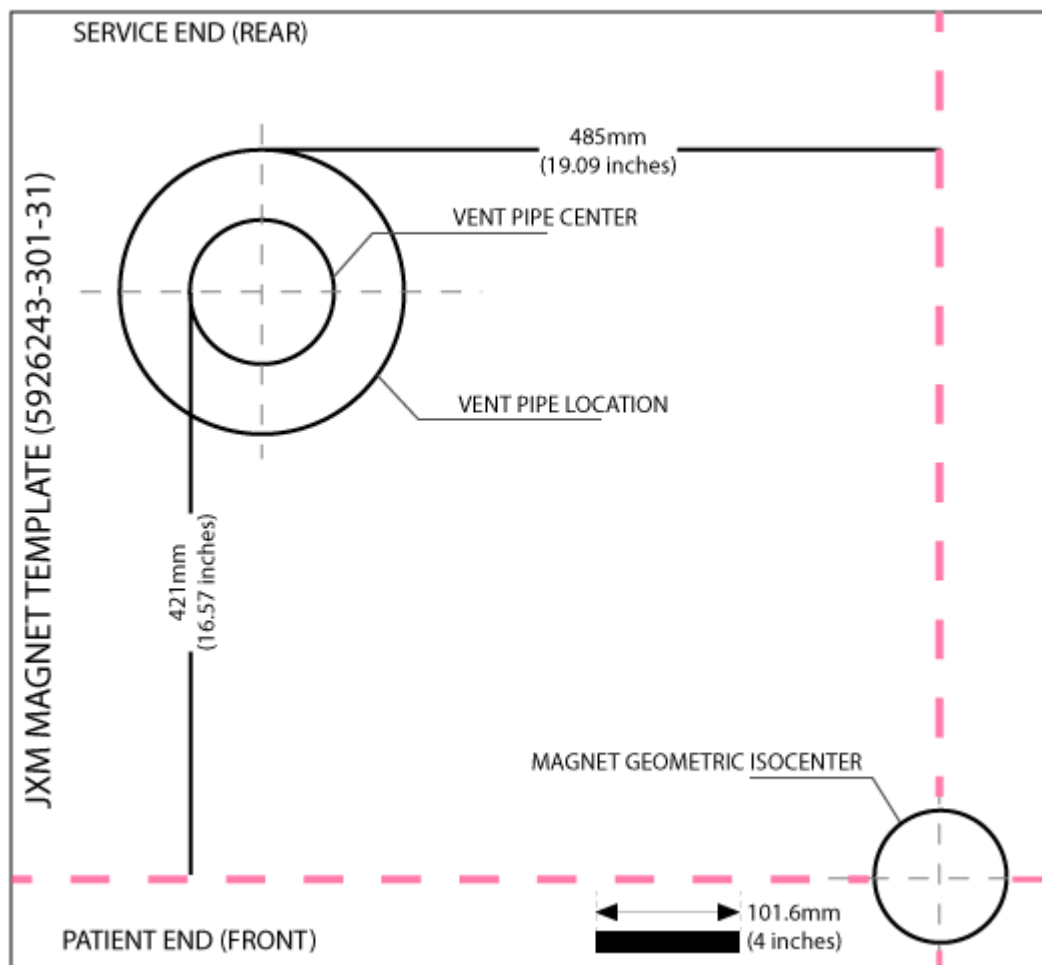


1	Vent center
2	Magnet isocenter

2. If needed, transcribe the marks from the ceiling to the floor with tape from the kit, using laser levels as needed.
3. Align the positioning template (5926243-301-31) in accordance to the markings, using one of the following strategies:
 - Align the center-line of the positioning template with the marked z-axis vector.

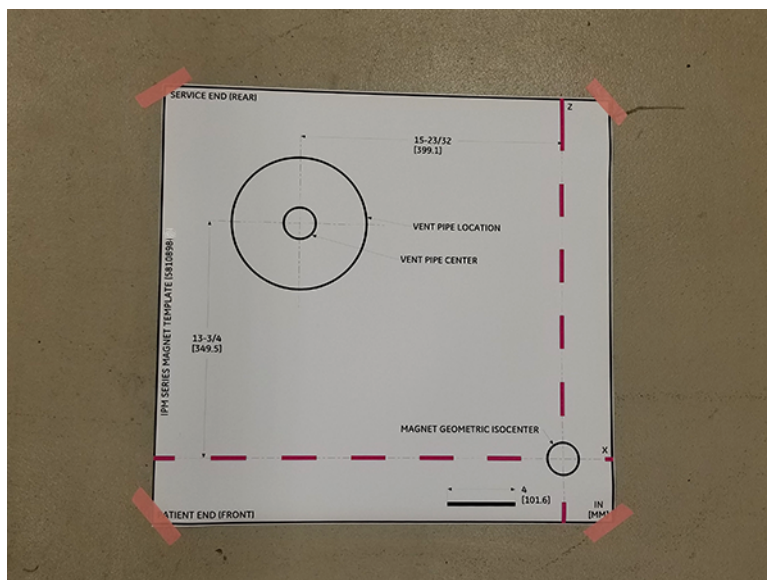
- Align the template to magnet isocenter and vent stack location.

Figure 5-3 JXM Series Magnet Positioning Template (5926243-301-31)



- Use tape from the kit to secure the positioning template to the floor.

Figure 5-4 Securing the positioning template to the floor



5. Make sure the first laser level from the kit is in self-leveling mode.

**NOTE**

In self-leveling mode, if the laser level tilts $> 4^\circ$ in any direction, the lasers do not show. When the laser level tilt is $\leq 4^\circ$, the level indicator LED is green and the lasers show.

Figure 5-5 Laser level in self-leveling mode (unlocked position)

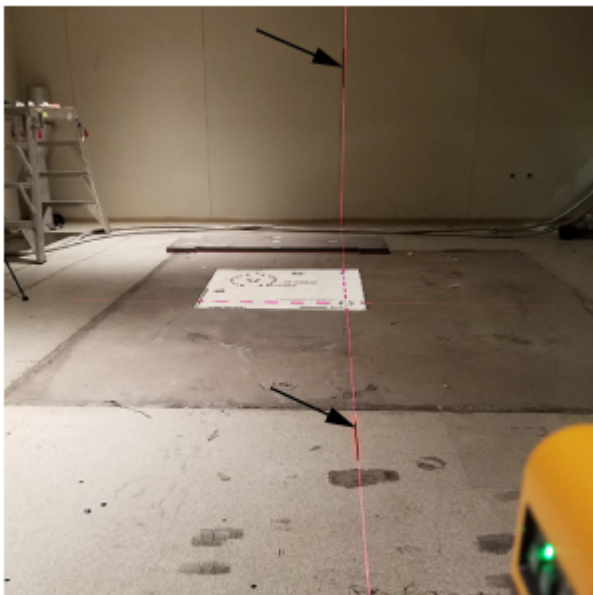


6. Position the first laser level on a tripod in the front of the room.
7. Adjust the laser line passes through the isocenter along the z-axis vector.
8. Mark at least two reference locations - one close to the laser and one far away - on the floor, wall, or ceiling. Do **not** mark any references on the floor where the magnet will be set. Make sure that the reference marks will still be within site of the laser when the magnet is in position.

NOTE

Recommend to leave the farther mark at where can be seen through the bore.

Figure 5-6 First laser reference marks



Result

The reference marks will help you realign the laser level in the event of an accidental movement during the magnet installation.

9. Make sure the second laser level from the kit is in self-leveling mode.

NOTE

In self-leveling mode, if the laser level tilts $> 4^\circ$ in any direction, the lasers do not show. When the laser level tilt is $\leq 4^\circ$, the level indicator LED is green and the lasers show.

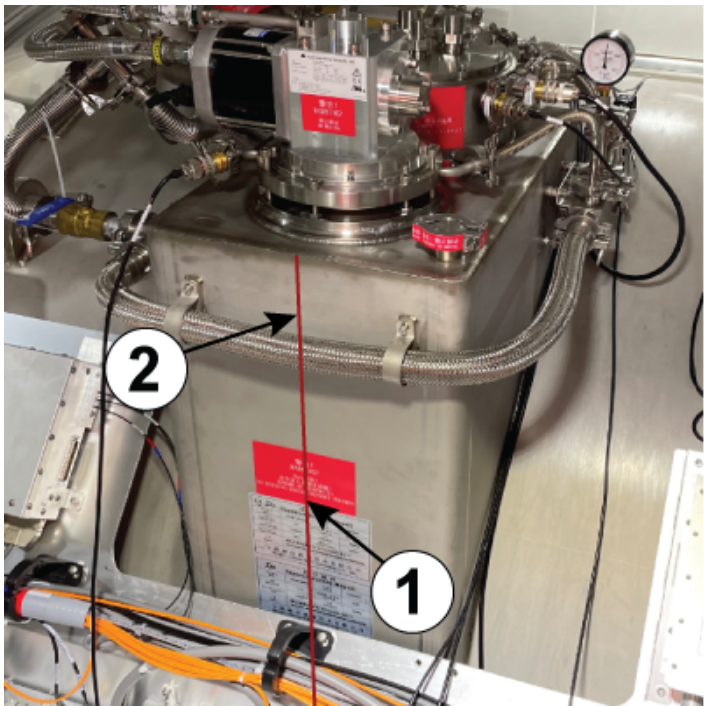
Figure 5-7 Laser level in self-leveling mode (unlocked position)



10. Position the second laser level on a tripod on the left side of the room (electronics side of the magnet).

11. Adjust the second laser level until the vertical laser line passes through the lateral (x-dir) centerline axis on the template and will be visible on the magnet's cryogen chamber when in position.

Figure 5-8 Laser line in the center of the cryogen chamber rib



1	Laser line
2	Cryogen chamber rib

12. Mark at least two reference locations - one close to the laser and one far away - on the floor, wall, or ceiling. Do **not** mark any references on the floor where the magnet will be set. Make sure that the reference marks will still be within site of the laser when the magnet is in position.

 **NOTE** Recommend to leave marks on the ceiling or the floor where is next to the magnet positioning region on the other side. Adjust the laser height to avoid magnet blocking the reference marks after it is positioned.

Figure 5-9 Second laser reference marks



Result

The reference marks will help you realign the laser level in the event of an accidental movement during the magnet installation.

- 13. Remove the template from the floor.
- 14. Continue to the appropriate chapter based on your type of installation.

Installation type	Next steps
Non-seismic	Chapter 6 Installing a nonseismic vibroacoustic damping mat and leveling the magnet on page 69
Seismic	Chapter 7 Installing a seismic vibroacoustic damping mat and leveling the magnet on page 91

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Chapter 6 Installing a nonseismic vibroacoustic damping mat and leveling the magnet

Prerequisites

Safety	
	NOTICE
	EQUIPMENT DAMAGE RISK
	Improper damping mat installation can result in damage to the magnet. Tasks in this section must be completed by riggers, not GE Service Personnel.

6.1 Equipment overview

Prerequisites

Tools and test equipment			
Item	Quantity	Part number	Manufacturer
Tape Measure	1	-	-

PH Alignment and Magnet Leveling Kit (5897979)			
Item	Quantity	Part number	Manufacturer
Laser Level PLS 180R Z	2	5831466	-
Tripod, Vanguard Alta Pro 263AB 100	2	5829559	-
Folding Ruler, 2 meters, FM-DELA.401.00	1	5897974	-
Vertical Alignment Tool	2	5831978	-

Accessories of JXM magnet (5926243-301)			
Item	Quantity	Part number	Manufacturer
Vibroacoustic damping mats, 20mm	4	5926243-301-37	-
Vibroacoustic damping mats, 10mm	4	5926243-301-38	-
Magnet aluminum shim pads, 3mm	12	5926243-301-39	-
Magnet aluminum shim pads, 1mm	16	5926243-301-40	-

Required conditions
Magnet and dock installation and placement are critical to image quality and hardware reliability.

6.2 Positioning a nonseismic mounted vibroacoustic damping mats

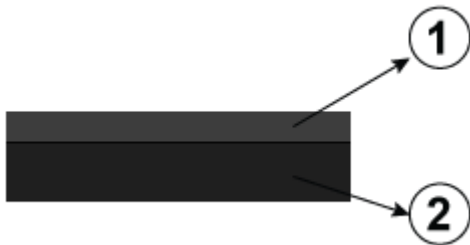
Procedure

1. Make sure that the vibroacoustic damping mat parts are listed in the following table.

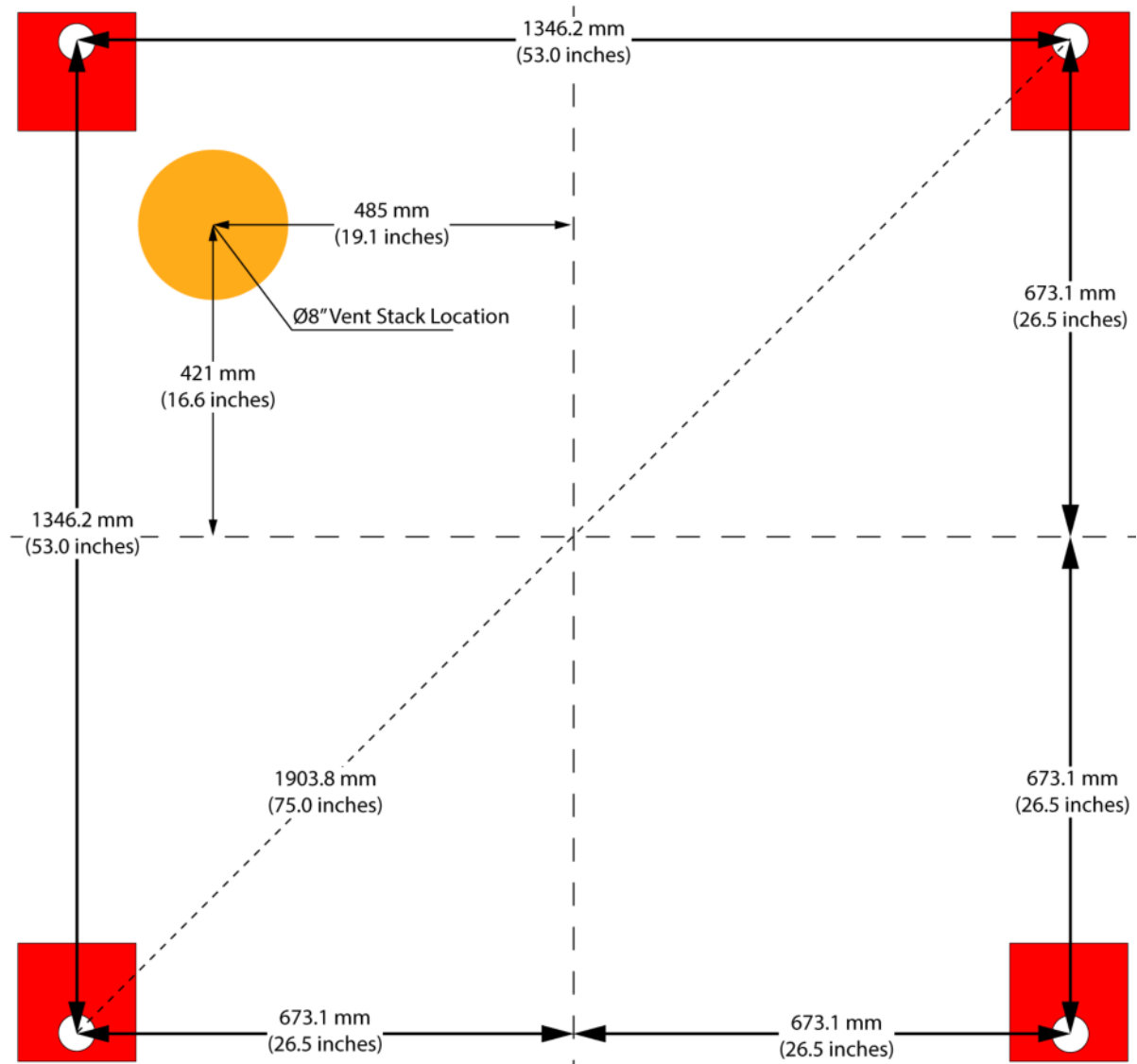
Accessories of JXM magnet (5926243-301)			
Item	Quantity	Part number	Manufacturer
Vibroacoustic damping mats, 20mm	4	5926243-301-37	-
Vibroacoustic damping mats, 10mm	4	5926243-301-38	-

2. Put the vibroacoustic damping mats on the floor, in the appropriate location where the magnet will be set.

Figure 6-1 Vibroacoustic damping mat, side view



Item	Description
1	Vibroacoustic damping mats, 10mm
2	Vibroacoustic damping mats, 20mm

Figure 6-2 Vibroacoustic damping mat placement (patient end is at the bottom)

6.3 Preparing to move the magnet

Prerequisites

Safety

Before working in any GE HealthCare MR suite or doing any GE HealthCare service procedure, you must:

- Have read and understood all hazard conditions and safety requirements in the latest revision of the GE HealthCare *MR Service Safety Manual* (5452735).
- Have successfully completed all relevant GE HealthCare Environmental Health and Safety (EHS) courses (or for non-GE HealthCare employees, equivalent workplace training courses).
- Comply with all site-specific training and workplace safety requirements.

If you have any safety concerns at any time, do not begin work or immediately stop work and move to a safe location. Immediately contact your supervisor or site safety officer for instructions on how to proceed.

Safety	
	⚠ DANGER POTENTIAL ASPHYXIATION HAZARD Loss of magnet vacuum will result in the rapid expulsion of helium gas, which can cause asphyxiation in enclosed areas. Make sure that hospital personnel are aware of this situation prior to the magnet being moved within the enclosed areas. Use extreme caution and do not contact or damage the vacuum vessel during magnet transit or siting.
	⚠ WARNING POTENTIAL INJURY HAZARD Moving the magnet improperly will cause personal injury or magnet damage. See Unloading and moving the magnet before moving the magnet using a forklift or crane.
	NOTICE EQUIPMENT DAMAGE RISK Improper transportation can result in damage to the magnet. Tasks in this section are to be done by riggers, not by GE HealthCare Personnel. Do not apply any force to the magnet's enclosures. Any floor anchors that are used to move the magnet must not penetrate the RF shield. Magnet and dock installation and placement are critical to image quality and hardware reliability. The magnet must be installed level with the isocenter to the specifications listed in this chapter.

About this task

After the magnet is moved to the building using a crane or forklift (see Unloading and moving the magnet), it needs to be moved to the magnet room. There are many methods to help move the magnet, including the use of a motorized tow vehicle, a come-along, or a chain jack. When you are moving the magnet, make sure that you are moving it in a smooth, controlled manner.



NOTE

Remove the bubble wrap at any point forward when the process allows. When removing the bubble wrap, do not nick or cut any electrical cables.

Procedure

- 1. Optionally, roller dollies are recommended for moving the magnet inside a building. If you use roller dollies, put steel floor plates along the magnet delivery route.

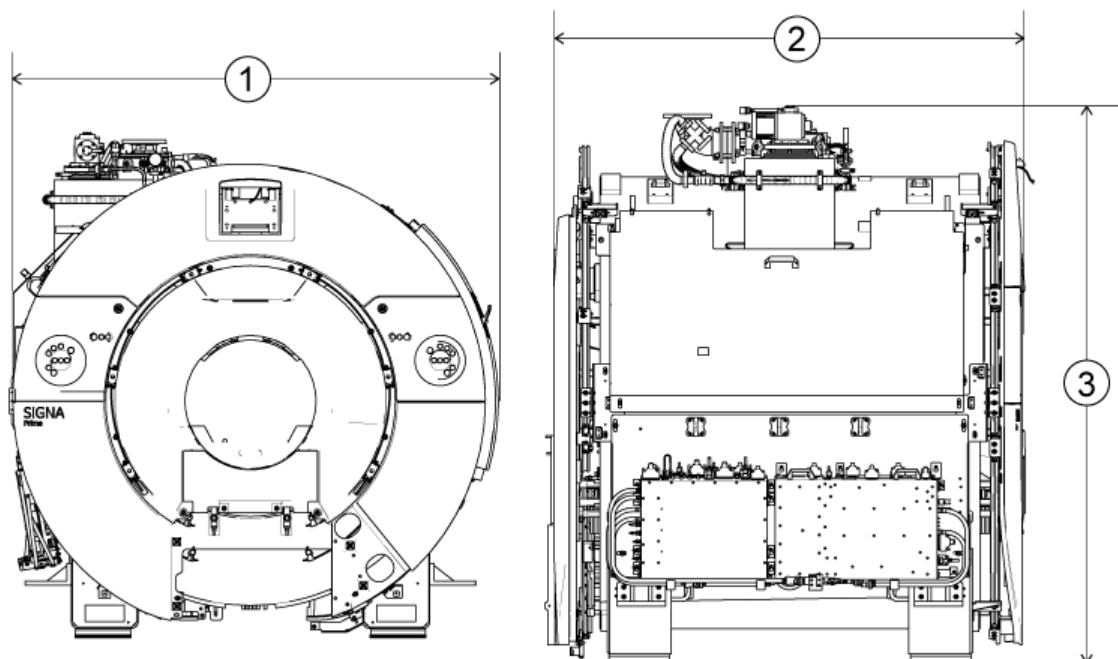
Figure 6-3 Magnet on roller dollies



1	Lifting angle brackets
2	Roller dollies

- 2. Have the rigger make sure that walls, floors, and so on along the transportation route are protected from potential damage.
- 3. Make sure that the magnet front-rear orientation is relative to the magnet room's front and rear.
- 4. Do a check of all clearances along the route that the magnet will move to get to the magnet room. Compare those clearances with the appropriate illustration below.

 **NOTE** Dimensions referenced are worst-case scenarios.

Figure 6-4 Clearance dimensions, magnet patient end as-shipped condition

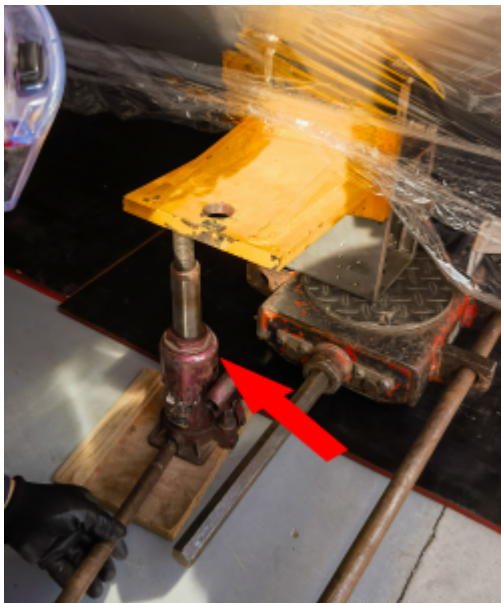
Items	Dimension
1	2330 mm (91.73 inches)
2	1984 mm (78.11 inches)
3	2267 mm (89.25 inches)

5. Compare the dimensions of the magnet on the moving fixtures being used with the clearances measured along the magnet delivery route. The height can be reduced by putting the moving fixtures directly under the lifting angle brackets.

6.4 Moving the magnet to the MR suite

1. If raising the magnet is required, the lifting portion of the jack must be fully below the lifting angle bracket.

Figure 6-5 Lifting portion



2. Avoid tilting or rotating the magnet while moving it into the MR suite.
3. From the magnet room entrance to the magnet's final position, put steel floor plates as needed to protect the magnet room floor. The rigger must take actions necessary to ensure that walls, floors, and so on along the transportation route/path are protected from potential damage.
4. Move the magnet to the magnet room.
5. Make sure that there is adequate distance between the service side of the magnet and the magnet room wall as specified in the appropriate *Preinstallation Manual*.

6.5 Preparations for centering the magnet

Prerequisites

Tools and test equipment			
Item	Quantity	Part number	Manufacturer
Nonmagnetic Titanium Service Tool Kit, Small Set	1	5113258	-

Safety

Before working in any GE HealthCare MR suite or doing any GE HealthCare service procedure, you must:

- Have read and understood all hazard conditions and safety requirements in the latest revision of the GE HealthCare *MR Service Safety Manual* (5452735).
- Have successfully completed all relevant GE HealthCare Environmental Health and Safety (EHS) courses (or for non-GE HealthCare employees, equivalent workplace training courses).
- Comply with all site-specific training and workplace safety requirements.

If you have any safety concerns at any time, do not begin work or immediately stop work and move to a safe location. Immediately contact your supervisor or site safety officer for instructions on how to proceed.

Removing the bridge cover

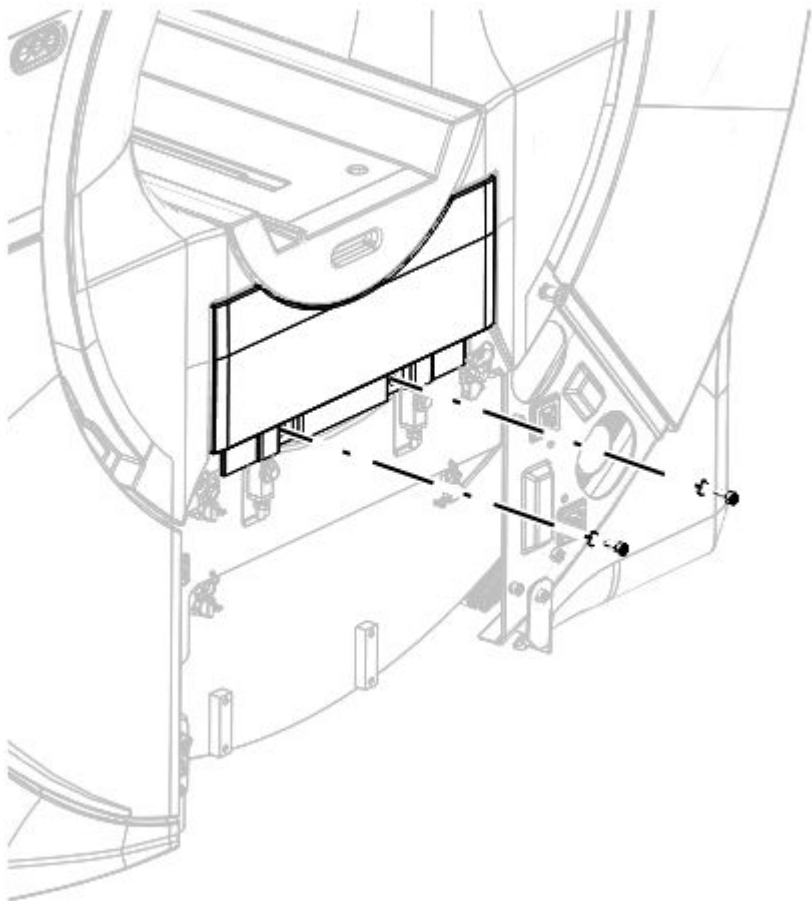
Prerequisites

Personnel requirements			
Required persons	Preliminary requirements	Procedure	Finalization
1	-	5 minutes	-

Procedure

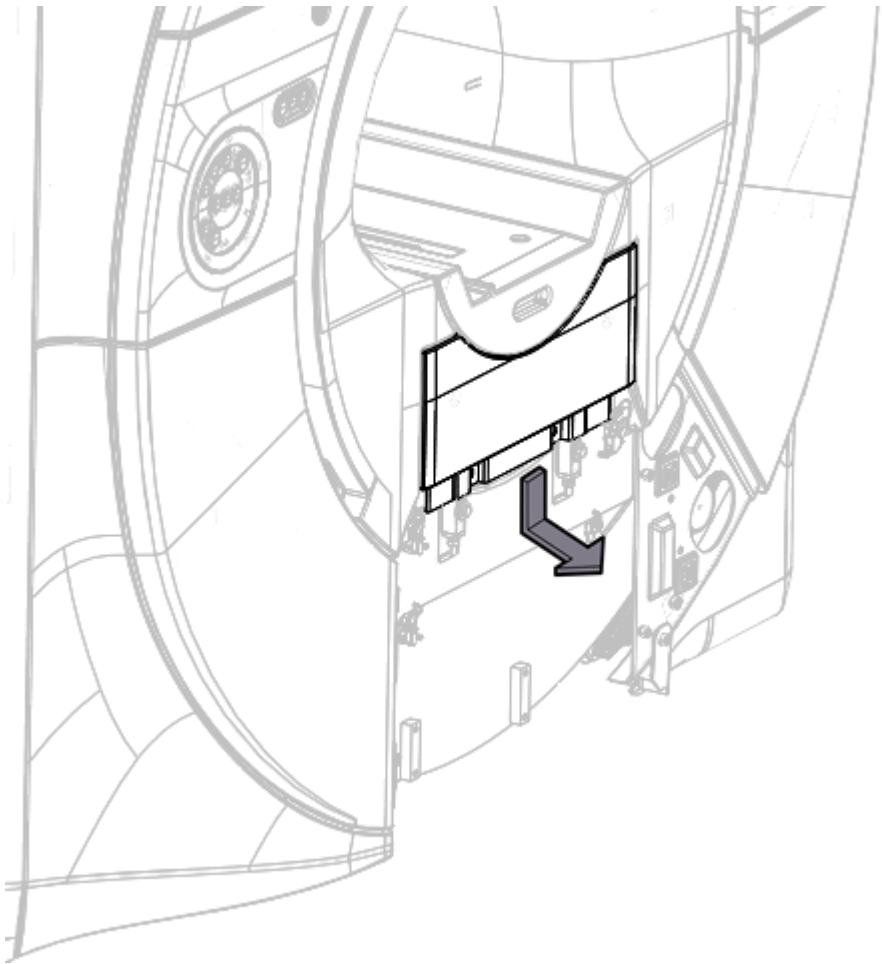
1. Remove two screws securing bridge cover to front endbell.

Figure 6-6 Front bridge screws



- 2. Slide the bridge cover downward until the tabs disengage from their tracks on the front endbell.
- 3. Remove the bridge cover.

Figure 6-7 Remove the bridge cover



Removing the fixed table front bridge

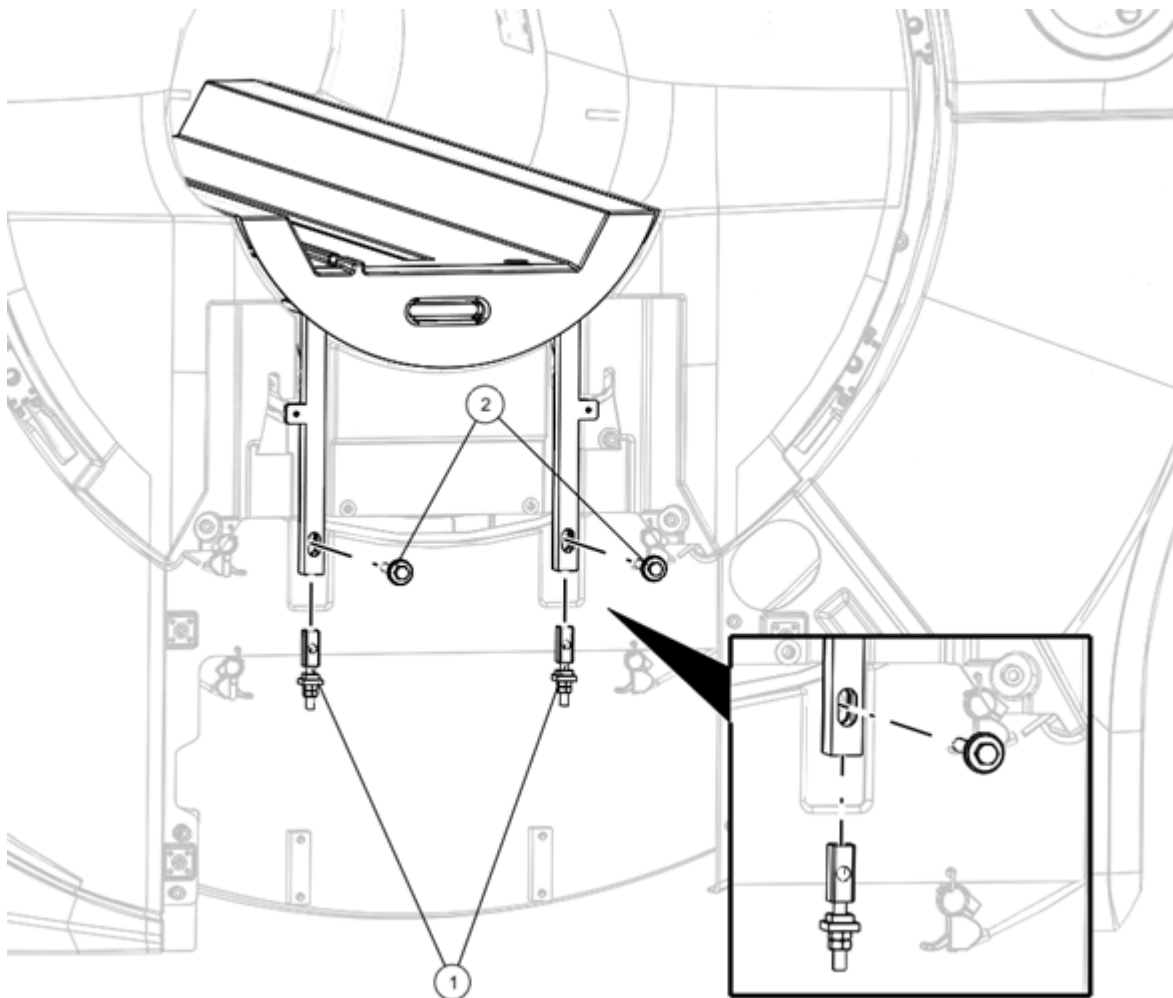
Prerequisites

Personnel requirements			
Required persons	Preliminary requirements	Procedure	Finalization
2	-	20 minutes	-

Procedure

1. Remove two screws securing to the magnet.

Figure 6-8 Front bridge screws



1	Adjuster
2	Screws

- 2.



POTENTIAL COMPONENT DAMAGE

The rear wheels of the bridge may be damaged.

Do not pull the front bridge out of the bore until one person has lifted the rear of the bridge and the rollers can enter the bore without damage.

Remove the front bridge.

Removing the front endbell

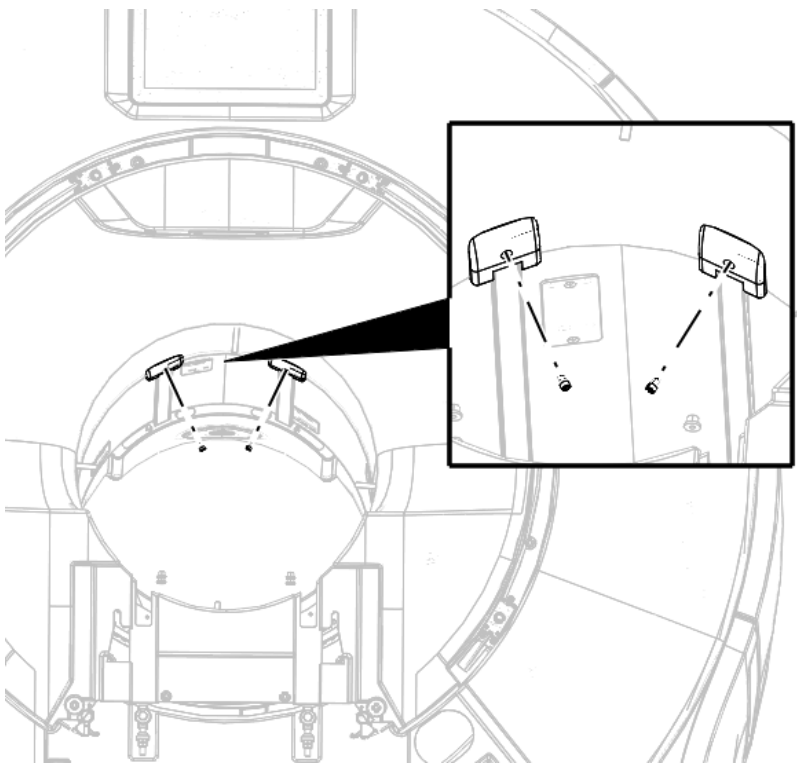
Prerequisites

Personnel Requirements			
Required persons	Preliminary requirements	Procedure	Finalization
1	-	10 minutes	-

Procedure

1. Remove the patient light end covers from the magnet bore upper side.

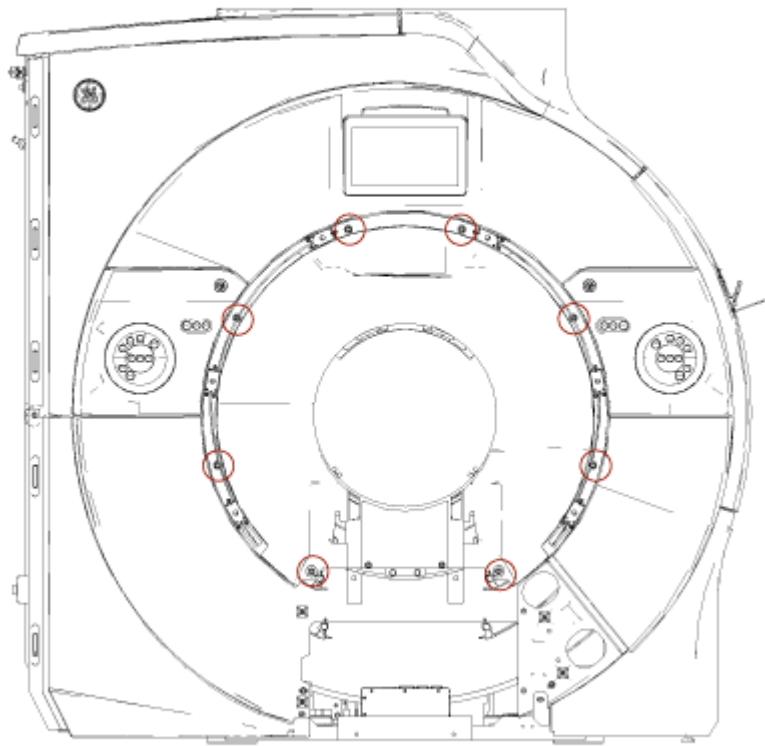
Figure 6-9 Patient light end covers



2. Disconnect sensor cable from J10 connector on the SRI (Scan Room Interface) assembly.

3. Remove eight screws securing the front endbell to the magnet.

Figure 6-10 Front endbell screws



4. Remove the front endbell from the magnet.

Removing the rear endbell

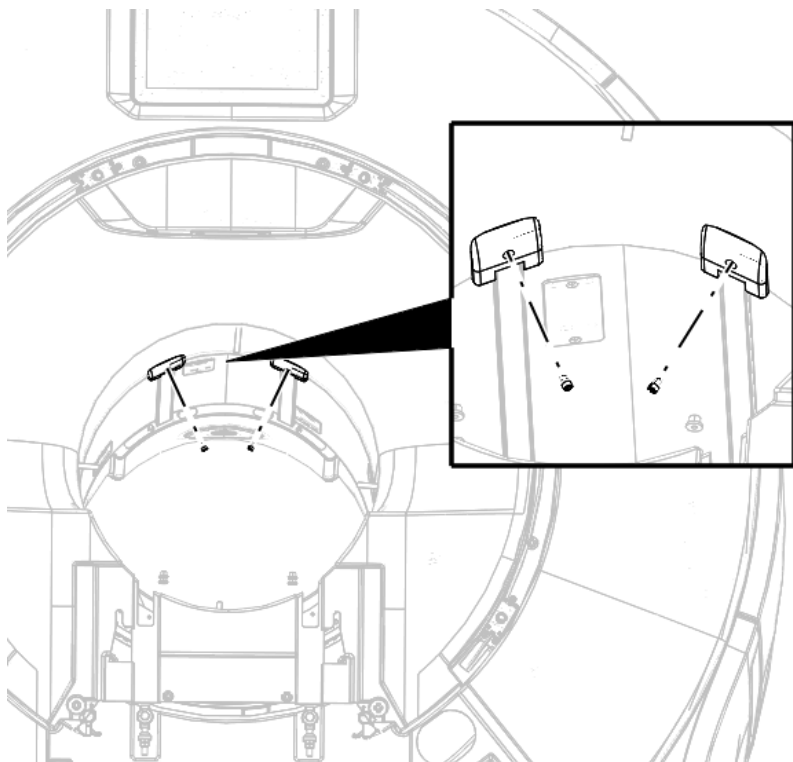
Prerequisites

Personnel requirements			
Required persons	Preliminary requirements	Procedure	Finalization
1	-	30 minutes	-

Procedure

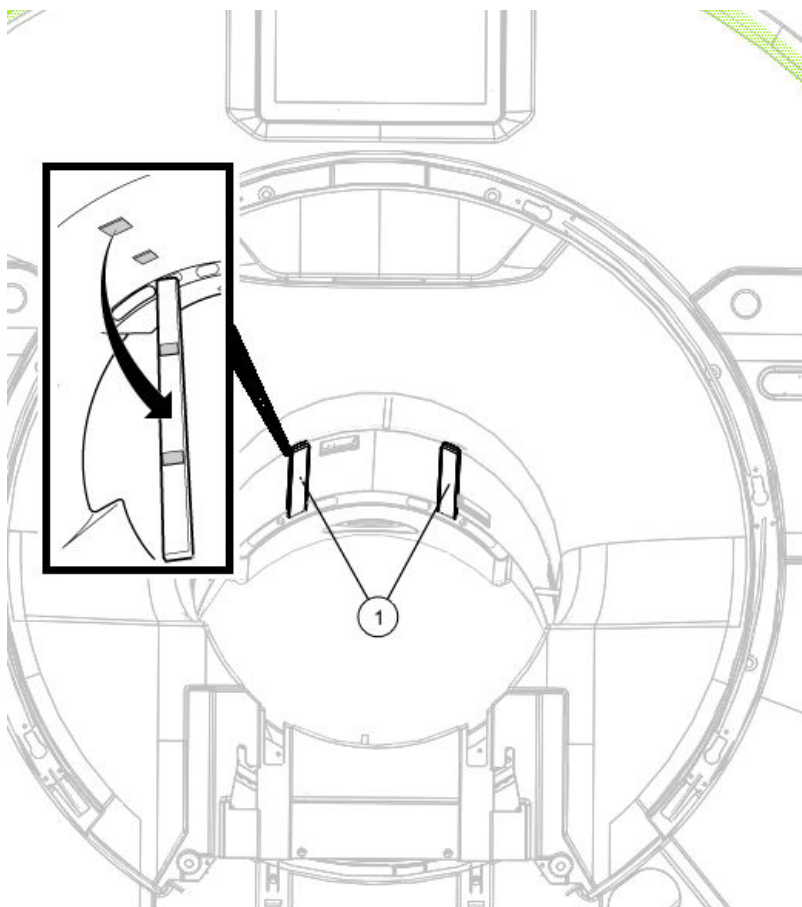
1. Remove the patient light end covers from the magnet bore upper side.

Figure 6-11 Patient light end covers



2. Remove the patient lights from the magnet bore upper side. The patient light is secured by Velcro tape.

Figure 6-12 Patient lights

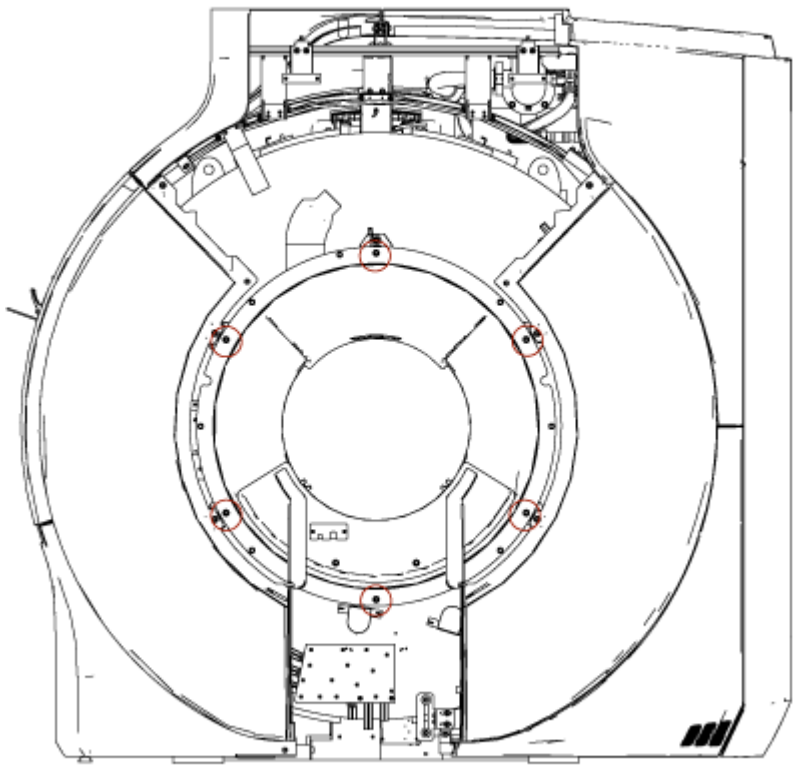


1

Patient lights

- 3. Remove the screws and washers securing the rear endbell to the magnet.

Figure 6-13 Rear endbell screws and washers

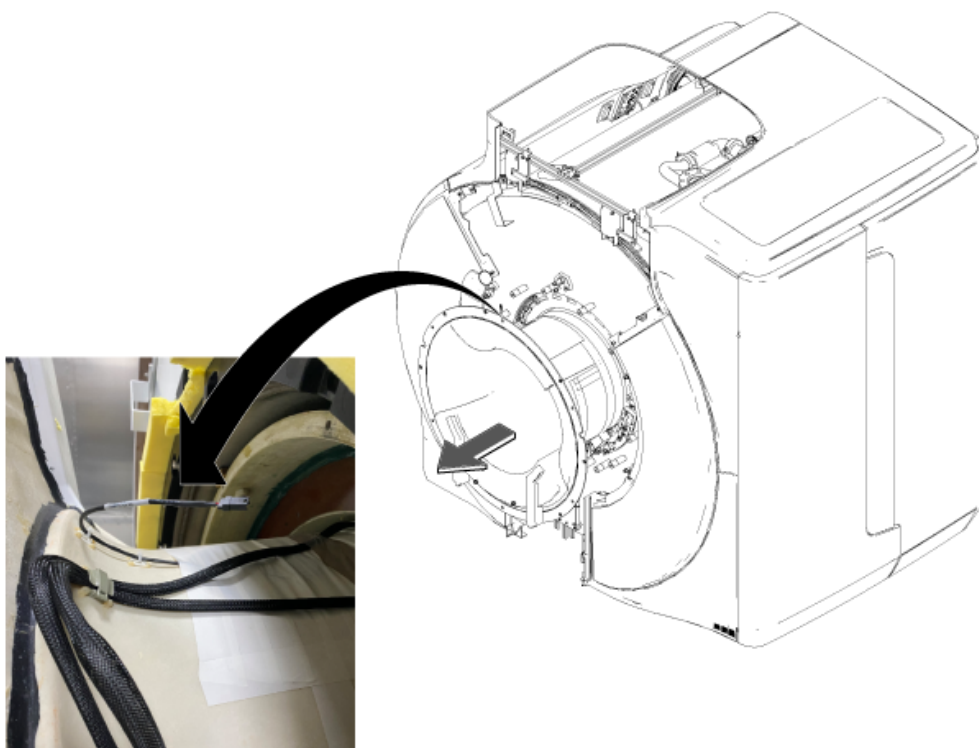


Item	Part number	Quantity	Manufacturer
M10 Screw	1000-M10C025-33	6	-
M10 Washer	2000-M10-30	6	-

- 4. Withdraw rear endbell until can access to speaker cable connector.

5. Disconnect the speaker power cable from the magnet upper side.

Figure 6-14 Speaker cable



6. Remove the rear endbell from the magnet.

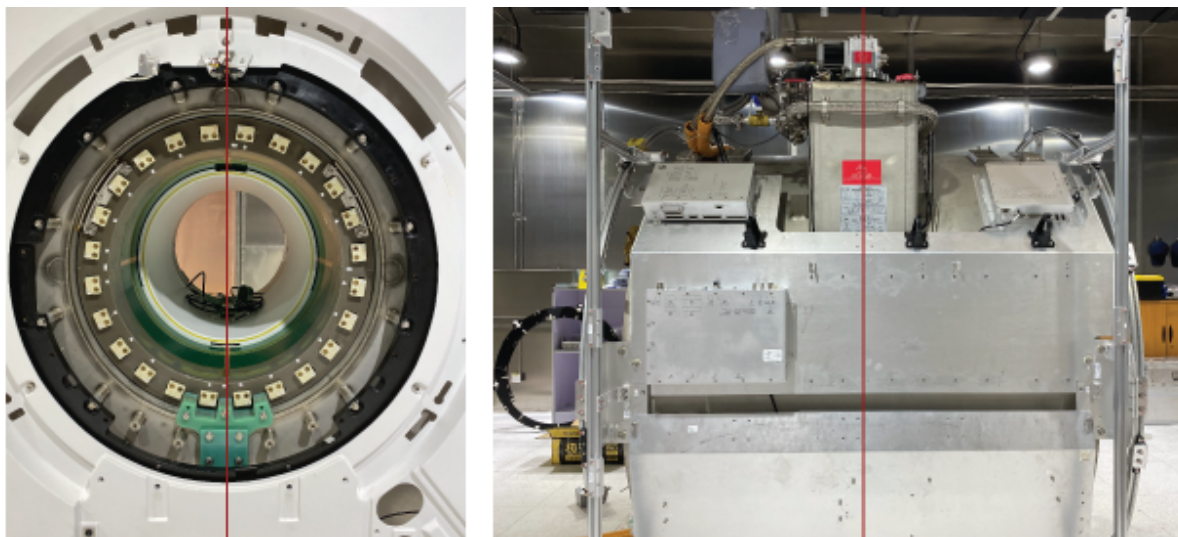
6.6 Centering the magnet in the MR suite

Procedure

1. Make sure the laser levels are still in position and activated. If they are not, see [5.1 Preparing the scan room for leveling magnet on page 61](#).
2. Carefully cut out the bubble wrap or plastic from the bore opening on the front and rear of the magnet, remove the bridge, the front and rear endbell to provide a clear view of the 6 and 12 o'clock marks. See [6.5 Preparations for centering the magnet on page 75](#) for detailed procedures.
3. Make sure the laser levels are in self-leveling mode.

4. Align the 2 marks of the front targets (12 and 6 o'clock) on the gradient coil with the vertical laser line, align the mark on the cryogen plenum rib and ceiling or floor marked before with the side vertical laser line for centering the magnet.

Figure 6-15 Centering the magnet



6.7 Lowering the magnet into position

Procedure

1. Cut and remove enough of the bubble wrap to allow access to the feet for leveling operations.

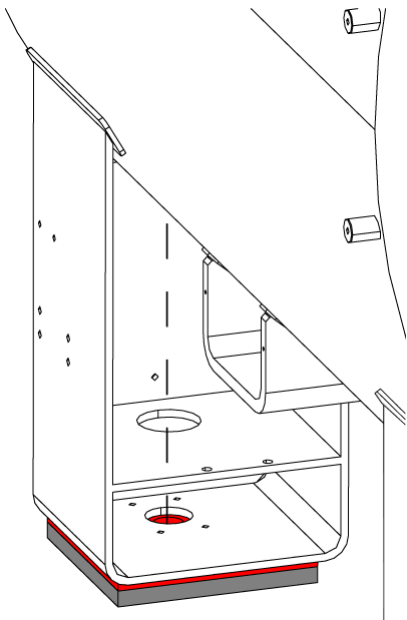


NOTE

When cutting off the bubble wrap, do not nick or cut any electrical cables.

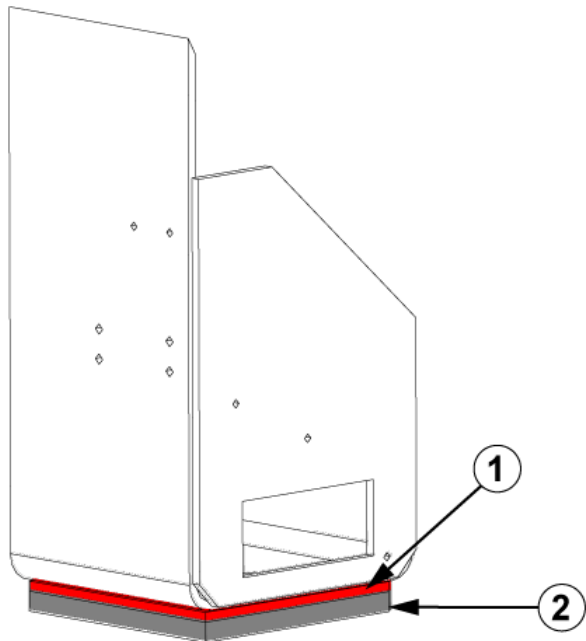
2. Align the holes in the magnet feet to the anchor holes in the vibroacoustic damping mats.
3. Make sure that the four holes, one per each magnet foot, align centered $\pm 3\text{mm}$ ($\pm 0.125\text{inches}$) over the anchor stud holes in the vibroacoustic damping mats.

Figure 6-16 Center the magnet foot to the anchor hole



- 4. Jack the magnet up sufficiently at the four angle brackets (four corners) and remove the moving fixtures.
- 5. Put the damping mats below the magnet feet, ensure that the magnet feet can be fully covered.

Figure 6-17 Magnet foot on a damping mat



1	10mm Damping mat
2	20mm Damping mat

- 6. Slowly lower the magnet onto the damping mats.

6.8 Leveling the magnet

Procedure

1. Make sure the laser level is in self-leveling mode.



NOTE

In self-leveling mode, if the laser level tilts $> 4^\circ$ in any direction, the lasers do not show. When the laser level tilt is $\leq 4^\circ$, the level indicator LED is green and the lasers show.

Figure 6-18 Laser level in self-leveling mode (unlocked position)

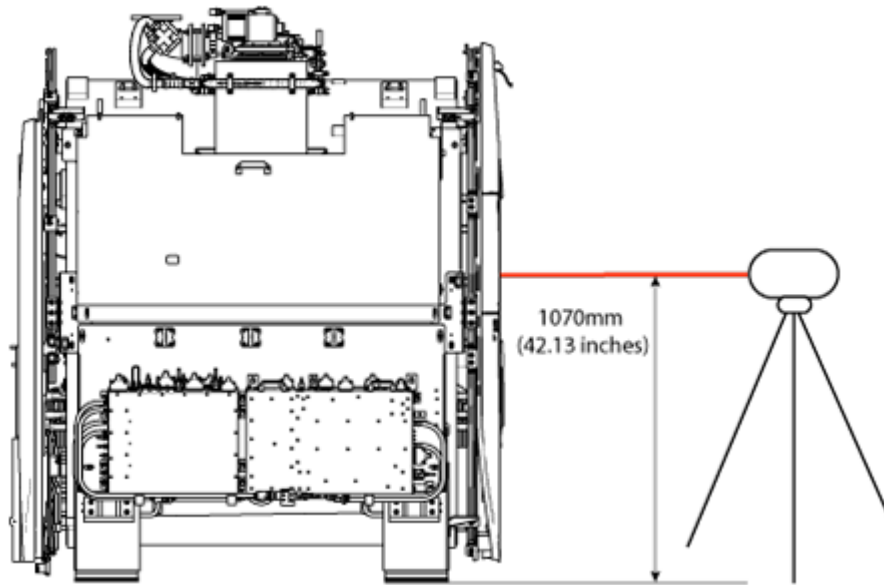


2. Set up laser on the front of magnet and adjust height to 1070mm (42.13 inches). Set 1070mm (42.13 inches) as the leveling target.



NOTE

If necessary, position a folding ruler vertically near the front center of the magnet, and adjust the tripod height until the horizontal laser line aligns on the ruler 1070 mm (42.13 inches) from the floor (the magnet isocenter should be 1070 ± 6 mm).

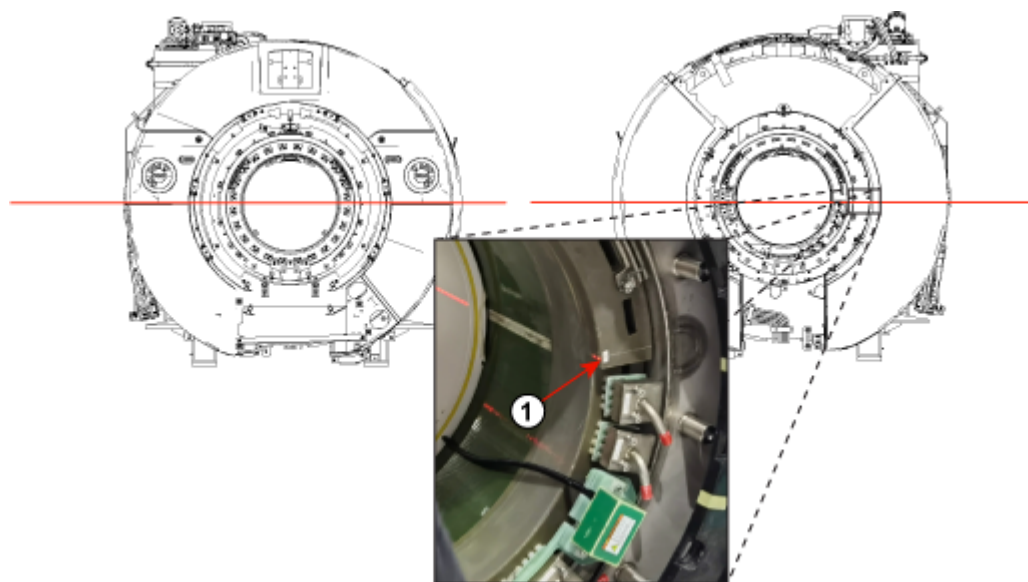
Figure 6-19 Setting the laser level

3. Check the four marks (3 and 9 o'clock) of the front and rear on the gradient coil with horizontal laser line.
 - 3.1. If the highest mark is above 1070mm (42.13 inches), and it is in the range of 1070mm (42.13 inches) ~1076mm (42.36 inches), take the measurement of the highest height as the leveling target.
 - 3.2. If all marks below the laser line, remain 1070mm (42.13 inches) as the leveling target.

**NOTE**

If the laser line is blocked by the body coil and cannot be seen on the rear side of the gradient coil, one can place the laser closer to the magnet.

Figure 6-20 Check the gradient coil marks with the horizontal laser line (left-rear mark as an example)



1	Mark on magnet
---	----------------

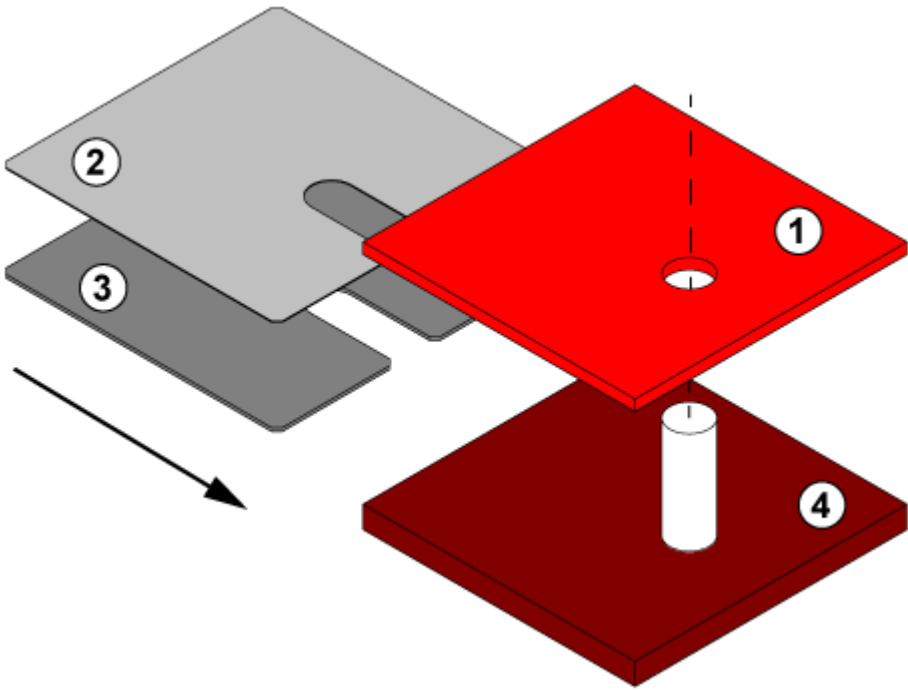
4. Leveling the magnet by inserting shim pads (there are 1mm shims and 3mm shims) between damping mats under the magnet feet. Make all four marks on magnet overlapped with the laser line.



NOTE

Make sure that the U-shaped holes align centered $\pm 3\text{mm}$ over the anchor stud holes in the vibroacoustic damping mats and the shim pads fully cover the magnet feet.

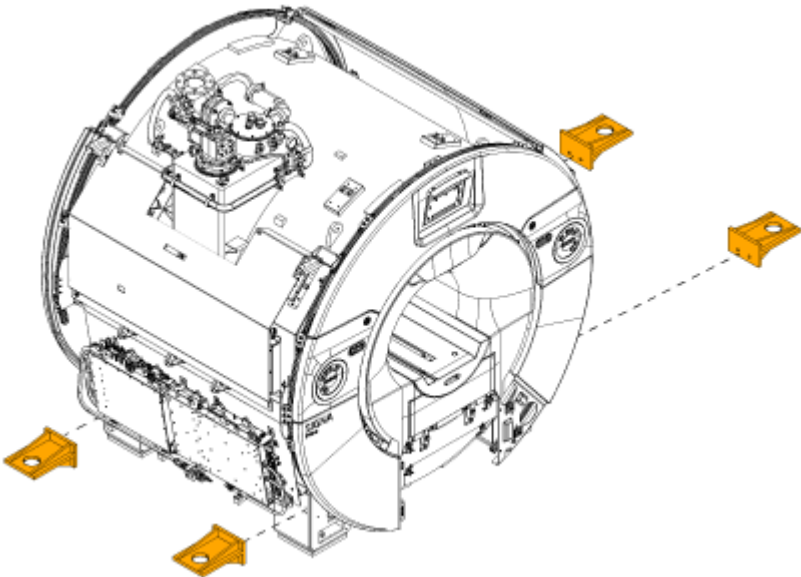
Figure 6-21 Shim pads



Item	Description	Item	Description
1	Vibroacoustic damping mats, 10mm	2	Magnet aluminum shim pads, 1mm
3	Magnet aluminum shim pads, 3mm	4	Vibroacoustic damping mats, 20mm

5. Recheck markers on the magnet, make sure all the markers not exceed 1076mm (42.36 inches) or under 1064mm (41.89 inches).
6. Test-fit the cryogen vent stack to make sure it aligns with the scan room vent pipe.
7. Remove the lifting angle brackets (5926243-302-23).

Figure 6-22 Lifting angle brackets



Chapter 7 Installing a seismic vibroacoustic damping mat and leveling the magnet

Prerequisites

Safety	
	NOTICE EQUIPMENT DAMAGE RISK Improper damping mat installation can result in damage to the magnet. Tasks in this section must be completed by riggers, not GE Service Personnel. The vibroacoustic damping mats used for seismic install must be installed over the seismic studs before moving the magnet into the magnet room.

7.1 Equipment overview

Prerequisites

Tools and test equipment			
Item	Quantity	Part number	Manufacturer
Tape Measure	1	-	-
Seismic Anchor Kit, Platform	1	M6001AH	-

PH Alignment and Magnet Leveling Kit (5897979)			
Item	Quantity	Part number	Manufacturer
Laser Level PLS 180R Z	2	5831466	-
Tripod, Vanguard Alta Pro 263AB 100	2	5829559	-
Folding Ruler, 2 meters, FM-DELA.401.00	1	5897974	-
Vertical Alignment Tool	2	5831978	-

Accessories of JXM magnet (5926243-301)			
Item	Quantity	Part number	Manufacturer
Vibroacoustic damping mats, 20mm	4	5926243-301-37	-
Vibroacoustic damping mats, 10mm	4	5926243-301-38	-
Magnet aluminum shim pads, 3mm	12	5926243-301-39	-

Accessories of JXM magnet (5926243-301)			
Item	Quantity	Part number	Manufacturer
Magnet aluminum shim pads, 1mm	16	5926243-301-40	-
Required conditions			
Magnet and dock installation and placement are critical to image quality and hardware reliability.			

7.2 Positioning a seismic mounted vibroacoustic damping mat

About this task



NOTE

The vibroacoustic damping mat must be installed on the floor before moving the magnet into the magnet room.

Procedure

1. Make sure that the customer has installed seismic threaded rods in the concrete to match the magnet foot pattern. The rods must meet seismic requirements and have a maximum diameter of 1.25 inches and be between 5.5 and 6 inches in height above the finished floor.



NOTE

Installing GE HealthCare magnets in seismic regions is the responsibility of the customer, and should be done by riggers, not by GE HealthCare Service Personnel. The customer is required to make the decision as to how seismic regulations are met in the region based off the architect's recommendations.

The diagram illustrates a square room layout with dimensions and a vent stack location. The room is defined by a square boundary with a side length of 1346.2 mm (53.0 inches). The layout includes a central dashed line representing the main axis, and a dashed diagonal line representing the main diagonal. The vent stack location is marked by an orange circle, with dimensions indicating its position: 485 mm (19.1 inches) from the left wall and 421 mm (16.6 inches) from the bottom wall. The vent stack is labeled "Ø8" Vent Stack Location". The room is divided into four quadrants by the dashed lines, with dimensions of 673.1 mm (26.5 inches) for each quadrant. The dimensions are labeled in millimeters (mm) and inches (inches).

- Put the vibroacoustic damping mats on the floor, over the seismic rods, according to the prescribed markings.

Figure 7-2 Vibroacoustic mat placement (patient end is at the bottom)

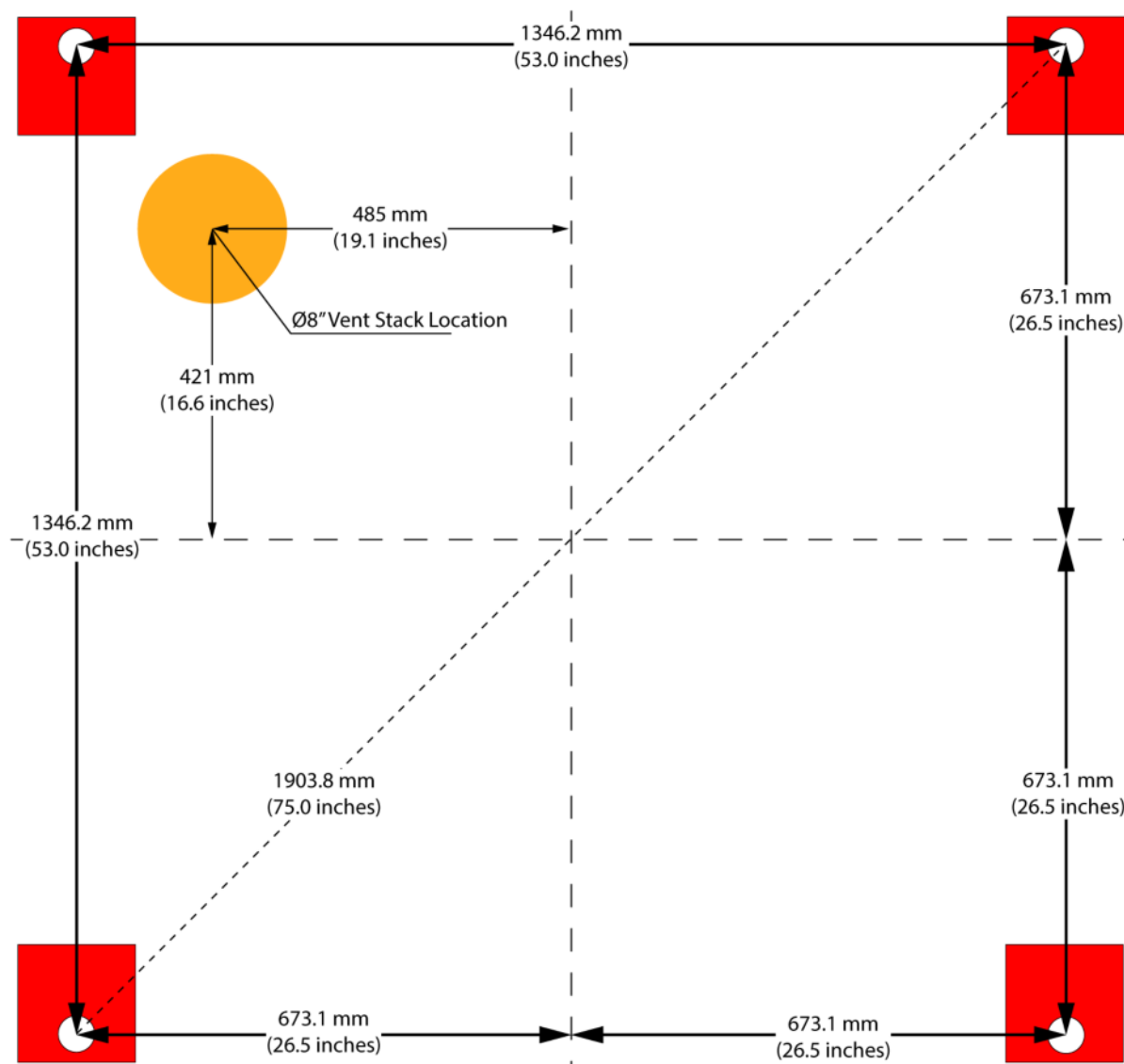
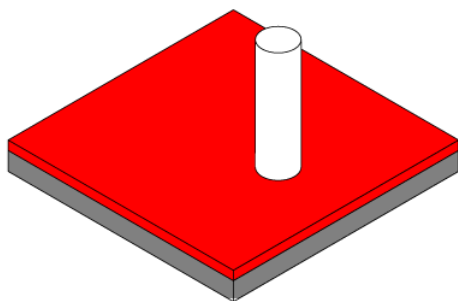


Figure 7-3 Vibroacoustic mat put over a seismic rod



7.3 Preparing to move the magnet

Prerequisites

Safety	
Before working in any GE HealthCare MR suite or doing any GE HealthCare service procedure, you must: • Have read and understood all hazard conditions and safety requirements in the latest revision of the GE HealthCare <i>MR Service Safety Manual</i> (5452735). • Have successfully completed all relevant GE HealthCare Environmental Health and Safety (EHS) courses (or for non-GE HealthCare employees, equivalent workplace training courses). • Comply with all site-specific training and workplace safety requirements. If you have any safety concerns at any time, do not begin work or immediately stop work and move to a safe location. Immediately contact your supervisor or site safety officer for instructions on how to proceed.	
	⚠ DANGER POTENTIAL ASPHYXIATION HAZARD Loss of magnet vacuum will result in the rapid expulsion of helium gas, which can cause asphyxiation in enclosed areas. Make sure that hospital personnel are aware of this situation prior to the magnet being moved within the enclosed areas. Use extreme caution and do not contact or damage the vacuum vessel during magnet transit or siting.
	⚠ WARNING POTENTIAL INJURY HAZARD Moving the magnet improperly will cause personal injury or magnet damage. See Unloading and moving the magnet before moving the magnet using a forklift or crane.
	NOTICE EQUIPMENT DAMAGE RISK Improper transportation can result in damage to the magnet. Tasks in this section are to be done by riggers, not by GE HealthCare Personnel. Do not apply any force to the magnet's enclosures. Any floor anchors that are used to move the magnet must not penetrate the RF shield. Magnet and dock installation and placement are critical to image quality and hardware reliability. The magnet must be installed level with the isocenter to the specifications listed in this chapter.

About this task

After the magnet is moved to the building using a crane or forklift (see Unloading and moving the magnet), it needs to be moved to the magnet room. There are many methods to help move the

magnet, including the use of a motorized tow vehicle, a come-along, or a chain jack. When you are moving the magnet, make sure that you are moving it in a smooth, controlled manner.

 **NOTE** Remove the bubble wrap at any point forward when the process allows. When removing the bubble wrap, do not nick or cut any electrical cables.

Procedure

1. Optionally, roller dollies are recommended for moving the magnet inside a building. If you use roller dollies, put steel floor plates along the magnet delivery route.

Figure 7-4 Magnet on roller dollies

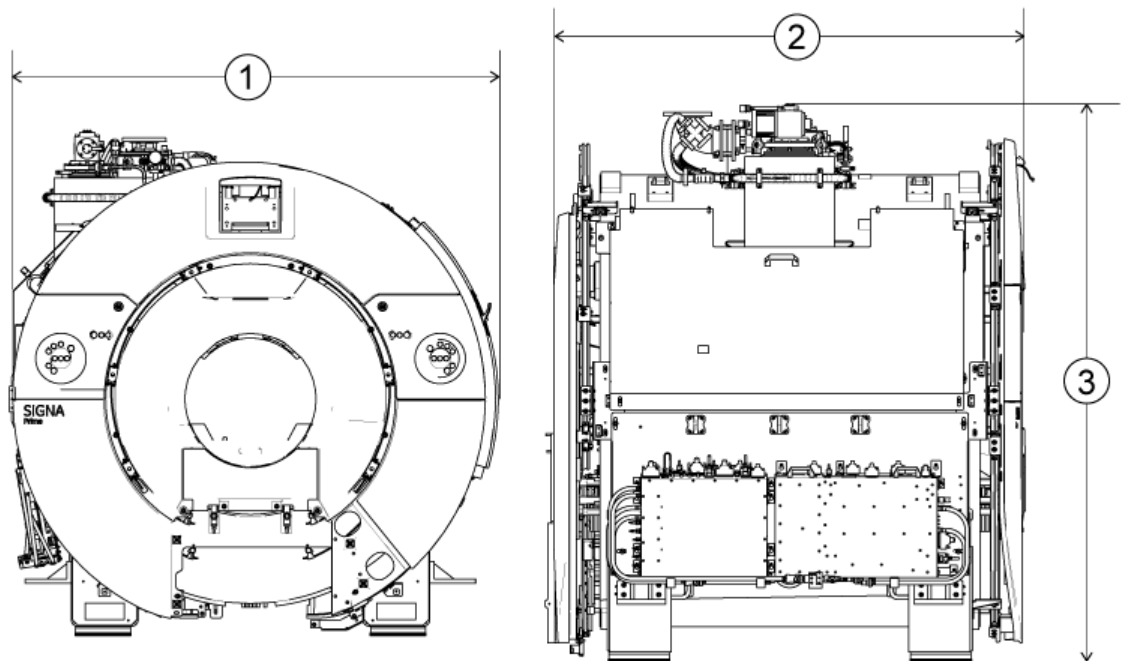


1	Lifting angle brackets
2	Roller dollies

2. Have the rigger make sure that walls, floors, and so on along the transportation route are protected from potential damage.
3. Make sure that the magnet front-rear orientation is relative to the magnet room's front and rear.
4. Do a check of all clearances along the route that the magnet will move to get to the magnet room. Compare those clearances with the appropriate illustration below.

 **NOTE** Dimensions referenced are worst-case scenarios.

Figure 7-5 Clearance dimensions, magnet patient end as-shipped condition



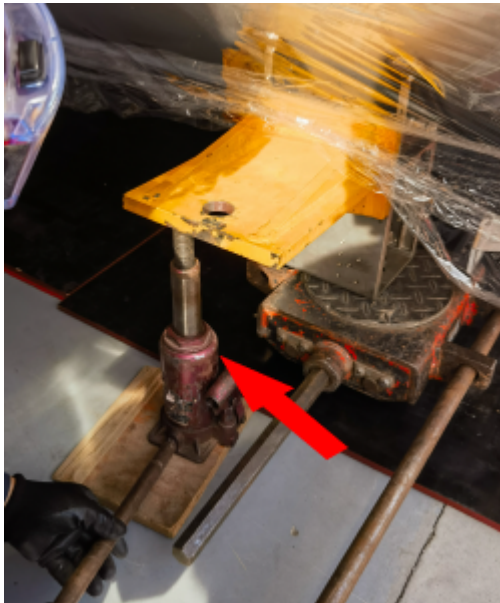
Items	Dimension
1	2330 mm (91.73 inches)
2	1984 mm (78.11 inches)
3	2267 mm (89.25 inches)

- 5. Compare the dimensions of the magnet on the moving fixtures being used with the clearances measured along the magnet delivery route. The height can be reduced by putting the moving fixtures directly under the lifting angle brackets.

7.4 Moving the magnet to the MR suite

1. If raising the magnet is required, the lifting portion of the jack must be fully below the lifting angle bracket.

Figure 7-6 Lifting portion



2. Avoid tilting or rotating the magnet while moving it into the MR suite.
3. From the magnet room entrance to the magnet's final position, put steel floor plates as needed to protect the magnet room floor. The rigger must take actions necessary to ensure that walls, floors, and so on along the transportation route/path are protected from potential damage.
4. Move the magnet to the magnet room.
5. Make sure that there is adequate distance between the service side of the magnet and the magnet room wall as specified in the appropriate *Preinstallation Manual*.

7.5 Lowering the magnet into position

Procedure

1. Cut open and remove enough of the bubble wrap to allow access to the feet and for leveling operations.



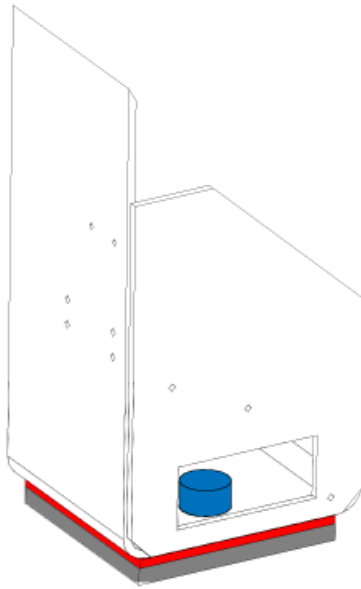
NOTE

When cutting off the bubble wrap, do not nick or cut any electrical cables.

2. Move the magnet to align the holes in the magnet feet to the anchor holes in the vibroacoustic damping mats.
3. Make sure that the four holes, one per each magnet foot, align centered ± 3 mm (± 0.125 inches) over the seismic anchor holes in the vibroacoustic mats.
4. Jack the magnet up sufficiently at the four lifting angle brackets, and remove the moving fixtures.

5. Slowly lower the magnet onto the vibroacoustic mats. Make sure the threaded seismic rods go through the holes in the magnet feet.

Figure 7-7 Magnet foot put on a vibroacoustic mat over a seismic rod



7.6 Adding leveling shims

About this task

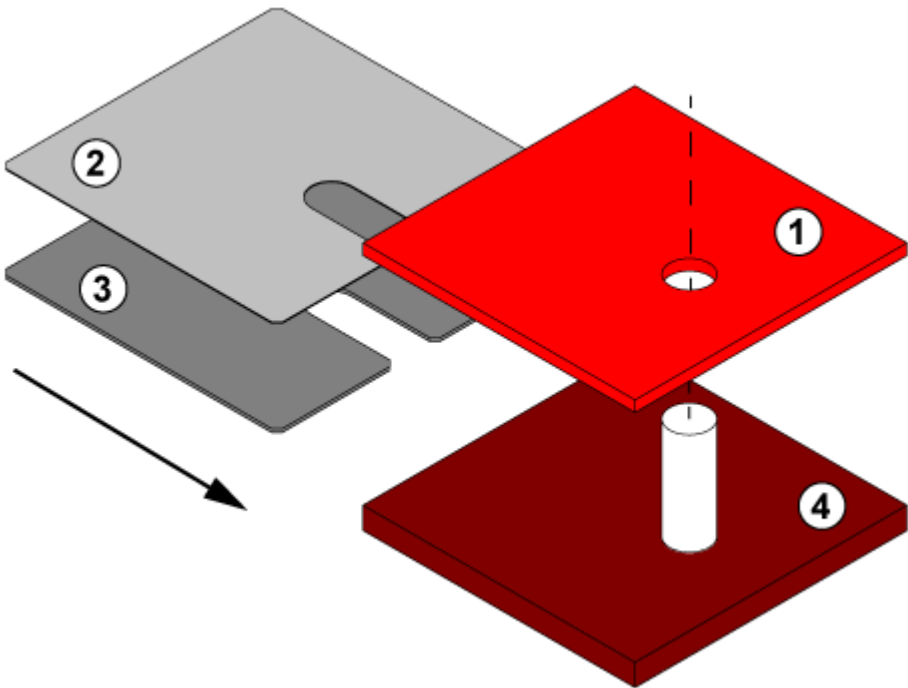
Review this section before you begin to level the magnet. If, while leveling the magnet, you need to insert shims under a magnet foot to achieve correct leveling, this section provides instructions for raising any one of the four magnet feet and properly placing shims.

Procedure

1. Use a jack to raise the lowest magnet foot.

- 2. Insert the appropriate thickness of aluminum shim pads under the low magnet foot along the vibroacoustic mat, adapt the U-shaped holes to the threaded seismic rods.

Figure 7-8 Adding shim pads



Item	Description	Item	Description
1	Vibroacoustic damping mats, 10mm	2	Magnet aluminum shim pads, 1mm
3	Magnet aluminum shim pads, 3mm	4	Vibroacoustic damping mats, 20mm

7.7 Leveling the magnet

Procedure

- 1. Make sure the laser level is in self-leveling mode.

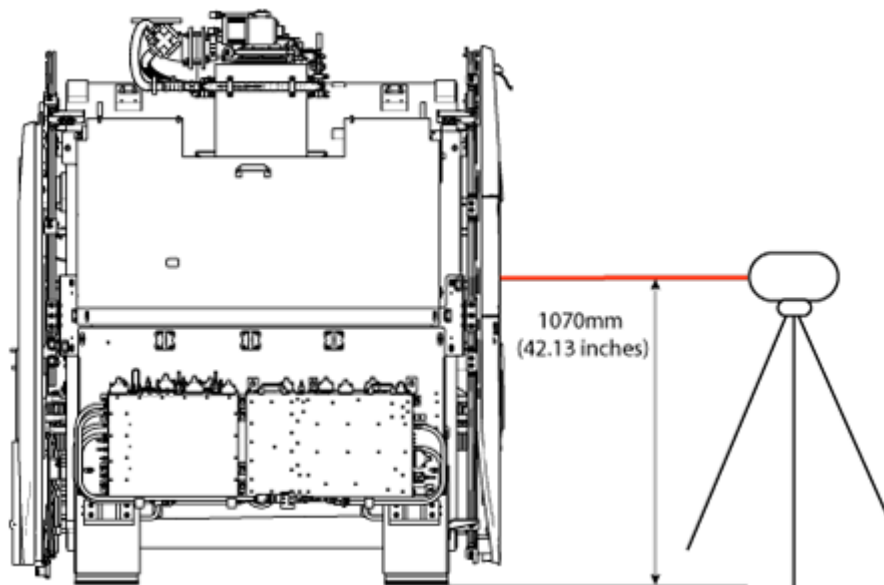
 **NOTE** In self-leveling mode, if the laser level tilts > 4° in any direction, the lasers do not show. When the laser level tilt is ≤ 4°, the level indicator LED is green and the lasers show.

Figure 7-9 Laser level in self-leveling mode (unlocked position)

2. Set up laser on the front of magnet and adjust height to 1070mm (42.13 inches). Set 1070mm (42.13 inches) as the leveling target.

**NOTE**

If necessary, position a folding ruler vertically near the front center of the magnet, and adjust the tripod height until the horizontal laser line aligns on the ruler 1070 mm (42.13 inches) from the floor (the magnet isocenter should be 1070 ± 6 mm).

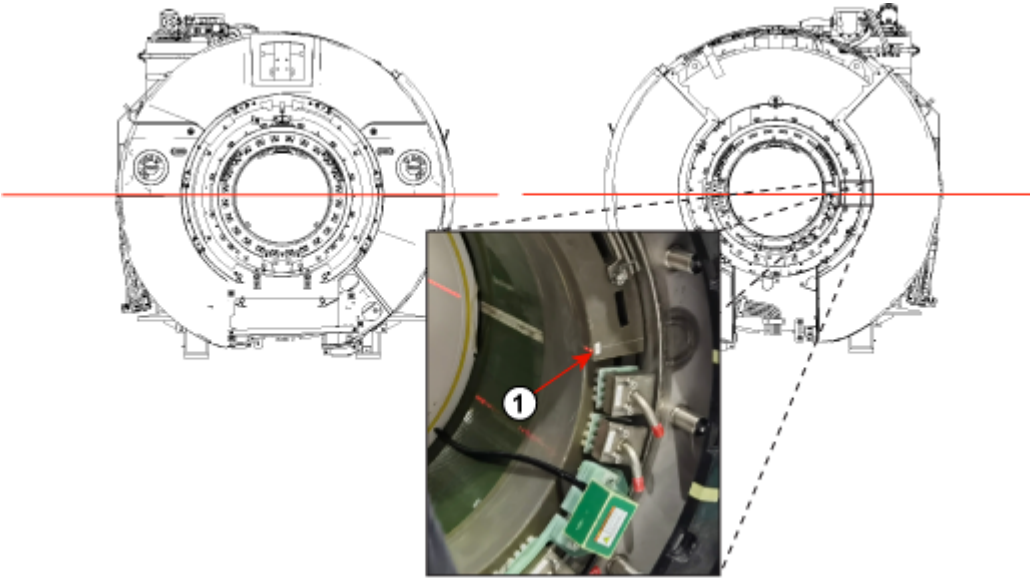
Figure 7-10 Setting the laser level

3. Check the four marks (3 and 9 o'clock) of the front and rear on the gradient coil with horizontal laser line.
 - 3.1. If the highest mark is above 1070mm (42.13 inches), and it is in the range of 1070mm (42.13 inches) ~1076mm (42.36 inches), take the measurement of the highest height as the leveling target.

3.2. If all marks below the laser line, remain 1070mm (42.13 inches) as the leveling target.

 **NOTE** If the laser line is blocked by the body coil and cannot be seen on the rear side of the gradient coil, one can place the laser closer to the magnet.

Figure 7-11 Check the gradient coil marks with the horizontal laser line (left-rear mark as an example)

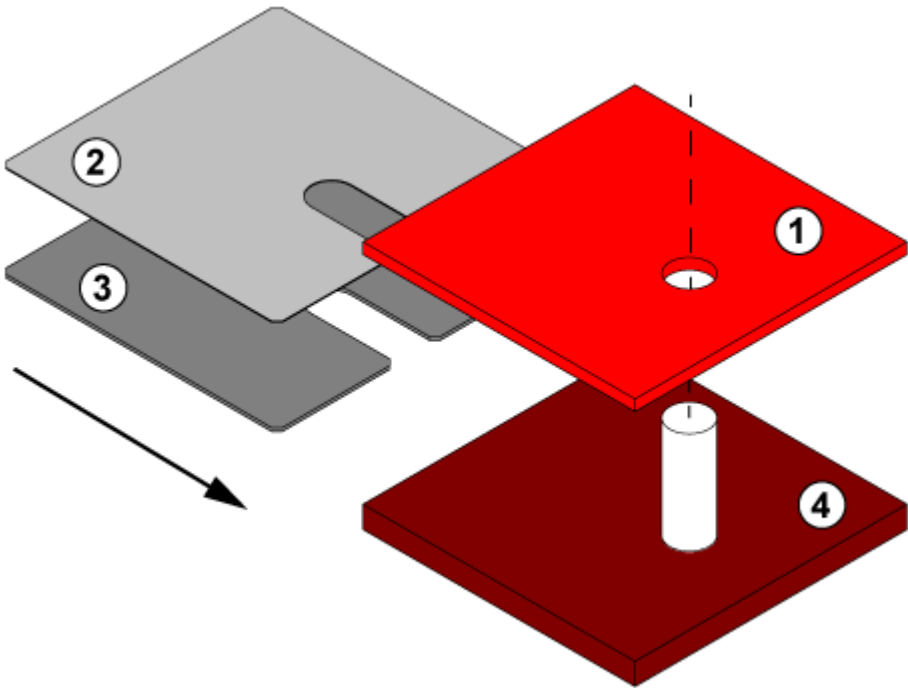


1	Mark on magnet
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4. Leveling the magnet by inserting shim pads (there are 1mm shims and 3mm shims) between damping mats under the magnet feet. Make all four marks on magnet overlapped with the laser line.

 **NOTE** Make sure that the U-shaped holes align centered $\pm 3\text{mm}$ over the anchor stud holes in the vibroacoustic damping mats and the shim pads fully cover the magnet feet.

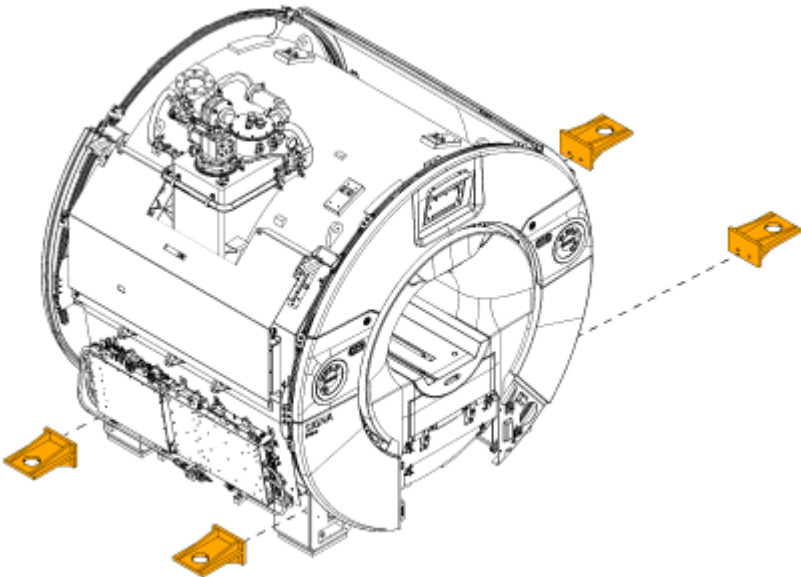
Figure 7-12 Shim pads



Item	Description	Item	Description
1	Vibroacoustic damping mats, 10mm	2	Magnet aluminum shim pads, 1mm
3	Magnet aluminum shim pads, 3mm	4	Vibroacoustic damping mats, 20mm

5. Recheck markers on the magnet, make sure all the markers not exceed 1076mm (42.36 inches) or under 1064mm (41.89 inches).
6. Test-fit the cryogen vent stack to make sure it aligns with the scan room vent pipe.
7. Remove the lifting angle brackets (5926243-302-23).

Figure 7-13 Lifting angle brackets



7.8 Seismic mounting of a vibroacoustic damping mat to the floor

About this task

Installing GE HealthCare magnets in seismic regions is the customer's responsibility. This section provides the process of how to install magnets in seismic regions using GE HealthCare vibroacoustic mats. However, this is only one way to achieve seismic installation. The customer is required to make the decision as to how seismic regulations are met in the region based off the architect's recommendations.



NOTE

Tasks in this section are to be done by riggers, not by GE HealthCare Service Personnel. The vibroacoustic mats used for seismic install must be installed over the seismic studs before moving the magnet into the magnet room.



NOTE

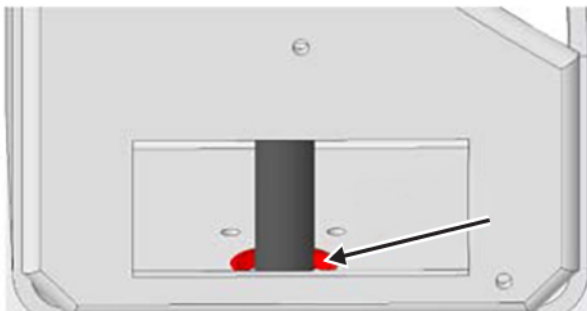
The seismic kit does not include studs, washers, and nuts. It is the customer's responsibility to buy the correct studs, washers, and nuts for their region based off the architect's recommendation.

The following mounting depiction is one of many ways to install magnets seismically and by no means is the mandatory method. Follow these steps for each magnet foot.

Procedure

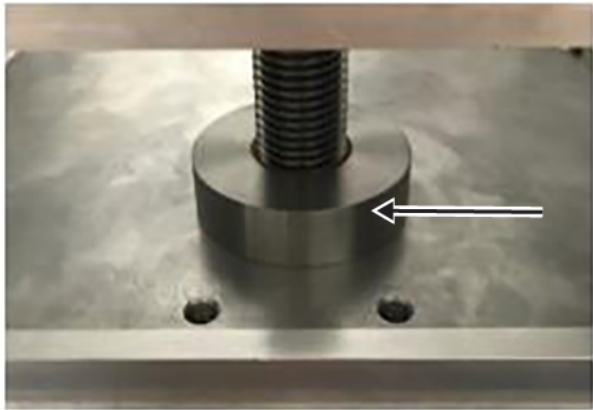
1. Install the rubber collar (5822743) from the seismic anchor kit (M6001AH) on the inner diameter of the magnet foot.

Figure 7-14 Rubber collar on the magnet foot



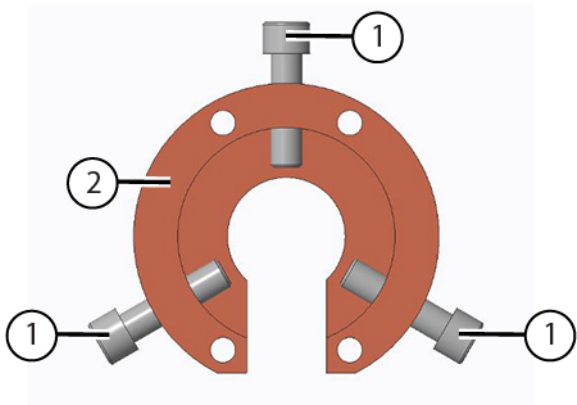
- 2. Put the stainless steel center ring (5817180) from the kit on the threaded seismic rod.

Figure 7-15 Center ring on a threaded seismic rod



- 3. Screw three of the M14 screws (1000-M14C050-31) from the kit into the stainless steel mounting plate (5817179) until the screws reaches the inner diameter of the plate, as shown below.

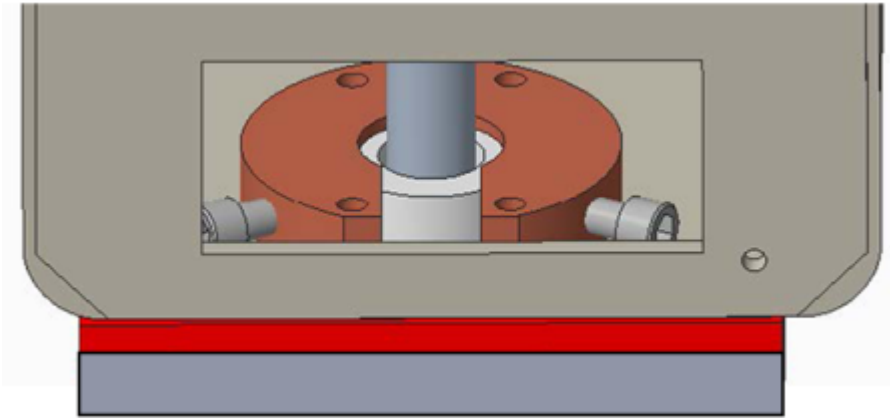
Figure 7-16 M14 screws installed in the mounting plate



1	M14 screws
2	Mounting plate

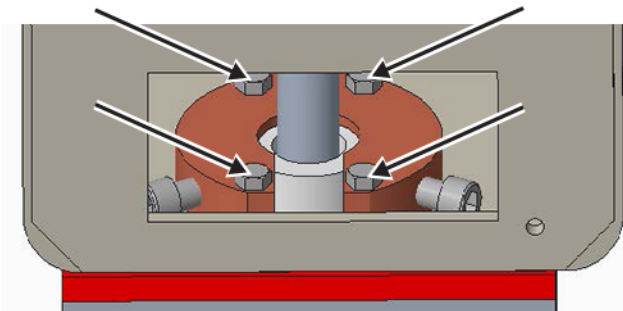
4. Put the mounting plate through the opening in the magnet foot, around the seismic rod and over the center ring, with the flush side of the plate pointed toward the ceiling. Make sure the flat edge of the mounting plate points toward the opening in the magnet foot.

Figure 7-17 Mounting plate around the seismic rod, over the center ring



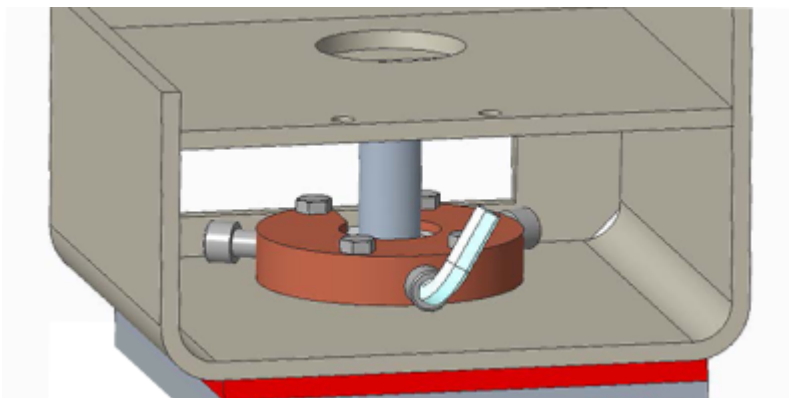
5. Screw four of the M10 screws (1000-M10C035-31) from the kit through the mounting bracket and engage the threads into the magnet foot. Using a ratchet or wrench, firmly tighten the M10 screws.

Figure 7-18 M10 screws



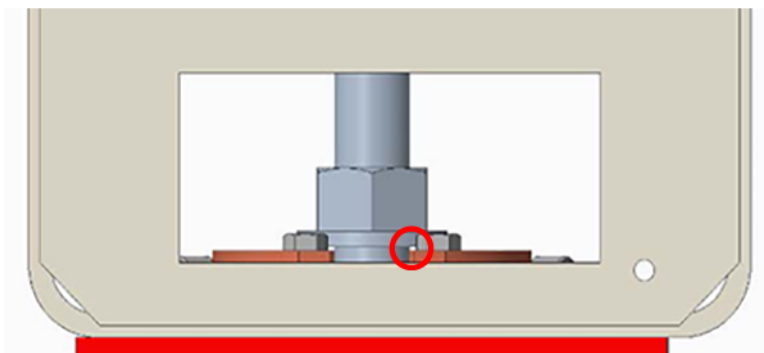
6. Hand-tighten the three M14 screws up to the center ring. If needed, use the hex tool (5822744) from the kit to help in tightening.

Figure 7-19 Using the hex tool to tighten M14 screws



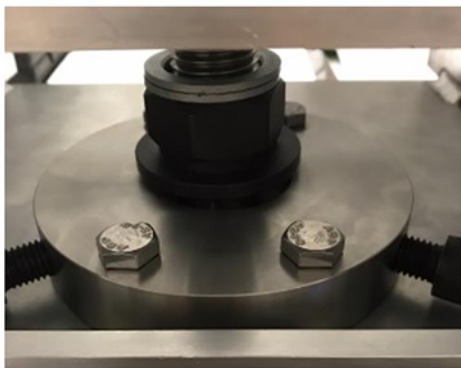
7. Put an appropriately sized flange nut on the seismic stud. Hand-tighten the flange nut toward the mounting plate; then back off one full turn to make sure there is a gap between the flange nut and the mounting plate.

Figure 7-20 Gap between flange nut and mounting plate



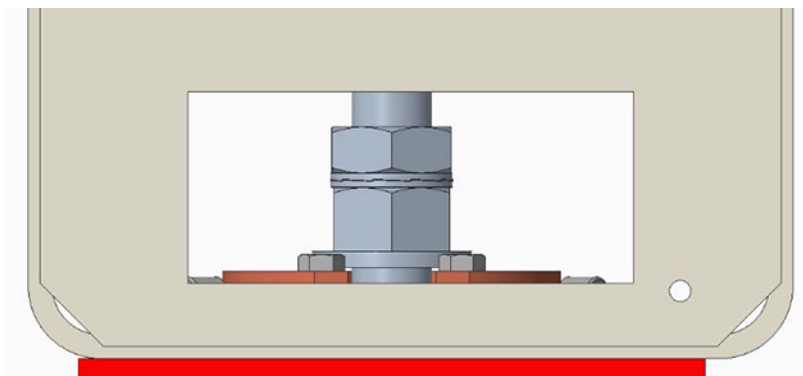
8. Put a Nord-Lock washer on top of the flange nut.

Figure 7-21 Nord-Lock washer on flange nut



9. Put a jam nut on top of the Nord-Lock washer and hand-tighten.

Figure 7-22 Jam nut on Nord-Lock washer



10. Using two wrenches, tighten the jam nut while keeping the flange nut in position. Make sure there is still a gap between the flange nut and the mounting plate.

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Chapter 8 Restoring the front and rear endbell

Prerequisites

Tools and test equipment			
Item	Quantity	Part number	Manufacturer
Nonmagnetic Titanium Service Tool Kit, Large Set, including Allen wrench to adjust body coil RL position	1	5112581	-

Safety
Before working in any GE HealthCare MR suite or doing any GE HealthCare service procedure, you must: • Have read and understood all hazard conditions and safety requirements in the latest revision of the GE HealthCare <i>MR Service Safety Manual</i> (5452735). • Have successfully completed all relevant GE HealthCare Environmental Health and Safety (EHS) courses (or for non-GE HealthCare employees, equivalent workplace training courses). • Comply with all site-specific training and workplace safety requirements. If you have any safety concerns at any time, do not begin work or immediately stop work and move to a safe location. Immediately contact your supervisor or site safety officer for instructions on how to proceed.

About this task

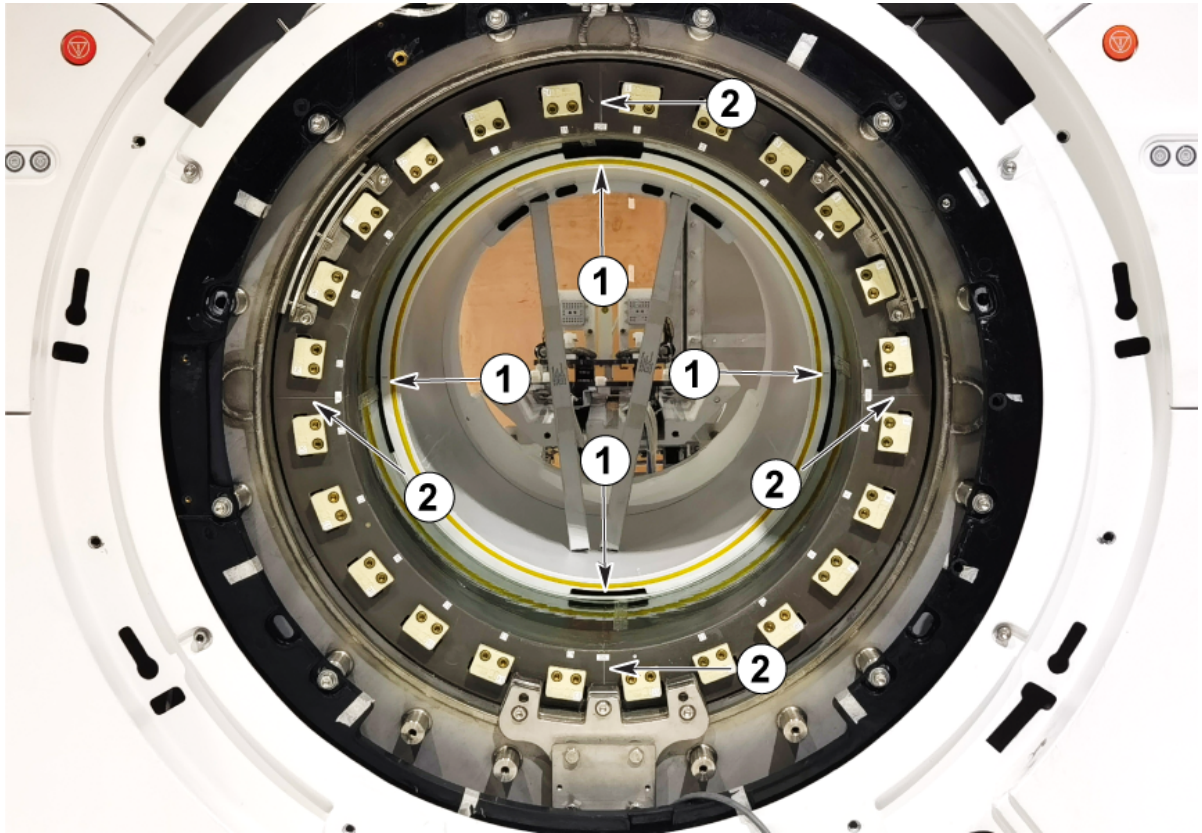
The following procedures are required after the magnet is positioned and centered in the MR suite.

Centering the body coil

Procedure

1. Position RF body coil aligning marks on it with gradient coil marks, with a tolerance of ± 2 mm.

Figure 8-1 Position the RF body coil

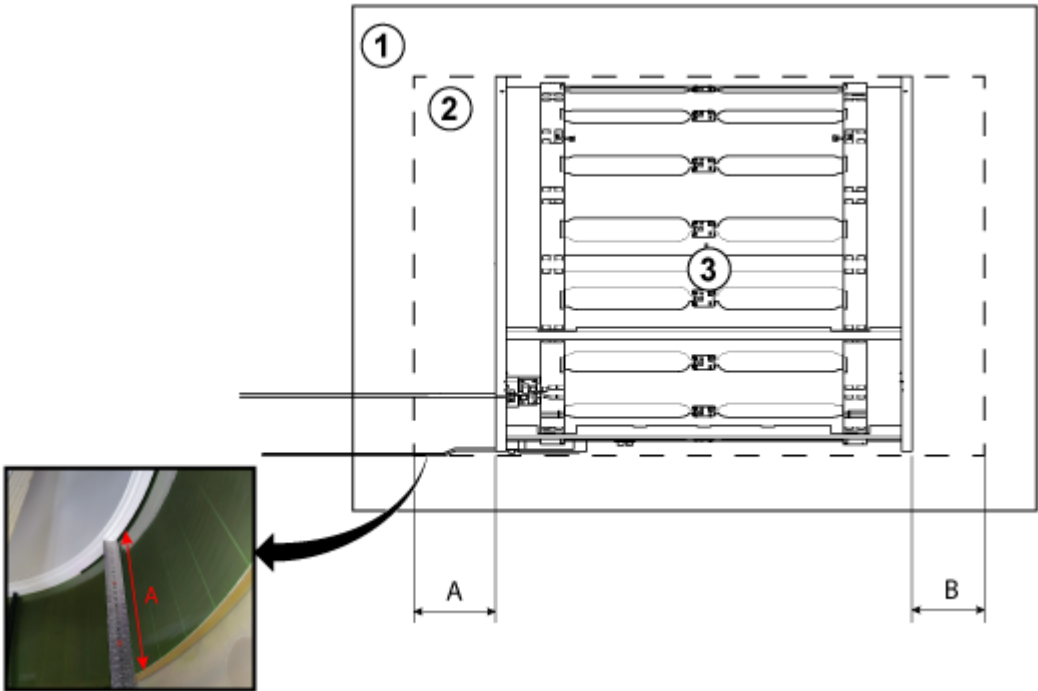


Item	Description
1	The marks on RF body coil
2	The marks on gradient coil

2. Center the body coil:
 - 2.1. Check the distance of "A" by ruler.
 - 2.2. Check the distance of "B" by ruler.

2.3. "A" - "B" ≤1mm, then the RF body coil be centered.

Figure 8-2 Center the RF body coil



Item	Description
1	Gradient coil
2	Gradient coil shielding
3	RF body coil

Installing the rear endbell

Prerequisites

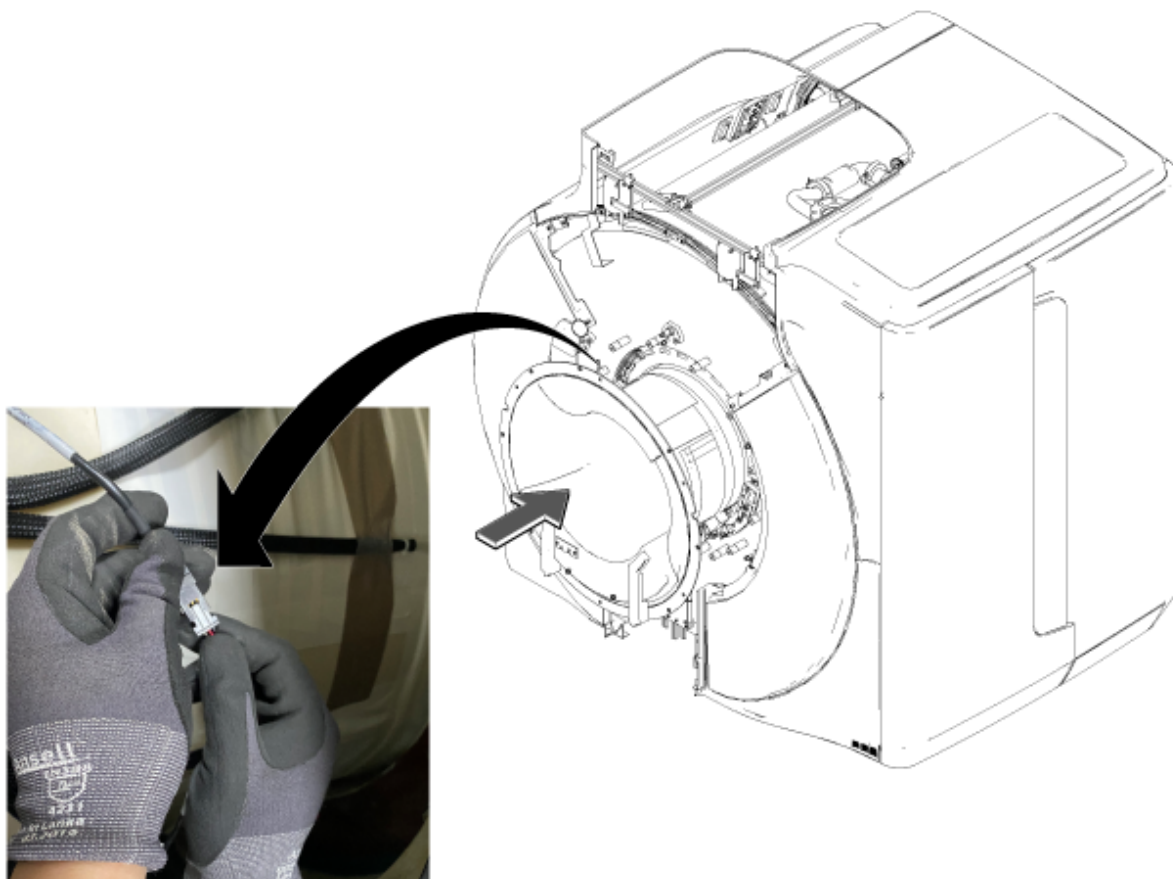
Personnel requirements			
Required persons	Preliminary requirements	Procedure	Finalization
1	-	30 minutes	-

Procedure

1. Align rear endbell with rear of magnet, partially insert the rear endbell.

2. Connect the speaker power cable. Route the speaker power cable out of the endbell.

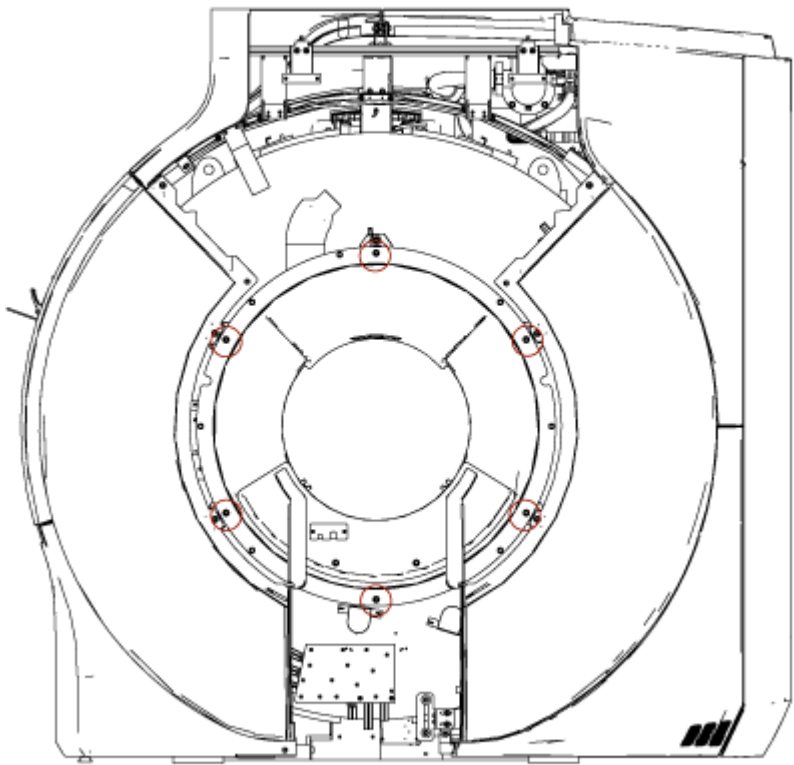
Figure 8-3 Connecting speaker cable



3. Install the rear endbell onto the magnet.

- 4. Fully install rear endbell by six M10 x 25 Socket Head Cap Screws.

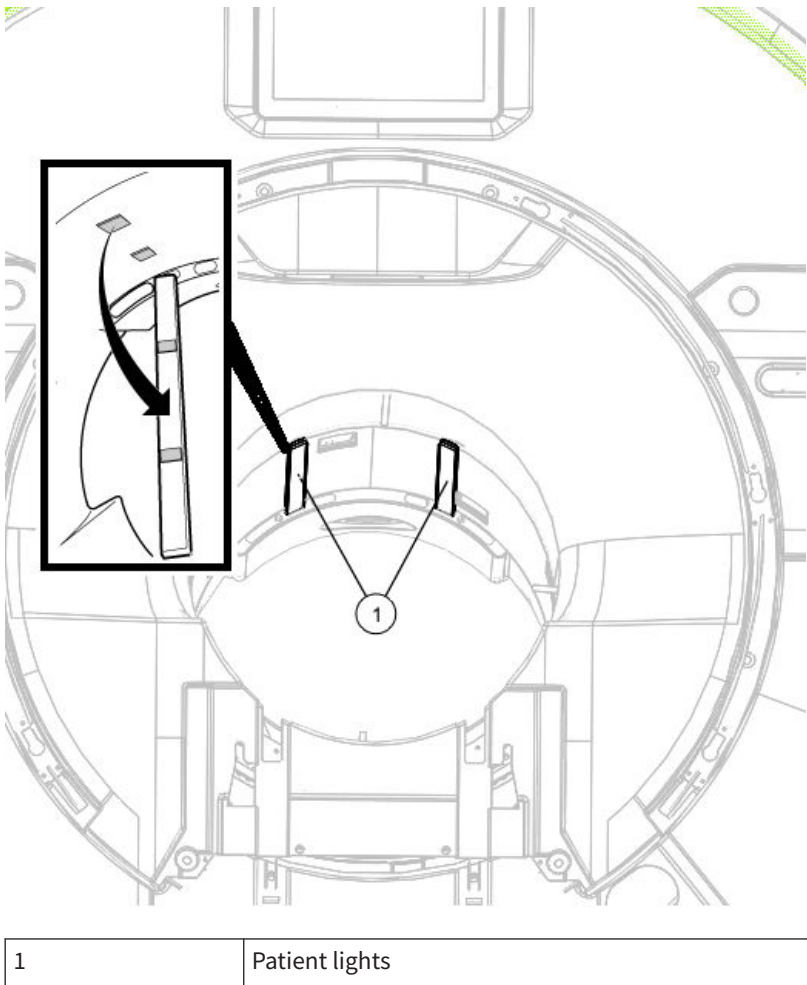
Figure 8-4 Rear endbell screws



Item	Quantity	Part Number	Manufacturer
M10 Screw	6	1000-M10C025-33	-
M10 Washer	6	5146169	-

5. Install the patient lights to the magnet bore upper side. The patient light is secured by Velcro tape.

Figure 8-5 Patient lights



1	Patient lights
---	----------------

Installing the front endbell

Prerequisites

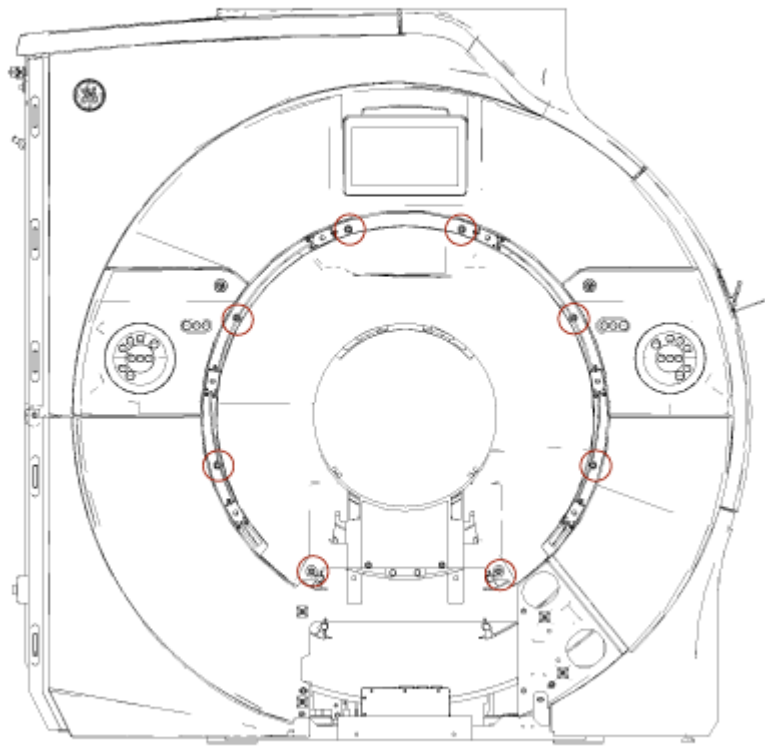
Personnel requirements			
Required persons	Preliminary requirements	Procedure	Finalization
1	-	10 minutes	-

Procedure

1. If installed, lower the patient table to the **down-limit** position.
2. Connect the laser light cable.
3. Connect the sensor cable (Run# 1260) to J10 connector which on SRI (Scan Room interface) assembly.

- 4. Install eight M10 x 25 Socket Head Cap Screws (1000-M10C025-33) and M10 Nylon Washer (2000-M10-30) securing the front endbell to the magnet.

Figure 8-6 Front endbell screws



Installing the fixed table front bridge

Prerequisites

Personnel requirements			
Required persons	Preliminary requirements	Procedure	Finalization
2	-	20 minutes	-

Procedure

1.



POTENTIAL PINCH HAZARD

Fingers caught between the front and rear bridge or the front bridge and the magnet bore can be pinched.

Lower the front bridge into place as close to the rear bridge as safely possible, then slide the front bridge into place.

Install the front bridge into the magnet bore.

2.



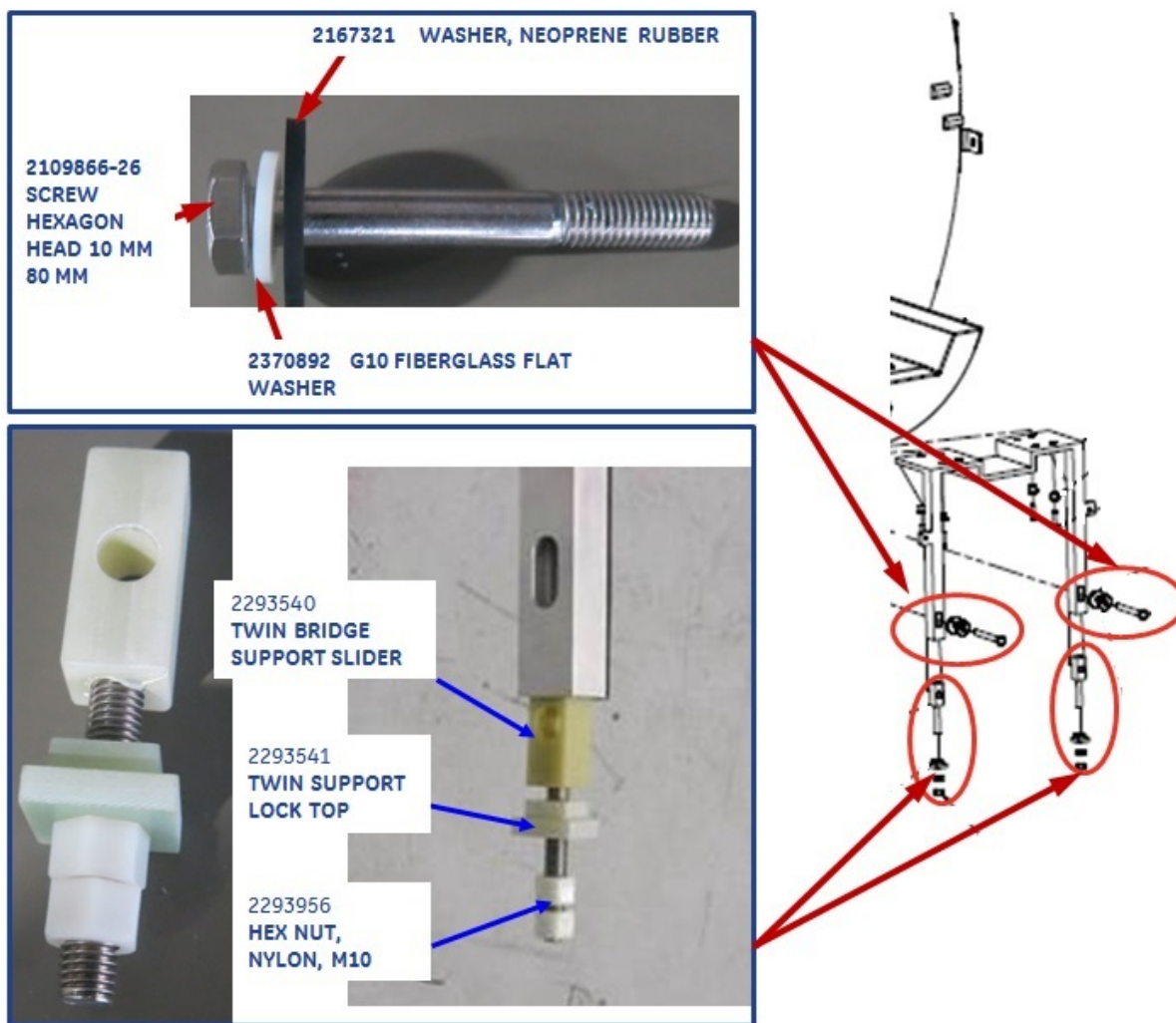
THE BRIDGE WEIGHT IS APPROX 23KG.

The bridge removal needs two people to damage prevention for enclosure bore.

When the installing the bridge, care should be taken to ensure the drive belt comes along with the bridge.

Install two screws securing to the magnet.

Figure 8-7 Install Bridge



Installing the bridge cover

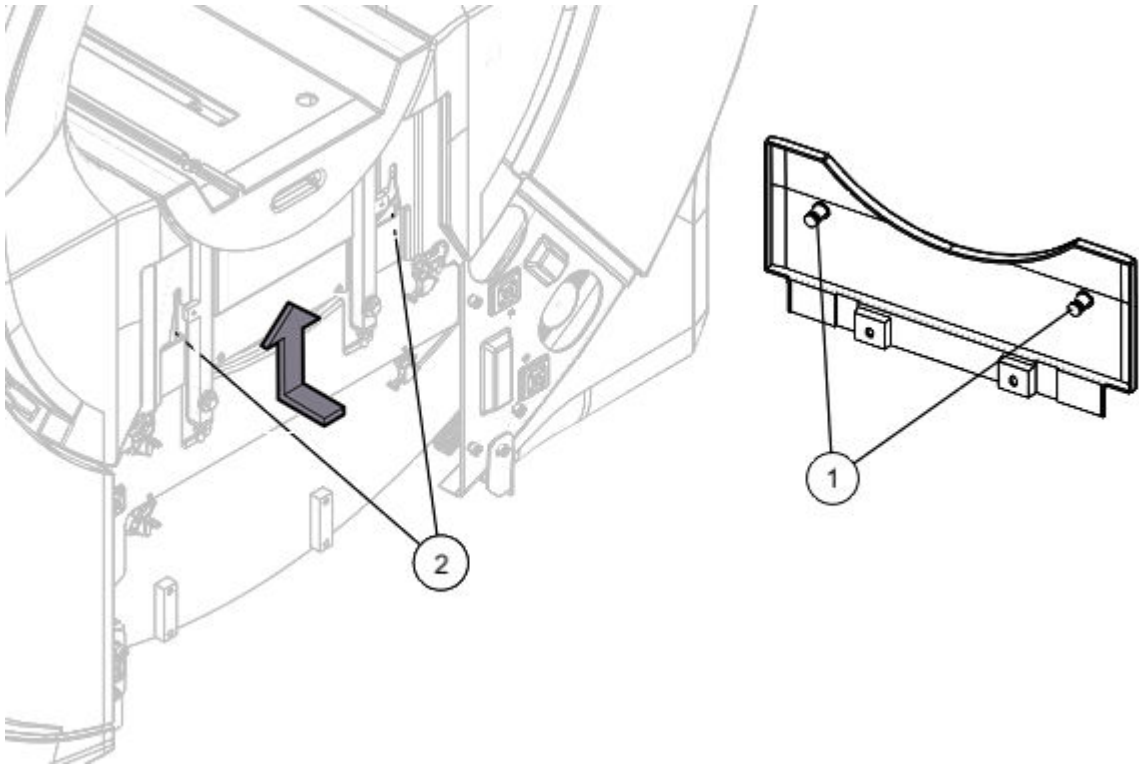
Prerequisites

Personnel requirements			
Required persons	Preliminary requirements	Procedure	Finalization
1	-	5 minutes	-

Procedure

- 1. Align the tabs on the back of the bridge cover with the tracks on the front endbell.

Figure 8-8 Bridge cover alignment

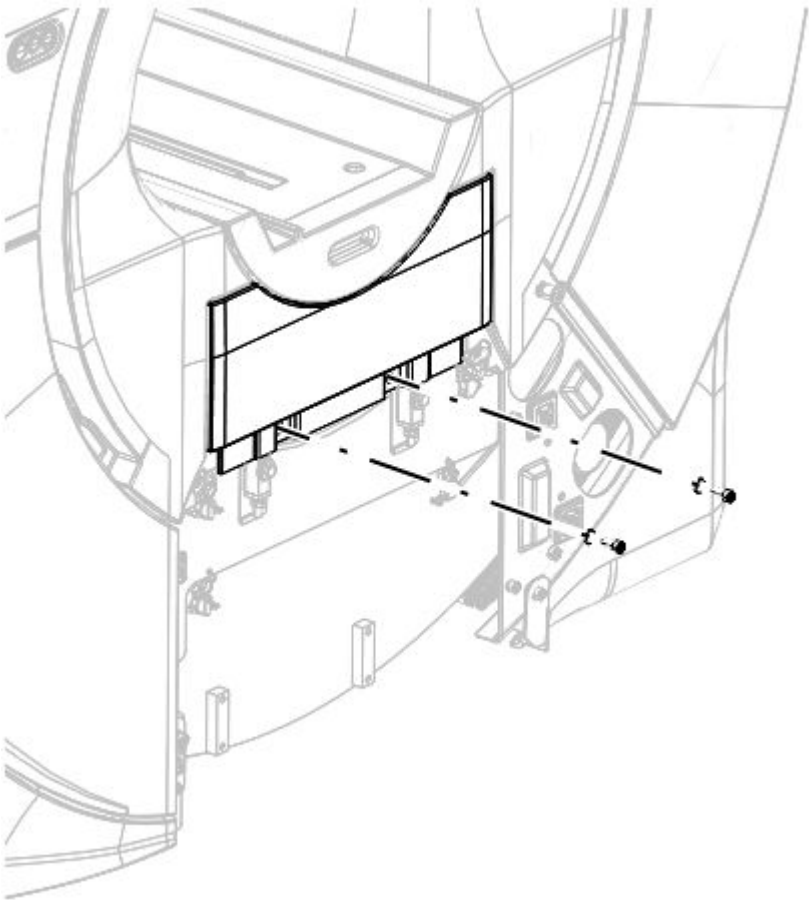


Item	Description
1	Tabs
2	Tracks

- 2. Slide the bridge cover upward engaging the tabs into the tracks.

- 3. Install the two screws securing the bridge cover to the front endbell.

Figure 8-9 Bridge cover screws



Item	Quantity	Part Number	Manufacturer
M10 Screw	2	1000-M10C025-33	-
Washer	2	2000-M10-10	-

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Revision history

Revision	Date	Description
Controlled document for English is posted as DOC2907271.		
3	February, 2026	- Chapter 2 Unloading and moving the magnet: - Section 2.2 <i>Equipment requirements</i>, corrected the forklift minimum length and inside distances. - Section 2.4 <i>Handling the crated magnet in transit with a crane</i>, changed the requirement for using crane to clear access to the top of the magnet. - Section 2.6 <i>Removing the shipping cage</i>, updated the image and table for Figure 2-6 Magnet B, C, D, E, and P cables to remove Magnet C. - Added section 3.1 <i>Converting an air magnet shipment to warehouse configuration</i> to the manual. - Section 6.8 <i>Leveling the magnet</i>, updated the image and table for Figure 6-21 Shim pads. - Chapter 7 Installing a seismic vibroacoustic damping mat and leveling the magnet: - Section 7.6 <i>Adding leveling shims</i>, updated the image and table for Figure 7-8 Adding shim pads. - Section 7.8 <i>Seismic mounting of a vibroacoustic damping mat to the floor</i>, added the part number for Seismic Anchor Kit, Platform. - Chapter 8 Restoring the front and rear endbell, removed not-used sections and updated the image for Figure 8-2 Center the RF body coil.

Revision	Date	Description
2	March, 2025	• All references to damping mats were updated to vibroacoustic damping mats. • All content related to JXM small magnet foot was updated with content applied to large magnet foot. • All content related to lifting rail was updated with content applied to the lifting angle bracket. • Section 1.3 <i>Truck loading specifications</i>, the options of Airline and Boat or ocean-going ship were added to the Table 1-2 Magnet loading specifications. • Chapter 2 Unloading and moving the magnet, was added with following sections: ◦ 2.3 <i>Removing subsystem crates</i> ◦ 2.4 <i>Handling the crated magnet in transit with a crane</i> ◦ 2.5 <i>Handling the crated magnet in transit with a forklift</i> ◦ 2.7 <i>Removing the shipping crate</i> • Section 2.6 <i>Removing the shipping cage</i>, all content related to shipping beams was removed. • Section 5.1 <i>Preparing the scan room for leveling magnet</i>, remove step 7 and Figure 5-6 Magnet isocenter height. • Chapter 6 Installing a nonseismic vibroacoustic damping mat and leveling the magnet: ◦ Section 6.1 <i>Equipment overview</i>, update the Part Number of vibroacoustic damping mats, 20mm, vibroacoustic damping mats, 10mm, Aluminum shims, 3mm, and Aluminum shims, 1mm, under the table of Accessories of JXM magnet (5926243-301). ◦ Section 6.2 <i>Positioning a nonseismic mounted vibroacoustic damping mats</i>, update Figure 6-2 Vibroacoustic damping mat placement (patient end is at the bottom). ◦ Section 6.5 <i>Preparations for centering the magnet</i>, remove the step of 4.1. • Chapter 7 Installing a seismic vibroacoustic damping mat and leveling the magnet, was added to the manual. • Chapter 8 Finalization, was added to the manual with two sections: ◦ 8.1 <i>Centering the table dock bolt</i> ◦ 8.2 <i>Restoring the front and rear endbell</i>
1	April, 2024	Initial release.



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